Understanding the Counterexample Construction

Accompanying answers

178 steps in twelve modules

30 October 2026

Purpose. Reconstruct the two main proofs in L. Cao's *Hierarchical Laminates Cannot Reproduce All Effective Properties of Planar Two-Phase Composites: A Counterexample with a 16-Component Potential* [1]. The exercises lead to Theorems 1.1 and 6.1 of the manuscript.

How to use the pair. Read a module's setup, then work in order. Use results from earlier steps when cited. Keep the answer book closed until you have tried a step; afterward compare the reasoning, not just the last formula. The answer book uses the same step numbers and repeats every question, so it can also serve as a self-contained guided reading of the proof. Three to five steps make a useful study session.

Difficulty key. Every question and its answer carry the same colored text label. The rating measures the work in that step, using the module's prerequisites and supplied background facts.

D1: Direct	A definition, calculation, or immediate application.
D2: Short proof	A short proof combining a few stated facts.
D3: Guided technical	A more delicate transition, with an explicit route or substeps in the question and a detailed answer.

The words and D1–D3 codes remain meaningful in grayscale. The definition finder, per-step lookup links, and background index are available in both books.

Prerequisites. Multivariable calculus, elementary real analysis, linear algebra including symmetric matrices, basic Hilbert spaces and weak derivatives, and introductory Fourier series. The supplied facts below cover the substantial general theory used by the manuscript. Prove their applications, rather than reproving the general theorems.

Background supplied as an input	Used in
Periodic Sobolev facts and Lax–Milgram (B1, B2)	Module 1
Separated-scale composition (B4)	Modules 2, 9
Fourier completeness and Parseval (B5, B6)	Modules 6, 10
The Cauchy determinant identity (B7)	Module 7
Weak derivatives and interface gluing (B8)	Module 8
Uniform higher integrability (B9)	Module 9
Smoothing and quasiconvex testing (B1, B10)	Module 12

Conventions used throughout. The cell is $Y = (0, 2\pi)^2$ and $\langle f \rangle = |Y|^{-1} \int_Y f$. A gradient is a $2 \times m$ matrix, with spatial derivatives in its rows. The vectors F_x, F_y are the transposed rows. The dot is the Frobenius inner product; $\|A\|$ for a matrix is its operator norm, whereas $|F|$ is the Frobenius norm of a gradient. $A \preceq B$ means $v \cdot Av \leq v \cdot Bv$ for all v. Energies have no factor $1/2$. Horizontal layers have normal $(0, 1)$. Constants in $O_\rho(N^{-1})$ may depend on the fixed contrast ρ but are independent of N.

Reading map

Module	Steps	Manuscript
1	1.1–1.12	Section 2.1
2	2.1–2.12	Sections 2.2–2.3, 3.1
3	3.1–3.14	Appendix A.1
4	4.1–4.12	Sections 3.1–3.3
5	5.1–5.17	Section 3.3, Appendix A.2
6	6.1–6.22	Sections 4.1–4.2, Appendix B
7	7.1–7.23	Section 4.3
8	8.1–8.18	Sections 5.1–5.2
9	9.1–9.12	Section 5.3
10	10.1–10.13	Section 6.1, planar duality
11	11.1–11.11	Section 6.2
12	12.1–12.12	Section 6.3, synthesis

The first theorem follows from Modules 1–9. The second theorem uses that binary witness and Modules 10–12. The essential chain is

Explicit matrix identities and a compact constitutive family
$\Downarrow$
An energy equality for every hierarchical laminate and its limits
$\Downarrow$
A convergent Fourier calculation gives a positive fifth-order difference
$\Downarrow$
A flux correction transfers it to finite binary stripes
$\Downarrow$
Exact prescribed fractions and an actual periodic witness
$\Downarrow$
Complementary energy and a rank-one convex minorant
$\Downarrow$
A strict gap between the two envelopes

Module 2 uses the composition theorem for the G-closure. Module 9 derives its exact-fraction correction from separated-scale recovery and the higher-integrability estimate.

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8 Replace the smooth medium by binary stripes 50

9 Prescribed fractions and more potentials 58

10 Complementary energy and planar duality 63

11 A rank-one convex lower bound 68

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Notation and definition finder

The lookup line below each step links to its module guide and setup paragraphs. The "Uses" list links to earlier results needed for the argument. The table below locates the main definitions.

Notation	Meaning and first useful location
$Y, \langle f \rangle, \|f\|_p$	Periodic cell and normalized integral/norm; Setup 1.1; normalized L^p in Setup 9.2.
$H^1_{\mathrm{per}}, \mathcal{H}$	Periodic Sobolev space and its zero-mean subspace; Setups 1.1–1.2, B1.
$E \cdot J, \|E\|, \|A\|$	Frobenius product/norm for gradients and operator norm for tensors; Setup 1.2.
$A \preceq B$	Order of symmetric quadratic forms; Setup 1.2, B3.
u_F, e_F, j_F	Cell minimizer, $e_F = F + \nabla u_F$, and $j_F = L e_F$; Step 1.4–Step 1.6.
$B = L_*$	Effective tensor: $BF = \langle j_F \rangle$ and $F \cdot BF = \min\langle (F+\nabla v) \cdot L(F+\nabla v) \rangle$; Step 1.9–Step 1.10.
n, λ, z, F_1, F_2	Layer normal, first-material layer weight, compatible jump, and layer gradients; Step 2.1–Step 2.3.
$\chi, \theta, G_\theta, G^{\mathrm{L}}_\theta$	Binary indicator, its mean, and the two exact-fraction tensor closures; Setup 2.3, Step 2.11–Step 2.12.
O_j, Γ_j, Ω	Explicit 4×4 blocks, 8×8 generators, and $\Omega = \Gamma_1\Gamma_2\Gamma_3\Gamma_4$; Step 3.1–Step 3.6.
$Q(h), \mathcal{S}$	$Q(h) = \sum_j h_j \Gamma_j$ and $\mathcal{S} = \mathrm{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$; Setup 3.3, Step 3.7–Step 3.9.
P_j, P, S	Four row blocks, their stack $P \in \mathbb{R}^{16\times 8}$, and $S = P^\top P$; Step 3.11–Step 3.14.
$D, L(H), \mathcal{K}, \rho$	Constant vertical block, constitutive family, compact parameter family, and contrast; Setup 4.1, Step 4.2–Step 4.4.
$Q = PS^{-1}, \xi, \eta, \ell$	Lifting (a different Q from $Q(h)$), two detecting loads, and their energy difference; Step 4.6–Step 4.8.
$\mathcal{R}(k), T(k)$	$\mathcal{R}(k) = (k_x^2 I_{16} + k_y^2 D)^{-1}$ and its compressed multiplier $T(k)$; Step 4.9–Step 4.12. The convention at $k = 0$ is stated in Module 6.
$\widehat{f}(k), \|f\|_{\mathcal{A}}, \mathcal{T}$	Fourier coefficients, absolute-coefficient norm, and multiplier operator; Setups 6.1–6.2, Step 6.2–Step 6.5.
t, a, b, c, X, X_p, H_*	Smooth fraction field, its three frequencies, source, series terms, and mean source; Step 6.1–Step 6.8.
$d(M), K_j, \kappa$	Entry detector, partial frequency sums, and positive fifth-order constant; Step 7.1, Step 7.9, Step 7.20.
χ_N, h, t_k, w_N, r_N	Binary stripes, strip width, sampled fraction, zero-edge primitive, and constitutive residual; Step 8.1–Step 8.3, Step 8.11.
C_θ, B_χ, g_χ	Ideal tensor at fraction θ, selected actual periodic tensor, and $g_\chi = \ell(B_\chi) > 0$; Step 9.1–Step 9.3, Step 9.9.
$R_\perp, C^{\mathrm{d}}$	Row rotation and planar dual $R_\perp C^{-1} R_\perp^\top$; Setup 10.1, Step 10.2.

Notation	Meaning and first useful location
$F = (f, g), q_i, W$	Paired gradients, two quadratic branches, and their minimum; Setup 10.1, Step 10.9–Step 10.12.
$E_C, V, \mathrm{RC}\, W$	Paired energy, its minimum over $\mathcal{K}$, and rank-one convex envelope; Setup 11.1, Step 11.5–Step 11.8.
$\mathrm{QC}\, W, p, \overline{F}$	Quasiconvex envelope, $p = B_\chi \eta$, and mean paired gradient $(\xi, R_\perp p)$; Setup 12.1, B10.

Overloaded letters. A subscript $*$ denotes an effective object; H_* is an 8×8 mean source, whereas L_* is a tensor acting on gradients. In Module 9, A, B temporarily denote coefficient fields. In Step 12.3 they denote gradient matrices. Every such local reuse is specified in its setup. D always denotes the constant vertical block.

Background references. The keys B1–B12 link to the background index at the back, with precise hypotheses, use locations, and textbook or paper sections. The bibliography links to publisher or journal records where available.

1 The periodic cell problem, one step at a time

Prerequisites. Basic multivariable calculus and finite-dimensional inner products; no earlier course steps.

Definition lookup. The next two setup paragraphs define the periodic space, norms, weak equilibrium, corrector, and cell data. The effective tensor is defined in Step 1.9–Step 1.10.

Supplied background. B1, B2, B3. Each key links to its statement, scope, and reference.

Setup 1.1. Work on $Y = (0, 2\pi)^2$. Averages and L^2 norms are normalized: $\langle f \rangle = |Y|^{-1} \int_Y f$ and $\|f\|_2^2 = \langle |f|^2 \rangle$. A potential $v : Y \to \mathbb{R}^m$ has spatial-first gradient $\nabla v \in \mathbb{R}^{2\times m}$, with entries $(\nabla v)_{i\alpha} = \partial_i v_\alpha$. The dot denotes the Frobenius product. Let L be a measurable periodic self-adjoint coefficient field with $\mu I \preceq L(x) \preceq MI$ for almost every x, where $0 < \mu \le M$. Let $\mathcal{H} = \{v \in H^1_{\mathrm{per}}(Y; \mathbb{R}^m) : \langle v \rangle = 0\}$, $a(u, v) = \langle \nabla v \cdot L \nabla u \rangle$, and $b_F(v) = -\langle \nabla v \cdot LF \rangle$ for periodic H^1 arguments u, v and a constant mean gradient $F \in \mathbb{R}^{2\times m}$.

Setup 1.2. Definitions and background: H^1 consists of square-integrable functions with square-integrable weak first derivatives, and H^1_{per} is its closed subspace with matching opposite periodic traces (equivalently, the H^1 closure of smooth periodic functions). We use $\|v\|_{H^1}^2 = \|v\|_2^2 + \|\nabla v\|_2^2$. A Hilbert space is a complete inner-product space. The space H^1 with this norm is Hilbert, and its closed linear subspaces are Hilbert (B1). The Frobenius product is $E \cdot J = \sum_{i,\alpha} E_{i\alpha} J_{i\alpha}$; $\|A\| = \sup_{|z|=1} |Az|$ is the operator norm. Self-adjoint means $E \cdot LJ = J \cdot LE$, and $A \preceq B$ means $z \cdot Az \le z \cdot Bz$ for every z. Columnwise weak equilibrium $\operatorname{div} j = 0$ means $\langle \nabla \varphi \cdot j \rangle = 0$ for every smooth periodic vector test φ. A compatible field with mean F is $F + \nabla v$ with $v \in H^1_{\mathrm{per}}$. Smooth periodic density and the zero-mean Poincaré inequality $\|v\|_2 \le C_P \|\nabla v\|_2$ are supplied (B1). Lax–Milgram supplies a unique u solving $a(u, v) = b(v)$ on a Hilbert space when a is bounded and coercive and b is bounded linear (B2); coercive means $a(v, v) \ge c\|v\|_{H^1}^2$ for a constant $c > 0$. These theorems may be used throughout the exercises.

D2: Short proof

Answer 1.1. The mean of a periodic gradient

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

Task. Show that $\langle \nabla v \rangle = 0$ for every $v \in H^1_{\mathrm{per}}(Y; \mathbb{R}^m)$. First treat smooth v by integration, then use periodic smooth density and $|\langle g \rangle| \le \|g\|_2$ to pass to H^1.

Solution. For a smooth scalar component and fixed y, periodicity gives $\int_0^{2\pi} \partial_x v_\alpha(x, y)\, dx = v_\alpha(2\pi, y) - v_\alpha(0, y) = 0$. Integrate in y; the same argument works for the y derivative. This proves the statement componentwise for smooth periodic potentials. For general periodic H^1 potentials, choose smooth periodic $v_n \to v$ in H^1. Since $|\langle \nabla v_n - \nabla v \rangle| \le \|\nabla v_n - \nabla v\|_2 \to 0$, their averages converge, and every $\langle \nabla v_n \rangle$ is zero. Thus adding a periodic gradient preserves the prescribed mean.

D1: Direct

Answer 1.2. Coercivity

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

Task. Using the stated Poincaré inequality, find $c > 0$ such that $a(v, v) \ge c\|v\|_{H^1}^2$ on $\mathcal{H}$.

Solution. Pointwise ellipticity and averaging give $a(v, v) \ge \mu \|\nabla v\|_2^2$. On $\mathcal{H}$, Poincaré gives

$$\|v\|_{H^1}^2 = \|v\|_2^2 + \|\nabla v\|_2^2 \le (1 + C_P^2)\|\nabla v\|_2^2.$$

Thus one can take $c = \mu/(1 + C_P^2)$. Removing the constant part of the potential was essential: a nonzero constant has positive H^1 norm but zero gradient energy.

D1: Direct

Answer 1.3. Bounded data

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

Task. For $u, v \in \mathcal{H}$, show that $|b_F(v)| \le M|F|\|\nabla v\|_2$ and $|a(u,v)| \le M\|\nabla u\|_2\|\nabla v\|_2$.

Solution. A positive self-adjoint matrix satisfying $L \preceq MI$ has operator norm at most M. Therefore $|\nabla v \cdot LF| \le M|\nabla v||F|$. The normalized Cauchy–Schwarz inequality gives

$$|b_F(v)| \le M|F|\langle|\nabla v|\rangle \le M|F|\|\nabla v\|_2.$$

Likewise $|\nabla v \cdot L\nabla u| \le M|\nabla v||\nabla u|$, whose average is at most $M\|\nabla v\|_2\|\nabla u\|_2$. Since $\|\nabla v\|_2 \le \|v\|_{H^1}$, these inequalities give boundedness in the H^1 norm.

D2: Short proof

Answer 1.4. Existence of the corrector

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.2, 1.3.

Task. Apply Lax–Milgram to obtain a unique $u_F \in \mathcal{H}$ with $a(u_F, v) = b_F(v)$ for all $v \in \mathcal{H}$. Explain why the same equation holds for arbitrary periodic H^1 tests. For the Hilbert-space hypothesis, verify that $v \mapsto \langle v \rangle$ is continuous and that its zero set is closed.

Solution. The mean map is continuous because $|\langle v \rangle| \le \|v\|_2 \le \|v\|_{H^1}$. Thus its kernel, the zero-mean space $\mathcal{H}$, is a closed subspace of the periodic H^1 Hilbert space and is itself Hilbert. Step 1.2 supplies coercivity; Step 1.3 supplies boundedness of the bilinear form and functional. Lax–Milgram therefore gives a unique $u_F \in \mathcal{H}$ satisfying

$$\langle \nabla v \cdot L(F + \nabla u_F) \rangle = 0 \quad (v \in \mathcal{H}).$$

For an arbitrary periodic test v, the function $v - \langle v \rangle$ belongs to $\mathcal{H}$ and has the same gradient as v. Hence the equation is valid for all periodic H^1 tests. Uniqueness is for a zero-mean potential; without that normalization one may add a constant.

D2: Short proof

Answer 1.5. A weak solution minimizes energy

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4.

Task. For $v \in H^1_{\mathrm{per}}(Y;\mathbb{R}^m)$, put $E_F(v) = \langle (F + \nabla v) \cdot L(F + \nabla v) \rangle$. Show that $E_F(v) - E_F(u_F) = a(v - u_F, v - u_F)$, and deduce uniqueness of the minimizing gradient. Write $w = v - u_F$, expand the quadratic, and test the weak equation with w.

Solution. Write $w = v - u_F$ and $e_F = F + \nabla u_F$. Expand the quadratic:

$$\begin{aligned} E_F(v) &= \langle (e_F + \nabla w) \cdot L(e_F + \nabla w) \rangle \\ &= E_F(u_F) + 2\langle \nabla w \cdot Le_F \rangle + \langle \nabla w \cdot L\nabla w \rangle. \end{aligned}$$

The middle term is zero by Step 1.4. The last term is $a(w,w) \ge 0$, proving both the identity and minimality. If v is another minimizer, ellipticity gives $\mu\|\nabla w\|_2^2 \le 0$, so the two gradients agree. A zero-mean potential is then unique by Poincaré. This comparison will later identify explicitly constructed fields with actual cell minimizers.

D1: Direct

Answer 1.6. Equilibrium

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4.

Task. Set $e_F = F + \nabla u_F$ and $j_F = Le_F$. Show that $\operatorname{div} j_F = 0$ in distributions, where divergence is taken in each column.

Solution. Since L is bounded and $e_F \in L^2$, the flux j_F belongs to L^2. For a smooth periodic vector test φ, the definition of distributional divergence is

$$\langle \operatorname{div} j_F, \varphi \rangle = -\langle \nabla\varphi \cdot j_F \rangle.$$

Here the brackets denote distributional pairing on the torus, with the same normalization as the cell average. The right side is zero by Step 1.4, because $j_F = L(F + \nabla u_F)$. Thus each column has zero divergence. This is the weak equilibrium equation for the flux.

D2: Short proof

Answer 1.7. Linearity in the mean gradient

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4.

Task. Use uniqueness in Step 1.4 to show $u_{\alpha F+\beta G} = \alpha u_F + \beta u_G$ for all constant $F, G \in \mathbb{R}^{2\times m}$ and real α, β.

Solution. The candidate $w = \alpha u_F + \beta u_G$ is periodic and zero-mean. Linearity of a and b gives, for every test v,

$$a(w, v) = \alpha b_F(v) + \beta b_G(v) = b_{\alpha F+\beta G}(v).$$

Hence w solves the defining equation for $u_{\alpha F+\beta G}$. Its uniqueness from Step 1.4 proves the claim. Consequently the local gradient, the local flux, and its average all depend linearly on the mean gradient.

D1: Direct

Answer 1.8. The mean gradient really is prescribed

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.1, 1.6.

Task. Show that $\langle e_F \rangle = F$.

Solution. By definition $e_F = F + \nabla u_F$. The constant F has average F, and Step 1.1 gives $\langle \nabla u_F \rangle = 0$. Therefore $\langle e_F \rangle = F$. The affine part of the full potential carries the imposed mean gradient; its periodic corrector only redistributes that gradient within the cell.

D2: Short proof

Answer 1.9. Symmetry of the mean-flux map

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4, 1.6, 1.7.

Task. Define $BF = \langle j_F \rangle$, which is linear by Step 1.7. Show that $G \cdot BF = \langle e_G \cdot Le_F \rangle = F \cdot BG$ for all constant $F, G \in \mathbb{R}^{2\times m}$, hence B is self-adjoint. Use u_G as a test for u_F and u_F as a test for u_G.

Solution. The linear map $BF = \langle j_F \rangle$ is well-defined by Step 1.7. Use u_G as a weak test in the equation for u_F:

$$\langle \nabla u_G \cdot Le_F \rangle = 0.$$

Adding the constant G inside the first factor gives

$$G \cdot BF = \langle G \cdot Le_F \rangle = \langle e_G \cdot Le_F \rangle.$$

Self-adjointness of L changes the last expression to $\langle e_F \cdot Le_G \rangle$. Testing the equation for u_G with u_F removes ∇u_F and gives $F \cdot BG$. Thus B is self-adjoint.

D1: Direct

Answer 1.10. The cell energy

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4, 1.5, 1.9.

Task. Show that $E_F(u_F) = F \cdot BF$. Thus the mean-flux definition and the minimum-energy definition give the same effective tensor $B = L_*$. Test the weak equation with u_F to remove the fluctuating part of the energy.

Solution. Use u_F as a test in its own equation to obtain $\langle \nabla u_F \cdot j_F \rangle = 0$. Hence

$$E_F(u_F) = \langle e_F \cdot j_F \rangle = \langle F \cdot j_F \rangle + \langle \nabla u_F \cdot j_F \rangle = F \cdot \langle j_F \rangle = F \cdot BF.$$

Step 1.5 says this is the minimum of E_F. Thus the minimum-energy and mean-flux descriptions are identical. Since B is self-adjoint by Step 1.9, the quadratic energy $q(F) = F \cdot BF$ determines every pairing by $2G \cdot BF = q(F+G) - q(F) - q(G)$, and hence determines the effective tensor.

D2: Short proof

Answer 1.11. Effective ellipticity

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.1, 1.5, 1.10.

Task. Show that $\mu|F|^2 \le F \cdot BF \le M|F|^2$. For the upper bound, try $v = 0$; for the lower bound, use $\langle \nabla v \rangle = 0$.

Solution. For any periodic v, Step 1.1 gives

$$\langle |F + \nabla v|^2 \rangle = |F|^2 + 2F \cdot \langle \nabla v \rangle + \|\nabla v\|_2^2 \ge |F|^2.$$

Ellipticity then yields $E_F(v) \ge \mu|F|^2$; taking the minimum proves the lower bound. For the upper bound, the admissible trial $v = 0$ gives $F \cdot BF \le \langle F \cdot LF \rangle \le M|F|^2$. The bounds retain the same ellipticity constants as L.

D2: Short proof

Answer 1.12. A useful gradient–flux identity

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

Task. If $j \in L^2_{\mathrm{per}}(Y; \mathbb{R}^{2\times m})$ is divergence-free and $\langle j \rangle = p$, show that $\langle (F + \nabla v) \cdot j \rangle = F \cdot p$ for every constant $F \in \mathbb{R}^{2\times m}$ and every periodic H^1 potential v. Extend the weak equilibrium identity from smooth tests by density.

Solution. The weak equation $\operatorname{div} j = 0$ means $\langle \nabla\varphi \cdot j \rangle = 0$ for all smooth periodic φ. Approximate v strongly in periodic H^1 by smooth v_n. Since $j \in L^2$, Cauchy–Schwarz implies $\langle \nabla v_n \cdot j \rangle \to \langle \nabla v \cdot j \rangle$, so the latter is also zero. Finally

$$\langle (F + \nabla v) \cdot j \rangle = F \cdot \langle j \rangle + 0 = F \cdot p.$$

The constant part of a gradient pairs only with the mean flux; the periodic part is orthogonal to an equilibrium flux.

2 Layers, the common block, and the two closures

Prerequisites. Cell minimization and testing: Step 1.4–Step 1.5, Step 1.10–Step 1.12.

Definition lookup. Layer coordinates and compatible fields are defined below; common spatial blocks are defined before Step 2.8; both closures and phase fraction are defined before Step 2.11.

Supplied background. B1, B4, B8. Each key links to its statement, scope, and reference.

Setup 2.1. Let B_1, B_2 be constant self-adjoint positive-definite tensors on $\mathbb{R}^{2\times m}$; positive definite means $E \cdot B_i E > 0$ for $E \neq 0$. Two planar layers have unit normal $n \in \mathbb{R}^2$, proportions $\lambda, 1-\lambda$ with $0 < \lambda < 1$, and mean gradient F. A unit tangent t satisfies $t \cdot n = 0$. The notation $n \otimes z$ means the matrix with entries $n_i z_\alpha$; the normal row $n^\top J$ is identified with a vector of $\mathbb{R}^m$. A compatible field means $F + \nabla v$ with periodic H^1 potential v, and weak equilibrium means $\langle \nabla\varphi \cdot j \rangle = 0$ for smooth periodic vector tests. For Steps 2.4–2.6 use the aligned period cell $\{sn + rt : 0 < s, r < 1\}$, with normalized averages and $s = n \cdot x$, $r = t \cdot x$. Module 1's variational proofs work on this cell as well. Periodicity on the aligned cell means invariance under translations by n and t. The resulting finite-dimensional minimum-energy formula defines the abstract laminate operation for every real unit normal (B4).

D1: Direct

Answer 2.1. The compatible jump

Lookup: module 2 guide; setups 2.1. *Uses:* local setup only.

Task. Continuity of a piecewise affine potential gives $t^\top(F_1 - F_2) = 0$. Show that $F_1 - F_2 = n \otimes z$ for some $z \in \mathbb{R}^m$.

Solution. Write $J = F_1 - F_2$. Each column of J is a vector in $\mathbb{R}^2$ orthogonal to t, because $t^\top J = 0$. The orthogonal complement of t is the one-dimensional span of n. Thus column α equals $z_\alpha n$ for a scalar z_α. Collecting these scalars gives $J = n \otimes z$, where $z = J^\top n$. Conversely, $J = n \otimes z$ gives $t^\top J = (t \cdot n) z = 0$, so this form characterizes the tangential compatibility condition.

D1: Direct

Answer 2.2. The two layer gradients

Lookup: module 2 guide; setups 2.1. *Uses:* 2.1.

Task. Solve $F_1 - F_2 = n \otimes z$ and $\lambda F_1 + (1-\lambda)F_2 = F$ to obtain $F_1 = F + (1-\lambda) n \otimes z$ and $F_2 = F - \lambda n \otimes z$.

Solution. Substitute $F_1 = F_2 + n \otimes z$ into $\lambda F_1 + (1-\lambda)F_2 = F$. This gives $F_2 + \lambda n \otimes z = F$, hence $F_2 = F - \lambda n \otimes z$. Adding the jump gives $F_1 = F + (1-\lambda) n \otimes z$. Direct substitution checks both equations. The factors $1-\lambda$ and $-\lambda$ ensure that the periodic gradient fluctuation has weighted mean zero.

D2: Short proof

Answer 2.3. The normal-flux equation

Lookup: module 2 guide; setups 2.1. *Uses:* 2.2.

Task. Let $\Phi(z) = \lambda F_1(z) \cdot B_1 F_1(z) + (1-\lambda) F_2(z) \cdot B_2 F_2(z)$, using Step 2.2. Differentiate in an arbitrary direction and show that its unique minimizer satisfies $n^\top B_1 F_1 = n^\top B_2 F_2$. To justify existence and uniqueness, define $K_i z = n^\top B_i (n \otimes z)$ and check that $(1-\lambda)K_1 + \lambda K_2$ is positive definite; the critical equation is a linear system with this matrix.

Solution. For any $w \in \mathbb{R}^m$, the directional derivative is

$$\begin{aligned} D\Phi(z)[w] &= 2\lambda(1-\lambda)(n \otimes w) \cdot B_1 F_1 - 2\lambda(1-\lambda)(n \otimes w) \cdot B_2 F_2 \\ &= 2\lambda(1-\lambda) w \cdot n^\top (B_1 F_1 - B_2 F_2). \end{aligned}$$

It is zero for every w exactly when the normal fluxes agree. The quadratic part of Φ in a nonzero increment w is

$$\lambda(1-\lambda)(n \otimes w) \cdot [(1-\lambda)B_1 + \lambda B_2](n \otimes w) > 0.$$

To obtain the minimizer, define $K_i w = n^\top B_i (n \otimes w)$. The identities $v \cdot K_i w = (n \otimes v) \cdot B_i (n \otimes w)$ show that K_i is symmetric positive definite. Substituting Step 2.2 into the normal-flux equation gives

$$[(1-\lambda)K_1 + \lambda K_2] z = -n^\top (B_1 - B_2) F.$$

The matrix on the left is positive definite and hence invertible. It gives a unique critical point; the positive quadratic part displayed above makes that point the unique global minimizer.

D2: Short proof

Answer 2.4. Construct a compatible layer potential

Lookup: module 2 guide; setups 2.1. *Uses:* 2.2, 2.3.

Task. For z from Step 2.3, construct a continuous periodic vector function $w(s)$ of period one with derivative $(1-\lambda)z$ on $0 < s < \lambda$ and $-\lambda z$ on $\lambda < s < 1$. Show that $F + \nabla[w(n \cdot x)]$ equals the two layer gradients on the aligned cell specified above. Start with $w(0) = 0$, integrate on each interval, and check the join at λ and the endpoint equality $w(1) = w(0)$.

Solution. For $s \in [0,1]$, define

$$w(s) = \begin{cases} (1-\lambda) s z, & 0 \le s \le \lambda, \\ \lambda(1-s) z, & \lambda \le s \le 1. \end{cases}$$

The two definitions agree at $s = \lambda$, and $w(0) = w(1) = 0$, so periodic extension is continuous and piecewise affine. Its derivative has exactly the two stated values. Since $\nabla[w(n \cdot x)] = n \otimes w'(n \cdot x)$, adding F gives F_1 and F_2 from Step 2.2. Continuity makes the integration-by-parts terms at the joins cancel, so the bounded piecewise derivative is also the weak derivative. Thus $w(n \cdot x)$ belongs to periodic H^1 on the aligned stripe cell (B8) and is an admissible corrector; a constant may be subtracted to normalize its mean without changing the gradient.

D3: Guided technical

Answer 2.5. Check equilibrium across the interfaces

Lookup: module 2 guide; setups 2.1. *Uses:* 2.3, 2.4.

Task. Let $j = B_i F_i$ in layer i. Use the normal-flux equality from Step 2.3 to show that $\operatorname{div} j = 0$ in distributions, including across the periodic join. In the aligned coordinates $x = sn + rt$, write $j \cdot \nabla\varphi = (n^\top j) \cdot \partial_s\varphi + (t^\top j) \cdot \partial_r\varphi$. The normal flux $n^\top j$ is the same in both layers; the tangential flux $t^\top j$ is independent of r. Integrate the two terms over $0 < s, r < 1$.

Solution. Use the aligned period cell $x = sn + rt$ from the setup. Since (n, t) is orthonormal, its area element is $ds\,dr$. Write $\varphi(s, r)$ for $\varphi(sn + rt)$. For a smooth periodic vector test φ,

$$\int_0^1 \int_0^1 j \cdot \nabla\varphi \, ds\, dr = \int_0^1 \int_0^1 \left[(n^\top j) \cdot \partial_s\varphi + (t^\top j) \cdot \partial_r\varphi\right] ds\, dr.$$

By Step 2.3 the normal flux $n^\top j = p$ is one constant vector in both layers. Integrating its term in s gives $p \cdot [\varphi(1, r) - \varphi(0, r)] = 0$. For each fixed s, $t^\top j$ is independent of r, so its term integrates to zero by periodicity in r. Thus $\langle j \cdot \nabla\varphi\rangle = 0$. Thus equilibrium holds across both the interior interface and the periodic join.

D2: Short proof

Answer 2.6. Prove global minimizing energy

Lookup: module 2 guide; setups 2.1. *Uses:* 1.12, 2.4, 2.5.

Task. Let e be the compatible field from Step 2.4 and let $L = B_i$ in layer i. Using Step 2.5, show $\langle (e + \nabla v) \cdot L(e + \nabla v)\rangle \geq \langle e \cdot Le\rangle$ for every periodic H^1 potential v. Expand the quadratic and use Step 1.12 to cancel the cross term; explain why this compares with every compatible field of mean F.

Solution. Let e be the compatible field in Step 2.4 and let $L = B_i$ in layer i. For any periodic H^1 potential v, expand the energy:

$$\langle (e + \nabla v) \cdot L(e + \nabla v)\rangle = \langle e \cdot Le\rangle + \langle \nabla v \cdot L\nabla v\rangle.$$

The omitted cross term is $2\langle \nabla v \cdot Le\rangle = 2\langle \nabla v \cdot j\rangle = 0$ by Step 2.5 and the H^1 testing argument in Step 1.12. The last displayed term is nonnegative by positive definiteness. Write $v_0(x) = w(n \cdot x)$ for the potential in Step 2.4, so $e = F + \nabla v_0$. Any other compatible field $F + \nabla v_1$ differs from e by the periodic H^1 gradient $\nabla(v_1 - v_0)$. Every compatible competitor therefore occurs in this comparison, which proves global cell minimality.

D1: Direct

Answer 2.7. The laminate energy bound

Lookup: module 2 guide; setups 2.1. *Uses:* 1.9, 1.10, 2.2, 2.6.

Task. By Step 2.4–Step 2.6, the cell tensor of the layers satisfies $F \cdot B_{\text{lam}} F = \min_z \Phi(z)$. Show $F \cdot B_{\text{lam}} F \leq \lambda F \cdot B_1 F + (1 - \lambda) F \cdot B_2 F$, and extend the definition by identifying B_{lam} at $\lambda = 0, 1$.

Solution. Steps 2.4–2.6 identify the minimum of Φ with the full cell minimum; Steps 1.9–1.10 therefore provide a self-adjoint tensor B_{lam} representing this quadratic energy. The value $z = 0$ is admissible and gives $F_1 = F_2 = F$. Therefore

$$F \cdot B_{\text{lam}} F = \min_z \Phi(z) \leq \lambda F \cdot B_1 F + (1 - \lambda) F \cdot B_2 F.$$

The actual layer fields may reduce this energy by redistributing the normal gradient. At $\lambda = 0$, only material 2 remains and the tensor is B_2; at $\lambda = 1$, only material 1 remains and it is B_1. The jump variable is unnecessary at these endpoints.

Setup 2.2. For Steps 2.8–2.10, assume the measurable periodic self-adjoint coefficient field L is uniformly elliptic, $\mu I \preceq L \preceq MI$ almost everywhere, and has spatial block form

$$L(x, y) = \operatorname{diag}(A(x, y), D).$$

Here $A(x, y), D \in \mathbb{R}^{m\times m}$, $0 < \mu \leq M$, and D is constant positive definite. Write the transposed spatial rows of a mean gradient as F_x, F_y.

D1: Direct

Answer 2.8. A vertical mean needs no corrector

Lookup: module 2 guide; setups 2.1, 2.2. *Uses:* 1.1, 1.4.

Task. For $F_x = 0$ and $F_y = q$ constant, show that $u_F = 0$ solves the weak cell equation. Determine the resulting mean flux.

Solution. At the candidate $u_F = 0$, the local gradient is constant and the flux is $j = (0, Dq)$. This flux is constant because D is constant. For every periodic test v,

$$\langle \nabla v \cdot j \rangle = \langle \partial_y v \rangle \cdot Dq = 0$$

by Step 1.1. Hence $u_F = 0$ solves the weak equation and is the zero-mean corrector by uniqueness. The mean flux is $(0, Dq)$. Notice that A can vary in both coordinates without affecting this calculation.

D2: Short proof

Answer 2.9. The vertical block survives homogenization

Lookup: module 2 guide; setups 2.1, 2.2. *Uses:* 1.1, 1.7, 1.9, 1.11, 2.8.

Task. For a horizontal mean gradient F with $F_y = 0$, show that the mean vertical flux is zero. Combine this with Step 2.8 to obtain $L_* = \operatorname{diag}(A_*, D)$ for some positive-definite A_*.

Solution. For a horizontal mean $F = (p, 0)$, the vertical flux is $D\partial_y u_F$. Its average is $D\langle \partial_y u_F \rangle = 0$ because D is constant and Step 1.1 applies. Thus the map from a horizontal mean to the vertical mean flux is zero. Step 2.8 says that a vertical mean produces no horizontal flux and gives the vertical flux Dq. By linearity in the mean gradient, these two statements determine the full tensor as $\operatorname{diag}(A_*, D)$. Self-adjointness and positive definiteness of A_* follow by restricting the corresponding properties from Steps 1.9 and 1.11 to horizontal means. The corrector for a horizontal mean may still depend on both x and y.

D1: Direct

Answer 2.10. Horizontal layers average arithmetically

Lookup: module 2 guide; setups 2.1, 2.2. *Uses:* 2.2, 2.3, 2.7.

Task. For $B_i = \operatorname{diag}(A_i, D)$ and $n = (0, 1)$, show that $z = 0$ minimizes Φ and that $B_{\text{lam}} = \lambda B_1 + (1 - \lambda)B_2$.

Solution. The compatible layer jump changes only the vertical row. The two horizontal rows stay equal to F_x, while the vertical rows are $F_y + (1 - \lambda)z$ and $F_y - \lambda z$. The weighted vertical energy is

$$F_y \cdot DF_y + \lambda(1 - \lambda)z \cdot Dz.$$

The linear terms in z cancel. Since D is positive definite, this is minimized at $z = 0$. The horizontal energy is already $\lambda F_x \cdot A_1 F_x + (1-\lambda) F_x \cdot A_2 F_x$. Therefore the tensor is exactly $\lambda B_1 + (1-\lambda) B_2$. The layers are horizontal because their normal is vertical.

Setup 2.3. Fix constant self-adjoint positive-definite original phases L_0, L_1 on $\mathbb{R}^{2\times m}$. A binary coefficient field is $L_\chi = (1-\chi)L_0 + \chi L_1$, where the phase indicator $\chi : Y \to \{0,1\}$ is measurable and periodic. Its phase-one fraction is the proportion $\theta = \langle \chi \rangle$ of the cell occupied by phase one. For $0 \le \theta \le 1$, let G_θ be the operator-norm closure of the effective tensors $(L_\chi)_*$ at exact fraction $\langle \chi \rangle = \theta$. A finite hierarchical laminate repeatedly layers earlier tensors, beginning with L_0, L_1, and records the original-phase fraction at every node. Let G_θ^{L} be the operator-norm closure of its tensors at exact fraction θ. A norm closure consists of all limits in operator norm of sequences from the original set. The following composition theorem may be used: for $C_i \in G_{\theta_i}$ and $0 \le \lambda \le 1$, layering C_1, C_2 at proportions $\lambda, 1-\lambda$ gives a tensor in $G_{\lambda\theta_1 + (1-\lambda)\theta_2}$, with exact-fraction periodic recovery (B4).

D1: Direct

Answer 2.11. Fraction bookkeeping

Lookup: module 2 guide; setups 2.1, 2.2, 2.3. *Uses:* local setup only.

Task. If the two materials in an outer laminate have phase-one fractions $\theta_1, \theta_2 \in [0,1]$ and outer proportions $\lambda, 1-\lambda$ with $0 \le \lambda \le 1$, compute the overall phase-one fraction.

Solution. A proportion λ of the total cell is occupied by material 1, of which fraction θ_1 is original phase 1. Its contribution is $\lambda\theta_1$. Material 2 contributes $(1-\lambda)\theta_2$. Adding the disjoint contributions gives

$$\theta = \lambda\theta_1 + (1-\lambda)\theta_2.$$

The outer proportions describe amounts of already composite materials; they need not equal the original-phase fractions within those materials.

D2: Short proof

Answer 2.12. Why every finite laminate lies in the G-closure

Lookup: module 2 guide; setups 2.1, 2.2, 2.3. *Uses:* 2.11.

Task. Use the composition theorem above and induction on the number of layering operations to show $G_\theta^{\mathrm{L}} \subseteq G_\theta$.

Solution. The original tensors L_0, L_1 belong to G_0, G_1, respectively, because the constant indicators $0, 1$ realize them. Suppose every earlier node of a finite laminate construction belongs to the G-closure with its recorded fraction. At a new layering node, the composition theorem puts its tensor in G at the weighted fraction from Step 2.11. Induction proves the statement at every finite node. In particular every finite laminate tensor at fraction θ belongs to G_θ. Taking its norm closure preserves membership because G_θ is closed by definition. This final passage gives $G_\theta^{\mathrm{L}} \subseteq G_\theta$. The inclusion also holds at $\theta = 0, 1$, where the constant original phases provide the initial nodes.

3 The explicit matrices and the eight auxiliary coordinates

Prerequisites. Matrix multiplication, transpose, and the real symmetric spectral theorem; no earlier course steps.

Definition lookup. The first setup defines O_j, Γ_j; $Q(h), \mathcal{S}$ are recorded before Step 3.7; P_j, P, S are recorded before Step 3.11. The lookup paragraph below recalls spectral and Gram terminology.

Supplied background. B3. Each key links to its statement, scope, and reference.

Setup 3.1. For $\beta = (\beta_0, \beta_1, \beta_2, \beta_3) \in \mathbb{R}^4$ define

$$O(\beta) = \begin{pmatrix} \beta_0 & -\beta_1 & -\beta_2 & -\beta_3 \\ \beta_1 & \beta_0 & -\beta_3 & \beta_2 \\ \beta_2 & \beta_3 & \beta_0 & -\beta_1 \\ \beta_3 & -\beta_2 & \beta_1 & \beta_0 \end{pmatrix}.$$

Set $O_1 = I_4$, $O_2 = O(0,1,0,0)$, $O_3 = O(0,0,1,0)$, $O_4 = O(0,0,0,-1)$, and $\Gamma_j = \begin{pmatrix} 0 & O_j \\ O_j^\top & 0 \end{pmatrix}$. Here $j \in \{1,2,3,4\}$, and all matrices are real.

Setup 3.2. Matrix lookup: I_d is the $d \times d$ identity and e_r is a coordinate vector. The anticommutator of A, B is $AB + BA$; δ_{ij} is 1 when $i = j$ and 0 otherwise. The spectral theorem says that a real symmetric matrix has an orthonormal eigenbasis; its trace, the sum of its diagonal entries, is also the sum of its eigenvalues with multiplicity. Its operator norm is the largest absolute eigenvalue, and $\sup_{|e|=1} e^\top A e$ is its largest eigenvalue. A Gram matrix is $P^\top P$, with $e^\top P^\top P e = |Pe|^2$; full column rank means $Pe = 0$ only for $e = 0$ (B3).

D1: Direct

Answer 3.1. The three signed permutation matrices

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* local setup only.

Task. Write O_2, O_3, O_4 explicitly from the displayed formula for $O(\beta)$.

Solution. Substituting each of the three four-tuples gives

$$O_2 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad O_3 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},$$

$$O_4 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

Each matrix permutes the coordinate vectors and possibly changes their signs. Subsequent identities can therefore be checked on four basis vectors instead of by multiplying long symbolic arrays.

D1: Direct

Answer 3.2. Transpose and square

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.1.

Task. Using the sparse matrices from Step 3.1, show $O_j^\top = -O_j$ and $O_j^2 = -I_4$ for $j = 2, 3, 4$. Deduce $O_j^\top O_j = I_4$.

Solution. Each displayed matrix has opposite entries on the two sides of its diagonal, so its transpose is its negative. Their actions on (e_1, e_2, e_3, e_4) are

$$\begin{aligned} O_2 &: (e_2, -e_1, e_4, -e_3), \\ O_3 &: (e_3, -e_4, -e_1, e_2), \\ O_4 &: (-e_4, -e_3, e_2, e_1). \end{aligned}$$

Applying the same list twice sends every e_i to $-e_i$, proving $O_j^2 = -I_4$. Therefore $O_j^\top O_j = (-O_j)O_j = -O_j^2 = I_4$. This orthogonality also holds for $O_1 = I_4$.

D1: Direct

Answer 3.3. One four-dimensional multiplication

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2.

Task. Verify $O_2O_3 = -O_4$ by comparing the two sides on the four coordinate vectors.

Solution. Using the lists in Answer 3.2, apply O_3 first and then O_2. The images are

$$(O_2O_3)e_1 = e_4, \quad (O_2O_3)e_2 = e_3, \quad (O_2O_3)e_3 = -e_2, \quad (O_2O_3)e_4 = -e_1.$$

These are exactly the negatives of the four images for O_4. Two linear maps that agree on a basis agree everywhere, so $O_2O_3 = -O_4$. This particular sign will fix the orientation of the later four-generator product.

D2: Short proof

Answer 3.4. The remaining anticommutations

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2, 3.3.

Task. Use Steps 3.2–3.3 to show $O_iO_j = -O_jO_i$ for distinct $i, j \in \{2, 3, 4\}$. First transpose $O_2O_3 = -O_4$, using $(AB)^\top = B^\top A^\top$; then substitute $O_4 = -O_2O_3$ in the remaining two pairs.

Solution. Transpose $O_2O_3 = -O_4$. The left side becomes $O_3^\top O_2^\top = O_3O_2$, and the right side becomes O_4. Thus $O_3O_2 = O_4 = -O_2O_3$. Since $O_4 = -O_2O_3$ and $O_2^2 = O_3^2 = -I$,

$$\begin{aligned} O_2O_4 &= -O_2^2O_3 = O_3, & O_4O_2 &= -O_2O_3O_2 = -O_3, \\ O_3O_4 &= -O_3O_2O_3 = -O_2, & O_4O_3 &= -O_2O_3^2 = O_2. \end{aligned}$$

These identities give the remaining two anticommutations.

D2: Short proof

Answer 3.5. The Clifford relations

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2, 3.4.

Task. By multiplying 2×2 blocks, show that $\Gamma_j^\top = \Gamma_j$ and $\Gamma_i\Gamma_j + \Gamma_j\Gamma_i = 2\delta_{ij} I_8$. Treat separately $i = j$, one index equal to 1, and distinct indices both larger than 1.

Solution. Block transposition immediately gives $\Gamma_j^\top = \Gamma_j$. Multiplying blocks gives

$$\Gamma_i\Gamma_j = \operatorname{diag}(O_i O_j^\top, O_i^\top O_j).$$

If $i = j$, orthogonality in Step 3.2 makes both blocks I_4. If $i = 1$ and $j > 1$, the first anticommutator block is $O_j^\top + O_j = 0$, and the second is the same sum. If $i, j > 1$ are distinct, the first block of the anticommutator is $-(O_iO_j + O_jO_i) = 0$, and likewise for the second block. Hence $\Gamma_i\Gamma_j + \Gamma_j\Gamma_i = 2\delta_{ij} I_8$.

D2: Short proof

Answer 3.6. The sign of the four-factor product

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2, 3.3.

Task. Use $O_2O_3 = -O_4$ to show $\Gamma_1\Gamma_2\Gamma_3\Gamma_4 = \operatorname{diag}(I_4, -I_4)$. Compute the two block-diagonal factors $\Gamma_1\Gamma_2$ and $\Gamma_3\Gamma_4$ before multiplying them.

Solution. Using the explicit form of Γ_1 and the skew-symmetry of the other O_j gives

$$\Gamma_1\Gamma_2 = \operatorname{diag}(-O_2, O_2), \qquad \Gamma_3\Gamma_4 = \operatorname{diag}(-O_3O_4, -O_3O_4).$$

Their product is $\operatorname{diag}(O_2O_3O_4, -O_2O_3O_4)$. Step 3.3 and $O_4^2 = -I_4$ give $O_2O_3O_4 = -O_4^2 = I_4$. Thus the desired product is $\operatorname{diag}(I_4, -I_4)$. Its first and fifth diagonal entries differ by 2, the two entries compared by the later energy detector.

Setup 3.3. Write $Q(h) = \sum_{j=1}^{4} h_j\Gamma_j$, $h \in \mathbb{R}^4$, and $\mathcal{S} = \{h_0 I_8 + Q(h) : h_0 \in \mathbb{R},\ h \in \mathbb{R}^4\}$.

D1: Direct

Answer 3.7. Cancellation of the mixed terms

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.5.

Task. Expand $Q(h)^2$ and prove $Q(h)^2 = |h|^2 I_8$.

Solution. Expand the square and collect the terms with unequal indices in pairs:

$$Q(h)^2 = \sum_j h_j^2\Gamma_j^2 + \sum_{i<j} h_ih_j(\Gamma_i\Gamma_j + \Gamma_j\Gamma_i).$$

The first sum is $\sum_j h_j^2 I_8$; the second is zero by Step 3.5. Hence $Q(h)^2 = |h|^2 I_8$.

D2: Short proof

Answer 3.8. The spectrum

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.5, 3.7.

Task. For $h \neq 0$, use Step 3.7 and $\operatorname{tr} Q(h) = 0$ to show that $h_0 I_8 + Q(h)$ has eigenvalues $h_0 \pm |h|$, each four times. State the spectrum for $h = 0$. First apply the symmetric spectral theorem to $Q(h)$; then determine the two multiplicities from their sum 8 and their weighted sum $\operatorname{tr} Q(h) = 0$.

Solution. The real matrix $Q(h)$ is symmetric, so it has an orthonormal eigenbasis. If ν is an eigenvalue, Step 3.7 gives $\nu^2 = |h|^2$. For $h \neq 0$, the possibilities are therefore $|h|$ and $-|h|$. Every Γ_j has zero diagonal blocks and hence zero trace, so $\operatorname{tr} Q(h) = 0$. If the positive eigenvalue has multiplicity r, the negative one has multiplicity $8 - r$, and zero trace says $(2r - 8)|h| = 0$. Thus $r = 4$. Adding $h_0 I_8$ shifts the eigenvalues to $h_0 \pm |h|$, each four times. If $h = 0$, the matrix is $h_0 I_8$, whose only eigenvalue is h_0 with multiplicity eight.

D1: Direct

Answer 3.9. The inverse stays in the same space

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.7, 3.8.

Task. For invertible $H = h_0 I_8 + Q(h)$, show

$$H^{-1} = \frac{h_0 I_8 - Q(h)}{h_0^2 - |h|^2}.$$

Use Step 3.8 to justify that the denominator is nonzero.

Solution. By Step 3.7,

$$(h_0 I_8 + Q(h))(h_0 I_8 - Q(h)) = h_0^2 I_8 - Q(h)^2 = (h_0^2 - |h|^2) I_8.$$

The eigenvalues from Step 3.8 show that H is invertible exactly when $h_0 + |h|$ and $h_0 - |h|$ are both nonzero, equivalently $h_0^2 - |h|^2 \neq 0$. Dividing the displayed identity by this nonzero scalar gives $H^{-1} = (h_0 I_8 - Q(h))/(h_0^2 - |h|^2)$. The result is still a linear combination of $I_8, \Gamma_1, \dots, \Gamma_4$, so it belongs to $\mathcal{S}$.

D2: Short proof

Answer 3.10. Inverses are different from products

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.5, 3.9.

Task. Show that $\Gamma_1 \Gamma_2$ is a nonzero skew-symmetric matrix and hence is not in $\mathcal{S}$. Conclude that $\mathcal{S}$ is not closed under ordinary multiplication.

Solution. Because the generators are symmetric and anticommute,

$$(\Gamma_1 \Gamma_2)^\top = \Gamma_2 \Gamma_1 = -\Gamma_1 \Gamma_2.$$

Thus the product is skew-symmetric. It is nonzero because each factor squares to the identity and is therefore invertible. Every member of $\mathcal{S}$ is symmetric, and a matrix that is both symmetric and skew-symmetric is zero. Hence this nonzero product cannot belong to $\mathcal{S}$.

Setup 3.4. Let $P_j = \frac{1}{2}(I_4 \;\; O_j)$, let P be their vertical stack, and set $S = P^\top P$. Thus P maps eight auxiliary coordinates to sixteen potential components.

D1: Direct

Answer 3.11. The four Gram factors

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.2.

Task. Show that $P_j^\top P_j = \frac{1}{4}(I_8 + \Gamma_j)$.

Solution. Multiply the two rectangular block matrices:

$$P_j^\top P_j = \frac{1}{4}\begin{pmatrix} I_4 \\ O_j^\top \end{pmatrix}\begin{pmatrix} I_4 & O_j \end{pmatrix} = \frac{1}{4}\begin{pmatrix} I_4 & O_j \\ O_j^\top & O_j^\top O_j \end{pmatrix}.$$

Orthogonality from Step 3.2 makes the lower-right block I_4. The result is exactly $(I_8 + \Gamma_j)/4$. This is a positive-semidefinite matrix because it is a Gram matrix, a fact used when bounding directional weights later.

D1: Direct

Answer 3.12. The Gram spectrum

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.8, 3.11.

Task. Sum Step 3.11 to obtain $S = I_8 + \frac{1}{4}\sum_j \Gamma_j$. Use Step 3.8 to show that its eigenvalues are $1/2$ and $3/2$, each four times.

Solution. For a vertically stacked matrix, $P^\top P$ is the sum of the Gram matrices of its block rows. Thus

$$S = \sum_{j=1}^{4} P_j^\top P_j = I_8 + \frac{1}{4}\sum_{j=1}^{4} \Gamma_j.$$

In Step 3.8 take $h_0 = 1$ and $h = (1/4, 1/4, 1/4, 1/4)$, whose norm is $1/2$. The two eigenvalues of S are $1 \pm 1/2$, namely $3/2$ and $1/2$, each four times. In particular S is positive definite.

D1: Direct

Answer 3.13. An explicit inverse of the Gram matrix

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.9, 3.12.

Task. Use Step 3.9 to show $S^{-1} = \frac{4}{3}I_8 - \frac{1}{3}\sum_j \Gamma_j$.

Solution. Apply Step 3.9 with $h_0 = 1$ and $h_j = 1/4$. The denominator is $1 - 4(1/4)^2 = 3/4$. Hence

$$S^{-1} = \frac{I_8 - (1/4)\sum_j \Gamma_j}{3/4} = \frac{4}{3}I_8 - \frac{1}{3}\sum_j \Gamma_j.$$

In particular both diagonal 4×4 blocks of S^{-1} are $(4/3)I_4$, because each Γ_j has zero diagonal blocks. All entries are rational.

D2: Short proof

Answer 3.14. Rank and operator norm

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.12.

Task. Use $P^\top P = S$ and Step 3.12 to prove that P has full column rank and $\|P\|^2 = 3/2$. Use $|Pe|^2 = e^\top S e$ for injectivity, and maximize this expression over unit e for the norm.

Solution. For $e \in \mathbb{R}^8$, $|Pe|^2 = e^\top S e \geq (1/2)|e|^2$. Therefore $Pe = 0$ implies $e = 0$, which is full column rank. Also

$$\|P\|^2 = \sup_{|e|=1} |Pe|^2 = \sup_{|e|=1} e^\top S e = \lambda_{\max}(S) = \frac{3}{2}.$$

The middle equality is the Rayleigh-quotient characterization of the largest eigenvalue of a real symmetric matrix. This bound will control the contrast needed to keep the physical horizontal block positive definite.

4 The phases, the detecting loads, and the directional matrix

Prerequisites. Common-block layers: Step 2.10; auxiliary matrices, spectrum, and norm: Step 3.5, Step 3.11–Step 3.14.

Definition lookup. The first setup defines $L(H), \mathcal{K}, D, \rho$ and recalls $\mathcal{S}, P, S$; the setups before Step 4.7 and Step 4.9 define the loads, detector, $\mathcal{R}(k)$, and $T(k)$.

Supplied background. B3. Each key links to its statement, scope, and reference.

Setup 4.1. Use $P, S, \Gamma_j, \mathcal{S}$ from Module 3 and set $D = \operatorname{diag}(1, 1/2, 1/3, 1/4) \otimes I_4$. Fix $0 < \rho < 2/3$ and define

$$L(H) = \operatorname{diag}(I_{16} - PHP^\top, D), \qquad \mathcal{K} = \{L(H) : H \in \mathcal{S},\ 0 \preceq H \preceq \rho I_8\}.$$

The blocks act on the horizontal and vertical rows of a 2×16 gradient. The notation $A \preceq B$ means $v^\top A v \le v^\top B v$ for every vector v.

Setup 4.2. Auxiliary lookup: $\mathcal{S} = \operatorname{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$; each Γ_j is symmetric with zero diagonal 4×4 blocks. The matrices $P_j = \frac{1}{2}(I_4 \ \ O_j)$ are stacked to form $P \in \mathbb{R}^{16\times 8}$, and $S = P^\top P = I_8 + \frac{1}{4}\sum_j \Gamma_j$. The identities needed here are $\frac{1}{2} I_8 \preceq S \preceq \frac{3}{2} I_8$, $\|P\|^2 = 3/2$, and $S^{-1} = \frac{4}{3} I_8 - \frac{1}{3}\sum_j \Gamma_j$ (Step 3.11–Step 3.14). Here $\operatorname{diag}(A, D)$ acts separately on the horizontal and vertical spatial rows, $\otimes I_4$ repeats each listed scalar as a 4×4 block, and all matrix norms are operator norms.

D1: Direct

Answer 4.1. The two diagonal blocks agree

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* local setup only.

Task. If $H = h_0 I_8 + \sum_j h_j \Gamma_j$, show that both diagonal 4×4 blocks of H equal $h_0 I_4$.

Solution. Each Γ_j has the block form $\begin{pmatrix} 0 & O_j \\ O_j^\top & 0 \end{pmatrix}$ and thus contributes nothing to either diagonal block. Only $h_0 I_8$ contributes there. Therefore

$$H = \begin{pmatrix} h_0 I_4 & \sum_j h_j O_j \\ \sum_j h_j O_j^\top & h_0 I_4 \end{pmatrix}.$$

Corresponding coordinate vectors in the two blocks have the same quadratic energy under H. This elementary equality is the algebraic feature that the two prescribed loads will detect.

D2: Short proof

Answer 4.2. A uniform ellipticity bound

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 3.14.

Task. Show that every $C \in \mathcal{K}$ satisfies $\mu_\rho I \preceq C \preceq I$, where $\mu_\rho = \min\{1 - 3\rho/2, 1/4\} > 0$. Bound $f^\top PHP^\top f$ by $\rho |P^\top f|^2$ and use $\|P^\top\| = \|P\|$.

Solution. For a horizontal vector f and $0 \preceq H \preceq \rho I_8$,

$$0 \le f^\top PHP^\top f \le \rho |P^\top f|^2 \le \rho \|P\|^2 |f|^2 = \frac{3\rho}{2} |f|^2.$$

Consequently $(1 - 3\rho/2) I_{16} \preceq I_{16} - PHP^\top \preceq I_{16}$. The eigenvalues of D lie between $1/4$ and 1. For a full gradient (f, g), add the two block estimates. This gives the stated lower constant $\mu_\rho = \min\{1 - 3\rho/2, 1/4\} > 0$ and upper constant 1. The strict inequality $\rho < 2/3$ guarantees positivity.

D2: Short proof

Answer 4.3. Compactness

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* local setup only.

Task. Show that $\{H \in \mathcal{S} : 0 \preceq H \preceq \rho I_8\}$ is closed and bounded, and deduce that $\mathcal{K}$ is compact. Pass each quadratic-form inequality to a matrix limit, use finite-dimensional closed-and-bounded compactness, and check continuity of $H \mapsto L(H)$.

Solution. The space $\mathcal{S}$ is a finite-dimensional linear subspace of the symmetric matrices, hence closed. If $H_n \to H$ and each $0 \preceq H_n \preceq \rho I_8$, then for every vector e the inequalities $0 \le e^\top H_n e \le \rho|e|^2$ pass to the limit. Thus the order interval is closed. Its members have operator norm at most ρ, so it is bounded. A closed bounded set in a finite-dimensional space is compact. Finally, $\|L(H) - L(\widetilde{H})\| \le \|P\|^2 \|H - \widetilde{H}\| = (3/2)\|H - \widetilde{H}\|$, so $H \mapsto L(H)$ is continuous. Its image of the compact parameter set is $\mathcal{K}$, hence compact.

D2: Short proof

Answer 4.4. The two original phases

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 2.10.

Task. Define $L_s = L(\rho s I_8)$ for $0 \le s \le 1$. Show $L_s = (1-s)L_0 + sL_1$ and $L_1 \preceq L_0$; identify the horizontal layering that realizes L_s.

Solution. By direct substitution,

$$L_s = \operatorname{diag}(I_{16} - \rho s P P^\top, D) = (1-s)\operatorname{diag}(I_{16}, D) + s \operatorname{diag}(I_{16} - \rho P P^\top, D).$$

These endpoint tensors are L_0, L_1, and their difference is $L_0 - L_1 = \operatorname{diag}(\rho P P^\top, 0) \succeq 0$. Hence $L_1 \preceq L_0$. Since they share D, Step 2.10 says that horizontal layers with proportions $1-s$ of phase 0 and s of phase 1 have precisely the arithmetic tensor L_s. At $s = 0, 1$ they are simply the pure phases. If ρ is rational, both original phases L_0, L_1 have rational entries, because P and D do.

D1: Direct

Answer 4.5. Lifting an auxiliary vector

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 3.12.

Task. For $e \in \mathbb{R}^8$, show that $f = PS^{-1}e$ satisfies $P^\top f = e$.

Solution. The definition $S = P^\top P$ and invertibility from Step 3.12 give

$$P^\top (PS^{-1}e) = SS^{-1}e = e.$$

Thus PS^{-1} is a right inverse of $P^\top$. It lifts a vector in the eight auxiliary coordinates to a horizontal mean gradient with exactly those reduced coordinates. The vector Pe instead has reduced coordinates Se.

D1: Direct

Answer 4.6. The reduced energy matrix

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 4.5.

Task. For a symmetric $H \in \mathbb{R}^{8\times 8}$, let $Q = PS^{-1} \in \mathbb{R}^{16\times 8}$ (a lifting matrix, distinct from the map $h \mapsto Q(h)$ in Module 3). First compute $P^\top Q$ and $Q^\top Q$, then show $Q^\top(I_{16} - PHP^\top)Q = S^{-1} - H$.

Solution. The matrices S and S^{-1} are symmetric. With $Q = PS^{-1}$,

$$Q^\top Q = S^{-1}P^\top PS^{-1} = S^{-1}SS^{-1} = S^{-1}.$$

Also, Step 4.5 gives $P^\top Q = I_8$, so $Q^\top P = I_8$ by transposition. Hence

$$Q^\top PHP^\top Q = (Q^\top P)H(P^\top Q) = H.$$

Subtracting gives the claimed identity. The calculation applies to every symmetric 8×8 matrix H.

Setup 4.3. Let $e_+ = e_1$ and $e_- = e_5$ be coordinate vectors of $\mathbb{R}^8$. Define horizontal mean gradients ξ, η by $\xi_x = PS^{-1}e_+$, $\eta_x = PS^{-1}e_-$, and $\xi_y = \eta_y = 0$. Define $\ell(B) = \eta \cdot B\eta - \xi \cdot B\xi$.

D2: Short proof

Answer 4.7. Equal energies at two distinct loads

Lookup: module 4 guide; setups 4.1, 4.2, 4.3. *Uses:* 3.13, 4.1, 4.6.

Task. For $H = h_0 I_8 + \sum_j h_j \Gamma_j$ with $L(H) \in \mathcal{K}$, use Steps 4.1 and 4.6 to show $\xi \cdot L(H)\xi = \eta \cdot L(H)\eta = 4/3 - h_0$. Deduce $|\xi|^2 = |\eta|^2 = 4/3$ and check that the loads are distinct. Use Step 3.13 to read the two needed diagonal entries of S^{-1}, and use $Q^\top Q = S^{-1}$ for the norms.

Solution. The loads have horizontal rows Qe_1, Qe_5 and zero vertical rows, where $Q = PS^{-1}$. Step 4.6 therefore gives

$$\xi \cdot L(H)\xi = e_1^\top(S^{-1} - H)e_1, \qquad \eta \cdot L(H)\eta = e_5^\top(S^{-1} - H)e_5.$$

Step 3.13 gives diagonal entries $4/3$ at positions 1 and 5 of S^{-1}; Step 4.1 gives h_0 at both positions of H. Both energies are consequently $4/3 - h_0$, and $\ell = 0$ on $\mathcal{K}$. Also $Q^\top Q = S^{-1}$ from Step 4.6, so $|\xi|^2 = e_1^\top S^{-1} e_1 = 4/3$ and likewise for η. The loads are distinct: if $Qe_1 = Qe_5$, applying $P^\top$ would contradict $e_1 \neq e_5$.

D1: Direct

Answer 4.8. Continuity of the detector

Lookup: module 4 guide; setups 4.1, 4.2, 4.3. *Uses:* 4.7.

Task. For linear maps B, C on $\mathbb{R}^{2\times 16}$, show $|\ell(B) - \ell(C)| \le (8/3)\|B - C\|$. In particular, $\ell = 0$ on every norm limit of tensors on which it vanishes.

Solution. For any linear map A and vector f, $|f \cdot Af| \le |f|\,|Af| \le \|A\||f|^2$. Apply this to $A = B - C$ and to each load:

$$\begin{aligned} |\ell(B) - \ell(C)| &\le |\eta \cdot (B - C)\eta| + |\xi \cdot (B - C)\xi| \\ &\le (|\eta|^2 + |\xi|^2)\|B - C\| = \frac{8}{3}\|B - C\|. \end{aligned}$$

Thus $B_n \to B$ in operator norm implies $\ell(B_n) \to \ell(B)$. In particular, if $\ell(B_n) = 0$ for every n, then $\ell(B) = 0$. An additional useful consequence is $\|B - C\| \ge (3/8)\ell(B)$ when $\ell(B) > 0$ and $\ell(C) = 0$.

Setup 4.4. For nonzero $k = (k_x, k_y)$, let $\mathcal{R}(k) = (k_x^2 I_{16} + k_y^2 D)^{-1}$ and $T(k) = k_x^2 P^\top \mathcal{R}(k) P$. These are defined because D is positive definite and $k \neq 0$.

D1: Direct

Answer 4.9. The cost of a normal gradient

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* local setup only.

Task. For a unit normal $n = (n_x, n_y) \in \mathbb{R}^2$, a vector $z \in \mathbb{R}^{16}$, and the baseline phase $L_0 = \operatorname{diag}(I_{16}, D)$, show $(n \otimes z) \cdot L_0(n \otimes z) = z^\top (n_x^2 I_{16} + n_y^2 D) z$.

Solution. The horizontal row of $n \otimes z$ is $n_x z$ and its vertical row is $n_y z$. The baseline tensor acts as the identity on the first and as D on the second. Therefore

$$\begin{aligned}(n \otimes z) \cdot L_0(n \otimes z) &= |n_x z|^2 + (n_y z) \cdot D(n_y z) \\ &= z^\top (n_x^2 I_{16} + n_y^2 D) z.\end{aligned}$$

The inverse of this cost matrix is $\mathcal{R}(n)$. The perturbation involving H couples only through the horizontal row, explaining the two factors of n_x in the reduced directional matrix $T(n)$.

D2: Short proof

Answer 4.10. The four directional weights

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* 3.11.

Task. Use the four diagonal blocks of D and Step 3.11 to derive

$$T(k) = \frac{1}{4} \sum_{j=1}^{4} \frac{j k_x^2}{j k_x^2 + k_y^2} (I_8 + \Gamma_j).$$

Solution. The jth 4×4 block of D is I_4/j. Thus the corresponding block of $k_x^2 I + k_y^2 D$ is $(k_x^2 + k_y^2/j) I_4$, and that of its inverse is $j I_4/(j k_x^2 + k_y^2)$. Multiplying with the stacked block rows of P gives

$$\begin{aligned}T(k) &= k_x^2 \sum_{j=1}^{4} \frac{j}{j k_x^2 + k_y^2} P_j^\top P_j \\ &= \frac{1}{4} \sum_{j=1}^{4} \frac{j k_x^2}{j k_x^2 + k_y^2} (I_8 + \Gamma_j),\end{aligned}$$

where Step 3.11 was used in the last line. The four different entries of D produce four different responses to the direction of k.

D2: Short proof

Answer 4.11. A uniform bound on the directional matrix

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* 3.11, 3.12, 4.10.

Task. Using weights between zero and one in Step 4.10, show $T(k) \in \mathcal{S}$ and $0 \preceq T(k) \preceq S \preceq (3/2) I_8$. For the order bound, rewrite the sum as $\sum_j w_j P_j^\top P_j$ with $0 \leq w_j \leq 1$; use the positive semidefinite Gram factors $P_j^\top P_j$.

Solution. Write $w_j = j k_x^2/(j k_x^2 + k_y^2)$; all denominators are positive for $k \neq 0$, so $0 \leq w_j \leq 1$. Step 4.10 is a linear combination of $I_8, \Gamma_1, \ldots, \Gamma_4$, proving $T(k) \in \mathcal{S}$. The first line in Answer

4.10 also gives $T(k) = \sum_j w_j P_j^\top P_j$. Each Gram matrix is positive semidefinite, and $S - T(k) = \sum_j (1 - w_j) P_j^\top P_j \succeq 0$. Hence

$$0 \preceq T(k) \preceq \sum_j P_j^\top P_j = S \preceq \frac{3}{2} I_8.$$

The last inequality is the largest-eigenvalue bound from Step 3.12. This is a bound uniform over every direction, including directions close to vertical.

D2: Short proof

Answer 4.12. Which directions can be inverted?

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* 3.14.

Task. Show that $T(k) = 0$ if $k_x = 0$, whereas $T(k)$ is positive definite if $k_x \neq 0$. Also observe directly from its definition that $T(-k) = T(k)$ and $T(ak) = T(k)$ for every nonzero real a.

Solution. If $k_x = 0$, its defining prefactor is zero, so $T(k) = 0$. If $k_x \neq 0$ and $e \neq 0$, the full column rank of P gives $Pe \neq 0$. The matrix $\mathcal{R}(k)$ is positive definite, so

$$e^\top T(k) e = k_x^2 (Pe)^\top \mathcal{R}(k) (Pe) > 0.$$

Thus $T(k)$ is invertible exactly in the nonvertical directions. Its formula contains only the squares k_x^2, k_y^2, giving $T(-k) = T(k)$. Replacing k by ak, $a \neq 0$, multiplies the inverse cost matrix by a^{-2} and the prefactor by a^2, so $T(ak) = T(k)$. Thus T depends only on the direction of k.

5 One layering step and the invariant

Prerequisites. The layer minimum: Step 2.2–Step 2.7; inverse closure: Step 3.9; phase ellipticity and directional matrices: Step 4.2, Step 4.11–Step 4.12.

Definition lookup. The setup below defines $L(H), \mathcal{K}, \overline{H}, \widetilde{H}, \mathcal{R}(n), T(n)$ and the layer fields. The lookup paragraph recalls matrix order and defines H_{lam}; Step 5.7 and Step 5.10 introduce the shifted matrices.

Supplied background. B3, B4. Each key links to its statement, scope, and reference.

Setup 5.1. Use the matrices P, D, Γ_j from Modules 3–4 and put $L(H) = \mathrm{diag}(I_{16} - PHP^{\top}, D)$. Thus $P \in \mathbb{R}^{16\times 8}$ has full column rank, $D \succeq I_{16}/4$ is constant, and $\|P\|^2 = 3/2$. The space $\mathcal{S} = \mathrm{span}\{I_8, \Gamma_1, \ldots, \Gamma_4\}$ consists of real symmetric matrices and is linear and closed under inversion of its invertible members. Define $\mathcal{K} = \{L(A) : A \in \mathcal{S},\ 0 \preceq A \preceq \rho I_8\}$. Fix $0 < \rho < 2/3$ and $H^{(1)}, H^{(2)} \in \mathcal{S}$ with $0 \preceq H^{(i)} \preceq \rho I_8$. Thus $L(H^{(i)}) \succeq \mu I$ for $\mu = \min\{1/4, 1 - 3\rho/2\} > 0$.

Let $n = (n_x, n_y)$ be a unit normal and $0 < \lambda < 1$. Set $\Delta H = H^{(1)} - H^{(2)}$, $\overline{H} = \lambda H^{(1)} + (1-\lambda)H^{(2)}$, and $\widetilde{H} = (1-\lambda)H^{(1)} + \lambda H^{(2)}$. For $F \in \mathbb{R}^{2\times 16}$ and $z \in \mathbb{R}^{16}$, the compatible layer gradients are $F_1 = F + (1-\lambda)n \otimes z$ and $F_2 = F - \lambda n \otimes z$. Their weighted energy is $\mathcal{E}_F(z) = \lambda F_1 \cdot L(H^{(1)})F_1 + (1-\lambda)F_2 \cdot L(H^{(2)})F_2$.

By Step 4.11–Step 4.12, $\mathcal{R}(n) = (n_x^2 I_{16} + n_y^2 D)^{-1}$ and $T(n) = n_x^2 P^{\top}\mathcal{R}(n)P \in \mathcal{S}$, with $0 \preceq T(n) \preceq (3/2)I_8$. It is positive definite when $n_x \neq 0$ and zero when $n_x = 0$. Matrix factors are multiplied in their displayed order. References: manuscript Section 3.3 and Appendix A.2.

Setup 5.2. Lookup: $A \preceq B$ means $v^{\top}(B-A)v \geq 0$ for every vector v; $A \succ 0$ means the inequality $v^{\top}Av > 0$ for every nonzero v. All order comparisons here are between real symmetric matrices. For $A \succ 0$, the inverse square root $A^{-1/2}$ is obtained by replacing each eigenvalue $a > 0$ with $a^{-1/2}$ in an orthogonal diagonalization of A. The operator norm is $\|A\| = \sup_{|v|=1} |Av|$, and $\mathrm{diag}(A, D)$ acts separately on the transposed spatial rows F_x, F_y. The matrix $n \otimes z$ has rows $n_x z, n_y z$. By Step 2.7, $\min_z \mathcal{E}_F(z) = F \cdot L_{\mathrm{lam}}F$ is the minimum energy over all compatible periodic fields on the aligned layer cell of Module 2. The calculation in Step 5.5 identifies $L_{\mathrm{lam}} = L(H_{\mathrm{lam}})$ and thereby defines the parameter H_{lam}. A finite hierarchical laminate is a finite binary tree of these layering operations, with the constituent phases at its leaves.

D1: Direct

Answer 5.1. Check the layer mean

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* local setup only.

Task. Show that $\lambda F_1 + (1-\lambda)F_2 = F$ and $F_1 - F_2 = n \otimes z$.

Solution. The coefficient of $n \otimes z$ in the weighted mean is $\lambda(1-\lambda) - (1-\lambda)\lambda = 0$. The coefficient in the difference is $(1-\lambda) + \lambda = 1$. Thus the mean is exactly F, and the compatible jump has rank at most one. The unknown $z \in \mathbb{R}^{16}$ changes the layer fields without changing the imposed mean.

D1: Direct

Answer 5.2. Find the constant energy term

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* local setup only.

Task. Evaluate $\mathcal{E}_F(0)$ and show that it equals $F \cdot L(\overline{H})F$.

Solution. When $z = 0$, both layer gradients are F. Since $L(H)$ is affine in H,

$$\begin{aligned}\lambda L(H^{(1)}) + (1-\lambda)L(H^{(2)}) &= \operatorname{diag}(I_{16} - P\overline{H}P^\top, D)\\ &= L(\overline{H}).\end{aligned}$$

Taking its quadratic form at F gives the constant term. The remaining terms describe the adjustment of the layer gradients.

D1: Direct

Answer 5.3. Find the linear term

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* local setup only.

Task. Expand only the terms linear in z in $\mathcal{E}_F(z)$. Show that their sum is

$$-2\lambda(1-\lambda)n_x z^\top P\Delta H P^\top F_x.$$

Solution. For a symmetric tensor C, the term linear in a in $(F+aG)\cdot C(F+aG)$ is $2aG\cdot CF$. Hence the two linear terms sum to

$$2\lambda(1-\lambda)(n\otimes z)\cdot[L(H^{(1)}) - L(H^{(2)})]F.$$

The vertical blocks, both D, cancel. The horizontal block is $-P\Delta H P^\top$, and the horizontal row of $n\otimes z$ is $n_x z$. Substitution gives the stated linear term.

D2: Short proof

Answer 5.4. Find and bound the quadratic matrix

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* 4.2.

Task. Show that the quadratic term is $\lambda(1-\lambda)z^\top M z$, where $M = n_x^2 I_{16} + n_y^2 D - n_x^2 P\widetilde{H}P^\top$. Prove $M \succeq \mu I_{16}$.

Solution. The quadratic contributions from the two layers have weights $\lambda(1-\lambda)^2$ and $(1-\lambda)\lambda^2$. Factoring out $\lambda(1-\lambda)$ leaves the tensor $(1-\lambda)L(H^{(1)}) + \lambda L(H^{(2)}) = L(\widetilde{H})$. Its energy on $n\otimes z$ is

$$n_x^2 z^\top(I_{16} - P\widetilde{H}P^\top)z + n_y^2 z^\top D z = z^\top M z.$$

Each constituent is bounded below by μI, so their convex combination is too. Therefore

$$z^\top M z \geq \mu|n\otimes z|^2 = \mu|z|^2,$$

where $|n| = 1$ was used. In particular M is invertible. The squared layer fluctuations give the reversed weights in $\widetilde{H}$.

D2: Short proof

Answer 5.5. Complete one quadratic square

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* 2.7, 5.2, 5.3, 5.4.

Task. Using steps 5.2–5.4, complete the square in z and identify the minimum energy as $F\cdot L(H_{\mathrm{lam}})F$, where

$$H_{\mathrm{lam}} = \overline{H} + \lambda(1-\lambda)\Delta H[n_x^2 P^\top M^{-1} P]\Delta H.$$

First put $b = n_x P\Delta H P^\top F_x$; the z-dependent part is $\lambda(1-\lambda)(z^\top M z - 2z^\top b)$.

Solution. Put $b = n_x P \Delta H P^\top F_x$. The energy is

$$\mathcal{E}_F(z) = F \cdot L(\overline{H})F + \lambda(1-\lambda)(z^\top M z - 2z^\top b).$$

Since $M \succ 0$,

$$z^\top M z - 2z^\top b = (z - M^{-1}b)^\top M (z - M^{-1}b) - b^\top M^{-1} b.$$

Thus $z = M^{-1}b$ is the unique minimizer. The subtracted term changes only the horizontal energy. Write it as

$$b^\top M^{-1} b = F_x^\top P \Delta H [n_x^2 P^\top M^{-1} P] \Delta H P^\top F_x.$$

The effective horizontal block is therefore $I_{16} - P H_{\mathrm{lam}} P^\top$ with the displayed H_{lam}. The vertical block is D and the spatial cross blocks are zero. By Step 2.7, this minimum is the effective energy among all compatible periodic fields on the aligned layer cell. Equality of these quadratic forms for every F therefore identifies the full symmetric effective tensor. The correction matrix $\Delta H [n_x^2 P^\top M^{-1} P] \Delta H$ is symmetric because its middle factor is symmetric.

D1: Direct

Answer 5.6. Handle horizontal layers separately

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* 5.5.

Task. Set $n_x = 0$ in step 5.5. Show $H_{\mathrm{lam}} = \overline{H}$, and explain why no inverse of $T(n)$ is needed in this case.

Solution. If $n_x = 0$, the vector b in the completed square is zero. The minimizer is $z = 0$, and the correction to $\overline{H}$ contains $n_x^2 = 0$. Hence $H_{\mathrm{lam}} = \overline{H}$. The layers are horizontal because their normal is vertical. Here $T(n) = 0$, and the formula follows directly from the energy minimum.

Setup 5.3. For Steps 5.7–5.12 assume $n_x \neq 0$, so $T(n)$ is invertible.

D2: Short proof

Answer 5.7. Check the shifted inverses

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 3.9, 4.11, 4.12.

Task. Assume $n_x \neq 0$ and set $\Lambda = T(n)^{-1}$. Show that $\Lambda \in \mathcal{S}$, $\Lambda \succeq (2/3) I_8$, and $\Lambda - H^{(i)}, \Lambda - \widetilde{H} \succ 0$.

Solution. The matrix $T(n)$ is positive definite and belongs to the inverse-closed space $\mathcal{S}$, so $\Lambda \in \mathcal{S}$. Every eigenvalue of $T(n)$ is at most $3/2$; its inverse therefore has eigenvalues at least $2/3$. Also $H^{(i)} \preceq \rho I_8$, and the convex combination $\widetilde{H}$ satisfies the same bound. Consequently all three differences are bounded below by $(2/3 - \rho) I_8 \succ 0$. In particular, these shifted matrices are invertible.

D1: Direct

Answer 5.8. Verify the reduction identity

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.4.

Task. Show that $M\mathcal{R}(n)P = P[I_8 - \widetilde{H} T(n)]$ by multiplying out the left side.

Solution. Let $C = n_x^2 I_{16} + n_y^2 D$, so $\mathcal{R}(n) = C^{-1}$ and $M = C - n_x^2 P \widetilde{H} P^\top$. Then

$$\begin{aligned} M\mathcal{R}(n)P &= P - n_x^2 P \widetilde{H} P^\top \mathcal{R}(n) P \\ &= P - P \widetilde{H} T(n) \\ &= P[I_8 - \widetilde{H} T(n)]. \end{aligned}$$

Associativity leaves $\widetilde{H}$ to the left of $T(n)$.

D2: Short proof

Answer 5.9. Reduce the inverse correction

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.4, 5.7, 5.8.

Task. Use step 5.8 and $I_8 - \widetilde{H}T(n) = (\Lambda - \widetilde{H})T(n)$ to prove $n_x^2 P^\top M^{-1} P = (\Lambda - \widetilde{H})^{-1}$. Set $J = I_8 - \widetilde{H}T(n)$ and solve the identity in step 5.8 for $M^{-1}P$; use $(AB)^{-1} = B^{-1}A^{-1}$.

Solution. Write $J = I_8 - \widetilde{H}T(n)$. Both factors in $J = (\Lambda - \widetilde{H})T(n)$ are invertible by step 5.7. From $M\mathcal{R}(n)P = PJ$, multiply first on the left by M^{-1} and then on the right by J^{-1} to obtain $M^{-1}P = \mathcal{R}(n)PJ^{-1}$. Hence

$$\begin{aligned} n_x^2 P^\top M^{-1} P &= T(n)J^{-1} \\ &= T(n)T(n)^{-1}(\Lambda - \widetilde{H})^{-1} \\ &= (\Lambda - \widetilde{H})^{-1}. \end{aligned}$$

The reverse order in the inverse of a product is essential in the second line.

D1: Direct

Answer 5.10. Factor the weighted inverse sum

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.7.

Task. Set $K_i = \Lambda - H^{(i)}$ and $K_\lambda = (1-\lambda)K_1 + \lambda K_2$. Prove $\lambda K_1^{-1} + (1-\lambda)K_2^{-1} = K_2^{-1} K_\lambda K_1^{-1}$.

Solution. Expand the right side without changing order:

$$\begin{aligned} K_2^{-1} K_\lambda K_1^{-1} &= (1-\lambda)K_2^{-1}K_1K_1^{-1} + \lambda K_2^{-1}K_2K_1^{-1} \\ &= (1-\lambda)K_2^{-1} + \lambda K_1^{-1}. \end{aligned}$$

Only adjacent inverse pairs were canceled. Therefore the identity holds even if $K_1K_2 \neq K_2K_1$. The matrices K_i are positive definite by step 5.7, and their convex combination K_λ is positive definite as well.

D1: Direct

Answer 5.11. Expand the harmonic mean in small steps

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.7, 5.10.

Task. Let $\Delta K = K_1 - K_2$. Substitute $K_1 = K_\lambda + \lambda\Delta K$ and $K_2 = K_\lambda - (1-\lambda)\Delta K$ into $K_1 K_\lambda^{-1} K_2$, and simplify.

Solution. Multiplication gives

$$\begin{aligned} K_1K_\lambda^{-1}K_2 &= (K_\lambda + \lambda\Delta K)K_\lambda^{-1}(K_\lambda - (1-\lambda)\Delta K) \\ &= K_\lambda + (2\lambda - 1)\Delta K - \lambda(1-\lambda)\Delta K K_\lambda^{-1}\Delta K. \end{aligned}$$

Also, substituting the same expressions for K_1, K_2 gives

$$\lambda K_1 + (1-\lambda)K_2 = K_\lambda + (2\lambda - 1)\Delta K.$$

Thus

$$K_1K_\lambda^{-1}K_2 = \lambda K_1 + (1-\lambda)K_2 - \lambda(1-\lambda)\Delta K K_\lambda^{-1}\Delta K.$$

Every term on the final right side is symmetric: in particular $(\Delta K K_\lambda^{-1}\Delta K)^\top = \Delta K K_\lambda^{-1}\Delta K$. This also explains why the initially asymmetric-looking product is symmetric.

D2: Short proof

Answer 5.12. Obtain the shifted harmonic formula

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.5, 5.9, 5.10, 5.11.

Task. Combine steps 5.5 and 5.9–5.11 to show $H_{\mathrm{lam}} = \Lambda - [\lambda K_1^{-1} + (1-\lambda)K_2^{-1}]^{-1}$.

Solution. Put $Q = \lambda K_1^{-1} + (1-\lambda)K_2^{-1}$. Both Q and K_λ are positive definite. Inverting the ordered factorization in step 5.10 gives $Q^{-1} = K_1 K_\lambda^{-1} K_2$. Also $\Delta K = -\Delta H$ and $K_\lambda = \Lambda - \widetilde{H}$. Therefore

$$\begin{aligned}\Lambda - [\lambda K_1^{-1} + (1-\lambda)K_2^{-1}]^{-1} &= \overline{H} + \lambda(1-\lambda)\Delta H(\Lambda - \widetilde{H})^{-1}\Delta H \\ &= H_{\mathrm{lam}},\end{aligned}$$

where the last equality is steps 5.5 and 5.9. Equivalently,

$$(\Lambda - H_{\mathrm{lam}})^{-1} = \lambda(\Lambda - H^{(1)})^{-1} + (1-\lambda)(\Lambda - H^{(2)})^{-1}.$$

At $\lambda = 0$ or 1, the formula returns the surviving material, as does the original energy minimum.

D2: Short proof

Answer 5.13. Preserve the matrix space

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 3.9, 5.6, 5.7, 5.10, 5.12.

Task. Use the formula in step 5.12 to show $H_{\mathrm{lam}} \in \mathcal{S}$. For $n_x = 0$, use step 5.6 instead.

Solution. For $n_x \neq 0$, both $K_i = \Lambda - H^{(i)}$ belong to $\mathcal{S}$. They are positive definite, so their inverses belong to $\mathcal{S}$. Their weighted average $Q = \lambda K_1^{-1} + (1-\lambda)K_2^{-1}$ belongs to $\mathcal{S}$ by linearity and is positive definite. Consequently $Q^{-1} \in \mathcal{S}$. Subtracting from $\Lambda \in \mathcal{S}$ gives $H_{\mathrm{lam}} = \Lambda - Q^{-1} \in \mathcal{S}$. For $n_x = 0$, H_{lam} is simply the convex combination $\overline{H}$. Only linear combinations and inverses were used.

D2: Short proof

Answer 5.14. Prove reversal of matrix order

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* local setup only.

Task. If $0 \prec A \preceq B$, show $B^{-1} \preceq A^{-1}$. Hint: diagonalize $A^{-1/2}BA^{-1/2}$.

Solution. If $A = U\,\mathrm{diag}(a_j)U^\top$ is its orthogonal spectral decomposition, with $a_j > 0$, define $A^{-1/2} = U\,\mathrm{diag}(a_j^{-1/2})U^\top$. Set $C = A^{-1/2}BA^{-1/2}$. For any vector v,

$$v^\top(C - I)v = (A^{-1/2}v)^\top(B - A)(A^{-1/2}v) \geq 0,$$

so $C \succeq I$. Since C is symmetric, the spectral theorem diagonalizes it orthogonally. Its eigenvalues are at least one, so the eigenvalues of C^{-1} are at most one. Thus $C^{-1} \preceq I$. Congruence preserves order: $X \preceq Y$ implies $R^\top X R \preceq R^\top Y R$, by testing the former inequality on Rv. Apply this with $R = A^{-1/2}$ to obtain

$$B^{-1} = A^{-1/2}C^{-1}A^{-1/2} \preceq A^{-1}.$$

The congruence reduces the proof to eigenvalue comparison for the single symmetric matrix C.

D2: Short proof

Answer 5.15. Preserve the order interval

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.6, 5.7, 5.10, 5.12, 5.14.

Task. For $n_x \neq 0$, start from $\Lambda - \rho I_8 \preceq K_i \preceq \Lambda$ and use step 5.14 twice to prove $0 \preceq H_{\mathrm{lam}} \preceq \rho I_8$. For $n_x = 0$, use step 5.6.

Solution. All matrices in the starting inequalities are positive definite by step 5.7. Invert them to obtain

$$\Lambda^{-1} \preceq K_i^{-1} \preceq (\Lambda - \rho I_8)^{-1}.$$

Averaging preserves these inequalities, so they also hold with K_i^{-1} replaced by $Q = \lambda K_1^{-1} + (1 - \lambda)K_2^{-1}$. Inverting again gives

$$\Lambda - \rho I_8 \preceq Q^{-1} \preceq \Lambda.$$

Since $H_{\mathrm{lam}} = \Lambda - Q^{-1}$, subtraction yields $0 \preceq H_{\mathrm{lam}} \preceq \rho I_8$. For $n_x = 0$, this follows directly from convexity of the interval $[0, \rho I_8]$. Together with step 5.13 this proves that every layering step preserves $\mathcal{K}$.

D2: Short proof

Answer 5.16. Induct over a finite hierarchy

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.13, 5.15.

Task. Assume $L_0, L_1 \in \mathcal{K}$. Use preservation of $\mathcal{K}$ under one layering step to prove that every finite hierarchical laminate belongs to $\mathcal{K}$.

Solution. A finite hierarchy is a finite binary tree. Its leaves are the original constituents L_0, L_1, which belong to $\mathcal{K}$ by assumption. If both children of an internal node are in $\mathcal{K}$, steps 5.13 and 5.15 put their laminate parent in $\mathcal{K}$. Induction from the leaves to the root therefore proves membership for every finite hierarchical laminate. Each node may have its own normal and proportion. A proportion of zero or one simply selects one child.

D1: Direct

Answer 5.17. Pass to tensor limits

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.16.

Task. Assume $\mathcal{K}$ is compact and a continuous functional ℓ vanishes on $\mathcal{K}$. Show that every limit of finite laminate tensors of L_0, L_1 satisfies $\ell = 0$, even if their phase fractions vary.

Solution. Let B_j be finite laminate tensors with $B_j \to B$ in tensor norm. Step 5.16 gives $B_j \in \mathcal{K}$. A compact subset of a finite-dimensional normed space is closed, so $B \in \mathcal{K}$. Thus $\ell(B) = 0$. Equivalently, continuity gives $\ell(B) = \lim_j \ell(B_j) = 0$. The argument applies regardless of the overall phase fraction of each B_j.

6 A convergent Fourier cell calculation

Prerequisites. The weak cell problem and mean flux: Step 1.5–Step 1.10; common vertical block: Step 2.8–Step 2.9; matrix rank and ellipticity: Step 3.14, Step 4.2; directional bound: Step 4.11–Step 4.12.

Definition lookup. The setup below defines $t, a, b, c, H, \mathcal{R}, T, \mathcal{T}$ and the Wiener norm; its lookup paragraph defines Fourier support, weak derivatives, and H^1_{per}. The series and averaged parameter are defined in Step 6.6–Step 6.8; σ_ζ is defined in Step 6.20.

Supplied background. B3, B4, B5, B6, B12. Each key links to its statement, scope, and reference.

Setup 6.1. Work on $Y = (0, 2\pi)^2$ with normalized averages. Define

$$t(x,y) = \frac{1}{2} + \frac{\cos x + \cos y + \cos(x+3y)}{6}, \qquad a = (1,0), \quad b = (0,1), \quad c = (1,3).$$

Use $H = \rho t I_8$, $0 < \rho < 2/3$, and $\overline{L} = L(H) = \operatorname{diag}(I_{16} - PHP^\top, D)$. Here $\mathcal{K} = \{L(A) : A \in \mathcal{S},\ 0 \preceq A \preceq \rho I_8\}$ and $\mathcal{S} = \operatorname{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$. The matrices P, D, Γ_j are fixed as in Modules 3–4: $D \succeq I_{16}/4$ is constant, P has full column rank, and $\|P\|^2 = 3/2$. For $k \neq 0$, set $\mathcal{R}(k) = (k_x^2 I_{16} + k_y^2 D)^{-1}$ and $T(k) = k_x^2 P^\top \mathcal{R}(k) P$; set $T(0) = 0$. Module 4 gives $T(-k) = T(k)$, $\|T(k)\| \le 3/2$, and $T(b) = 0$.

Our Fourier convention is

$$\widehat{M}(k) = \langle M(x,y) e^{-\mathrm{i}k\cdot(x,y)} \rangle, \qquad \widehat{\partial_\nu M}(k) = \mathrm{i}k_\nu \widehat{M}(k).$$

Define $\widehat{\mathcal{T}M}(k) = T(k)\widehat{M}(k)$ and $\|M\|_{\mathcal{A}} = \sum_{k\in\mathbb{Z}^2} \|\widehat{M}(k)\|$, using the matrix operator norm. Completeness of the space of absolutely summable coefficient sequences is supplied; it makes this Wiener space complete. A horizontal mean gradient ζ means $\zeta_y = 0$, with transposed first row $\zeta_x \in \mathbb{R}^{16}$. References: manuscript Sections 4.1–4.2 and Appendix B.

Setup 6.2. Fourier lookup: $\langle M \rangle = (2\pi)^{-2} \int_Y M$; the exponentials have $\langle e^{\mathrm{i}(k-l)\cdot(x,y)} \rangle = 1$ if $k = l$ and 0 otherwise. The Fourier support is the set of k with $\widehat{M}(k) \neq 0$. A Fourier multiplier acts on each coefficient separately and creates no new frequencies. The Wiener space $\mathcal{A}$ consists of absolutely summable coefficients, identified with their Fourier series as justified in Step 6.3. Its matrix norm also applies to complex coefficients, with the Euclidean norm on complex vectors. The same Wiener and supremum notation applies to scalar fields using absolute value and to vector fields using the Euclidean norm. Here $\|M\|_\infty = \sup_{(x,y)\in Y} \|M(x,y)\|$ for continuous fields. A weak derivative $g = \partial_\nu u$ satisfies $\langle g \cdot \varphi \rangle = -\langle u \cdot \partial_\nu \varphi \rangle$ for every smooth periodic test φ. The space H^1_{per} consists of periodic L^2 functions with both weak first derivatives in L^2. A corrector is the zero-mean weak solution from Step 1.4; its mean flux defines the effective tensor by Step 1.9.

D1: Direct

Answer 6.1. Bound the local fraction

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* local setup only.

Task. Show $0 \le t \le 1$ and $\langle t \rangle = 1/2$. Deduce that $\overline{L}(x,y)$ takes values in $\mathcal{K}$.

Solution. Each cosine lies between -1 and 1, so their sum lies between -3 and 3. Adding $1/2$ after division by six gives $0 \le t \le 1$. Each nonconstant cosine has zero integral over the periodic cell, hence $\langle t \rangle = 1/2$. Therefore $H = \rho t I_8$ belongs to $\mathcal{S}$ and satisfies $0 \preceq H \preceq \rho I_8$ pointwise. These are the defining conditions for $L(H) \in \mathcal{K}$.

D1: Direct

Answer 6.2. List the Fourier data

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* local setup only.

Task. Compute $\widehat{t}(k)$ and show $\|t\|_{\mathcal{A}} = 1$.

Solution. The identity $\cos(k \cdot x) = (e^{\mathrm{i}k\cdot x} + e^{-\mathrm{i}k\cdot x})/2$ gives

$$\widehat{t}(0) = \frac{1}{2}, \qquad \widehat{t}(\pm a) = \widehat{t}(\pm b) = \widehat{t}(\pm c) = \frac{1}{12}.$$

All other coefficients are zero because distinct exponential Fourier modes are orthogonal on Y. These seven frequencies are distinct. Their absolute values sum to $1/2 + 6/12 = 1$. This simple norm value controls every later product estimate.

D2: Short proof

Answer 6.3. Control a function by its coefficients

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* local setup only.

Task. Show $\|M\|_{\infty} \le \|M\|_{\mathcal{A}}$. Explain why the Fourier series converges uniformly when the right side is finite.

Solution. For every partial sum and every point (x, y), the triangle inequality gives

$$\left\| \sum_{k \in E} \widehat{M}(k) e^{\mathrm{i}k\cdot(x,y)} \right\| \le \sum_{k \in E} \|\widehat{M}(k)\|.$$

The same estimate for tails is independent of (x, y). Since the coefficient norms are summable, these tails tend uniformly to zero. Thus the series converges uniformly to a continuous matrix field, and taking the limit in the inequality proves the supremum bound. The estimate holds in operator norm and therefore controls every matrix entry. Integrating the uniformly convergent series against $e^{-\mathrm{i}l\cdot(x,y)}$ recovers its prescribed coefficient at l by Fourier orthogonality.

D2: Short proof

Answer 6.4. Bound one product

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3.

Task. Use $\widehat{MN}(k) = \sum_l \widehat{M}(l)\widehat{N}(k-l)$ to prove $\|MN\|_{\mathcal{A}} \le \|M\|_{\mathcal{A}}\|N\|_{\mathcal{A}}$.

Solution. Apply the operator-norm inequality $\|AB\| \le \|A\|\|B\|$ to each summand. Then

$$\begin{aligned} \|MN\|_{\mathcal{A}} &\le \sum_{k,l} \|\widehat{M}(l)\| \, \|\widehat{N}(k-l)\| \\ &= \left(\sum_{l} \|\widehat{M}(l)\| \right) \left(\sum_{j} \|\widehat{N}(j)\| \right). \end{aligned}$$

The nonnegative double sum can be reindexed by $j = k - l$; the resulting product is finite. Absolute convergence justifies multiplying the Fourier series and gives the convolution formula. The matrices inside each product retain the order M then N.

D1: Direct

Answer 6.5. Bound the multiplier

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.1, 4.11.

Task. Use $\|T(k)\| \le 3/2$ to show $\|\mathcal{T}M\|_{\mathcal{A}} \le (3/2)\|M\|_{\mathcal{A}}$. Why is $T(0)$ set to zero?

Solution. By definition,

$$\|\mathcal{T}M\|_{\mathcal{A}} = \sum_k \|T(k)\widehat{M}(k)\| \le \frac{3}{2}\sum_k \|\widehat{M}(k)\|.$$

A periodic gradient fluctuation has mean zero; it must therefore have no zero-frequency component. Setting $T(0) = 0$ implements this condition. This assigned value removes the constant mode; the other values of the symbol depend on direction.

D1: Direct

Answer 6.6. Bound each term of the source series

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.2, 6.4, 6.5.

Task. Let $X_1 = tI_8$ and $X_{p+1} = t\mathcal{T}X_p$. Prove $\|X_p\|_{\mathcal{A}} \le (3/2)^{p-1}$.

Solution. For $p = 1$, $\|X_1\|_{\mathcal{A}} = \|t\|_{\mathcal{A}}\|I_8\| = 1$. If the bound holds at p, steps 6.4 and 6.5 give

$$\|X_{p+1}\|_{\mathcal{A}} \le \|t\|_{\mathcal{A}}\|\mathcal{T}X_p\|_{\mathcal{A}} \le \frac{3}{2}(3/2)^{p-1} = (3/2)^p.$$

Induction proves the estimate. The profile factor costs one in this norm, and each multiplier costs at most $3/2$.

D2: Short proof

Answer 6.7. Solve the reduced source equation

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.4, 6.5, 6.6.

Task. Show that $X = \sum_{p\ge1} \rho^p X_p$ converges in the Wiener norm and is the unique Wiener-space solution of $X = H + H\mathcal{T}X$. First sum the norm bounds with $q = 3\rho/2 < 1$, then pass the bounded linear map $M \mapsto H\mathcal{T}M$ through the series. For uniqueness, bound the norm of the difference of two solutions.

Solution. Put $q = 3\rho/2 < 1$. Step 6.6 yields

$$\sum_{p\ge1} \|\rho^p X_p\|_{\mathcal{A}} \le \rho \sum_{p\ge1} q^{p-1} = \frac{\rho}{1-q}.$$

Completeness gives a limit X. Multiplication and $\mathcal{T}$ are bounded by steps 6.4–6.5, so the recurrence may be summed:

$$X = \rho t I_8 + \rho t\mathcal{T}\sum_{p\ge1} \rho^p X_p = H + H\mathcal{T}X.$$

For uniqueness, subtract two solutions. Their difference Z satisfies $\|Z\|_{\mathcal{A}} \le q\|Z\|_{\mathcal{A}}$. As $q < 1$, this forces $Z = 0$. Thus the series solves the source equation in the Wiener space.

D1: Direct

Answer 6.8. Bound the fifth-order remainder

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.6, 6.7.

Task. Set $H_* = \langle X \rangle$. Prove $\|H_* - \sum_{p=1}^{5} \rho^p \langle X_p \rangle\| \le (3/2)^5 \rho^6/(1 - 3\rho/2)$.

Solution. The average is the coefficient at frequency zero, so $\|\langle M \rangle\| \le \|M\|_{\mathcal{A}}$. In particular averaging is a continuous linear operation and may be taken term by term in the convergent series. Therefore the norm of the remainder is at most

$$\begin{aligned}\sum_{p\ge 6} \rho^p \|X_p\|_{\mathcal{A}} &\le \sum_{p\ge 6} \rho^p (3/2)^{p-1} \\ &= (3/2)^5 \rho^6 \sum_{j\ge 0} (3\rho/2)^j \\ &= \frac{(3/2)^5 \rho^6}{1 - 3\rho/2}.\end{aligned}$$

The index $p = 6$ determines the prefactor. This explicit bound will be used to control the sign of a fifth-order coefficient.

D1: Direct

Answer 6.9. Bound the Green multipliers

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 5.14.

Task. Prove $\|\mathcal{R}(k)\| \le 4/|k|^2$ for $k \ne 0$, and deduce $\|k_\nu k_x \mathcal{R}(k)\| \le 4$ for $\nu = x, y$.

Solution. Since $D \succeq I_{16}/4$,

$$k_x^2 I_{16} + k_y^2 D \succeq (k_x^2 + k_y^2/4) I_{16} \succeq \frac{|k|^2}{4} I_{16}.$$

Inversion gives the first inequality. For the second use $|k_\nu k_x| \le |k|^2$ and multiply the bounds. Thus the first derivative of the potential, whose Fourier formula contains two frequency factors and one $\mathcal{R}(k)$, is controlled uniformly in k.

D1: Direct

Answer 6.10. Recover the Fourier potential

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.7, 6.9.

Task. For horizontal ζ, solve $(\partial_{xx} + D\partial_{yy})u = \partial_x(PXP^\top \zeta_x)$ in Fourier coefficients with $\widehat{u}(0) = 0$.

Solution. At a nonzero frequency k, the differential equation is

$$-(k_x^2 I_{16} + k_y^2 D)\widehat{u}(k) = \mathrm{i} k_x P \widehat{X}(k) P^\top \zeta_x.$$

Multiply by the inverse of the positive matrix on the left. This gives

$$\widehat{u}(k) = -\mathrm{i} k_x \mathcal{R}(k) P \widehat{X}(k) P^\top \zeta_x.$$

At $k = 0$ the equation is $0 = 0$; prescribing $\widehat{u}(0) = 0$ selects the zero-mean potential. Differentiation then gives

$$\widehat{\partial_\nu u}(k) = k_\nu k_x \mathcal{R}(k) P \widehat{X}(k) P^\top \zeta_x,$$

because $(\mathrm{i})(-\mathrm{i}) = 1$.

D2: Short proof

Answer 6.11. Check that the potential is real

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 4.12, 6.6, 6.7, 6.10.

Task. Show that the coefficients in step 6.10 satisfy $\widehat{u}(-k) = \overline{\widehat{u}(k)}$.

Solution. The scalar function t is real, so its Fourier coefficients have conjugate symmetry. The matrices $T(k)$ are real and even in k. Convolution with t and multiplication by $T(k)$ preserve conjugate symmetry, so every X_p, and then X, satisfies $\widehat{X}(-k) = \overline{\widehat{X}(k)}$. The matrices P, $\mathcal{R}(k)$, and ζ_x are real, and $\mathcal{R}(-k) = \mathcal{R}(k)$. Consequently

$$\widehat{u}(-k) = \mathrm{i}k_x \mathcal{R}(k) P \overline{\widehat{X}(k)} P^{\top} \zeta_x = \overline{\widehat{u}(k)}.$$

Fourier modes at k and $-k$ therefore pair into real-valued contributions. The zero coefficient is real as well.

D3: Guided technical

Answer 6.12. Check admissibility of the potential

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3, 6.7, 6.9, 6.10, 6.11.

Task. Use steps 6.9–6.10 and $|k| \geq 1$ on $\mathbb{Z}^2 \setminus \{0\}$ to show $u \in H^1_{\mathrm{per}}(Y; \mathbb{R}^{16})$. Bound the absolute sums for u and its formal first derivatives. For the weak-derivative identity, integrate by parts for finite Fourier sums and pass to their uniform limits against a smooth periodic test.

Solution. Since $\|P\|^2 = 3/2$, each derivative coefficient in step 6.10 is bounded by $6\|\widehat{X}(k)\||\zeta_x|$. Its coefficients are therefore absolutely summable. The potential coefficients satisfy

$$|\widehat{u}(k)| \leq \frac{6}{|k|}\|\widehat{X}(k)\||\zeta_x| \leq 6\|\widehat{X}(k)\||\zeta_x|.$$

Let u_m be the finite Fourier sum over $|k_x|, |k_y| \leq m$, and let g_ν be the uniformly convergent series with coefficients from step 6.10. The absolute-sum bounds and step 6.3 give $u_m \to u$ and $\partial_\nu u_m \to g_\nu$ uniformly. For any smooth periodic vector test φ, integration by parts on the periodic cell gives

$$\langle \partial_\nu u_m \cdot \varphi \rangle = -\langle u_m \cdot \partial_\nu \varphi \rangle.$$

Uniform convergence permits passing to the limit in both integrals, so g_ν is the weak derivative of u. These fields are continuous on the compact torus, hence belong to L^2. This proves $u \in H^1_{\mathrm{per}}$; its zero Fourier coefficient fixes its mean at zero.

D1: Direct

Answer 6.13. Find the reduced horizontal derivative

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.5, 6.10, 6.12.

Task. Multiply the x-derivative formula in step 6.10 by $P^{\top}$ and prove $P^{\top}\partial_x u = (\mathcal{T}X)P^{\top}\zeta_x$.

Solution. At a nonzero frequency,

$$\begin{aligned}\widehat{P^{\top}\partial_x u}(k) &= k_x^2 P^{\top}\mathcal{R}(k)P\widehat{X}(k)P^{\top}\zeta_x \\ &= T(k)\widehat{X}(k)P^{\top}\zeta_x.\end{aligned}$$

Both sides have zero coefficient at $k = 0$: the left side is a derivative, and $T(0) = 0$ on the right. Hence their Fourier coefficients agree at every frequency. The uniformly convergent Fourier series therefore give the field identity. The order $T(k)\widehat{X}(k)$ agrees with the definition of $\mathcal{T}$.

D3: Guided technical

Answer 6.14. Verify the cell equilibrium equation

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.5, 4.2, 6.1, 6.7, 6.10, 6.12, 6.13.

Task. Use $X = H + H\mathcal{T}X$ and step 6.13 to prove $\operatorname{div}[\overline{L}(\zeta + \nabla u)] = 0$. Deduce that u is the corrector for $\overline{L}$. Write the two flux rows explicitly and use the baseline equation of step 6.10 in distributions. Extend the test identity from smooth tests to H^1 tests, and use Step 1.5 to identify the energy minimizer.

Solution. Step 6.13 and the source equation give

$$\begin{aligned} PHP^\top(\zeta_x + \partial_x u) &= PH[I_8 + \mathcal{T}X]P^\top\zeta_x \\ &= PXP^\top\zeta_x. \end{aligned}$$

The baseline equation solved in step 6.10 therefore becomes

$$\partial_{xx}u + D\partial_{yy}u = \partial_x\{PHP^\top(\zeta_x + \partial_x u)\}.$$

Indeed the flux rows are

$$j_x = \zeta_x + \partial_x u - PXP^\top\zeta_x, \qquad j_y = D\partial_y u.$$

Their distributional divergence is zero by the preceding baseline equation. To verify this distributional statement directly, truncate both X and u to the same frequency rectangle. The resulting finite sums satisfy the baseline equation by step 6.10. Test it against a smooth periodic vector field and integrate all derivatives onto the test. Uniform convergence of the truncated X and u then permits passage to the limit. Thus $\langle j \cdot \nabla v\rangle = 0$ first for smooth periodic v. Since $j \in L^2$, the bound

$$|\langle j \cdot \nabla(v_m - v)\rangle| \le \|j\|_2\|\nabla(v_m - v)\|_2$$

extends the identity to periodic H^1 tests by smooth density from Module 1. For any such competitor w, expansion and this test identity give

$$E_\zeta(w) - E_\zeta(u) = \langle \nabla(w-u) \cdot \overline{L}\nabla(w-u)\rangle \ge \mu\|\nabla(w-u)\|_2^2,$$

where $E_\zeta(v) = \langle(\zeta + \nabla v) \cdot \overline{L}(\zeta + \nabla v)\rangle$ and $\mu > 0$ is supplied by Step 4.2 and Step 6.1. This is the argument of Step 1.5. Hence u is a minimizing corrector; equality forces the same gradient, and zero mean makes the potential unique.

D2: Short proof

Answer 6.15. Average the flux

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.1, 1.7, 1.9, 2.8, 2.9, 6.8, 6.14.

Task. For $e = \zeta + \nabla u$, show $\langle \overline{L}e\rangle_x = \zeta_x - PH_*P^\top\zeta_x$ and $\langle \overline{L}e\rangle_y = 0$. Deduce $B = (\overline{L})_* = L(H_*)$. Use Step 1.1 for periodic derivatives and Step 2.8–Step 2.9 to include vertical mean gradients.

Solution. The constitutive source identity in step 6.14 gives horizontal flux $j_x = \zeta_x + \partial_x u - PXP^\top\zeta_x$. Since a periodic derivative has mean zero,

$$\langle j_x\rangle = \zeta_x - P\langle X\rangle P^\top\zeta_x = \zeta_x - PH_*P^\top\zeta_x.$$

The vertical flux is $D\partial_y u$, whose mean is zero because D is constant. Here both derivative averages vanish by Step 1.1. These identities hold for every horizontal ζ. By Step 2.8–Step 2.9, every constant vertical mean gradient has zero corrector and retains the block D, with zero horizontal mean flux. By linearity from Step 1.7, the two classes of loads determine the flux at every mean gradient. Thus the full effective tensor is $\operatorname{diag}(I_{16} - PH_*P^\top, D) = L(H_*)$. Its parameter is the averaged source $\langle X\rangle$.

D1: Direct

Answer 6.16. Prove symmetry of the averaged source

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.9, 3.14, 6.15.

Task. Use self-adjointness of $B = L(H_*)$ and $S = P^\top P \succ 0$ to show $H_*^\top = H_*$.

Solution. Self-adjointness was proved in Step 1.9. Taking the transpose of the horizontal block from step 6.15 therefore gives $P(H_* - H_*^\top)P^\top = 0$. Since P has full column rank, $v^\top S v = |Pv|^2 > 0$ for every nonzero v, so S^{-1} exists. The identities $S^{-1}P^\top P = I_8$ and $P^\top P S^{-1} = I_8$ show that $S^{-1}P^\top$ is a left inverse of P, and PS^{-1} is a right inverse of $P^\top$. Multiply the displayed identity on the left and right by these two matrices to obtain $H_* - H_*^\top = 0$. Thus the averaged source is symmetric.

D1: Direct

Answer 6.17. Check the first two averaged coefficients

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.2, 6.5, 6.6, 6.7, 6.8.

Task. Use $X_2 = t\mathcal{T}(tI_8)$ to show $H_* = (\rho/2)I_8 + (\rho^2/72)[T(a) + T(c)] + O(\rho^3)$. Here $O(\rho^3)$ means that the operator norm of the error is at most $C\rho^3$ for all sufficiently small positive ρ.

Solution. First, $\langle X_1 \rangle = \langle t \rangle I_8 = I_8/2$. For the next coefficient, convolution at frequency zero gives

$$\langle X_2 \rangle = \sum_k \widehat{t}(-k) T(k) \widehat{t}(k).$$

The $k = 0$ term vanishes because $T(0) = 0$. The terms at $\pm b$ vanish because $T(b) = 0$. Each of the four terms at $\pm a, \pm c$ has scalar factor $1/144$; evenness of T combines each pair. Hence $\langle X_2 \rangle = [T(a) + T(c)]/72$. The remaining terms satisfy $\| \sum_{p \geq 3} \rho^p \langle X_p \rangle \| \leq (9/4)\rho^3/(1 - 3\rho/2)$ by step 6.6. For $0 < \rho \leq 1/3$, this is at most $(9/2)\rho^3$, which proves the stated error bound.

D2: Short proof

Answer 6.18. Track the Fourier support

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.2, 6.6.

Task. Prove that every frequency in X_p satisfies $|k_x| \leq p$ and $|k_y| \leq 3p$.

Solution. For $p = 1$, the support is $\{0, \pm a, \pm b, \pm c\}$, whose coordinate sizes are at most one and three. A Fourier multiplier does not create new frequencies: it only multiplies each coefficient at its existing frequency. Multiplication by t convolves with this seven-point support. Thus each step from X_p to X_{p+1} adds at most one to the bound on $|k_x|$ and three to the bound on $|k_y|$. Induction proves the stated rectangle bounds. This also shows that every X_p is a trigonometric polynomial: its support is finite, contained in sums of p frequencies from the support of t.

D3: Guided technical

Answer 6.19. Differentiate the source series once

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3, 6.6, 6.7, 6.18.

Task. Use step 6.18 to prove $\|\partial_x X\|_{\mathcal{A}} \leq \rho/(1-q)^2$ and $\|\partial_y X\|_{\mathcal{A}} \leq 3\rho/(1-q)^2$, where $q = 3\rho/2$. Use $\sum_{p \geq 1} p q^{p-1} = (1-q)^{-2}$. To justify differentiation, integrate the uniformly convergent derivative series along one coordinate segment and compare with the original series at the segment endpoints.

Solution. Each derivative multiplies a Fourier coefficient by its frequency component. Conse-

quently step 6.18 and step 6.6 give

$$\|\partial_x X_p\|_{\mathcal{A}} \le p(3/2)^{p-1}, \qquad \|\partial_y X_p\|_{\mathcal{A}} \le 3p(3/2)^{p-1}.$$

Multiply by ρ^p and sum, using $\sum_{p\ge1} pq^{p-1} = (1-q)^{-2}$. For example, the scalar identity follows by counting the pairs of nonnegative integers with sum $p-1$ in $(\sum_{j\ge0} q^j)^2$. This gives the two bounds. To justify that these are the derivatives of X, write $S_m = \sum_{p=1}^{m} \rho^p X_p$. Both S_m and their first derivatives converge uniformly by step 6.3. For a fixed y and any coordinate segment, the fundamental theorem of calculus gives

$$S_m(x+h,y) - S_m(x,y) = \int_x^{x+h} \partial_x S_m(s,y)\, ds.$$

Pass to the uniform limits. The limiting derivative series is continuous, so another application of the fundamental theorem proves that it is $\partial_x X$. Apply the same argument in y. The derivative series therefore are the classical first derivatives of X.

D1: Direct

Answer 6.20. Bound the source field itself

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 3.14, 6.3, 6.7, 6.9, 6.10.

Task. Set $\sigma_\zeta = PP^\top(\zeta_x + \partial_x u)$ and $q = 3\rho/2$. Prove $\|\sigma_\zeta\|_\infty \le \frac{3}{2}[1 + 6\rho/(1-q)]|\zeta_x|$.

Solution. By steps 6.9–6.10, each Fourier coefficient of $\partial_x u$ has norm at most $4\|P\|^2\|\widehat{X}(k)\||\zeta_x| = 6\|\widehat{X}(k)\||\zeta_x|$. Summing and applying step 6.3 gives

$$\|\partial_x u\|_\infty \le 6\|X\|_{\mathcal{A}}|\zeta_x| \le \frac{6\rho}{1-q}|\zeta_x|,$$

using the sum in step 6.7. Also $\|PP^\top\| \le \|P\|^2 = 3/2$. Applying this operator norm to the definition of σ_ζ gives

$$\|\sigma_\zeta\|_\infty \le \frac{3}{2}\left(|\zeta_x| + \|\partial_x u\|_\infty\right) \le \frac{3}{2}\left(1 + \frac{6\rho}{1-q}\right)|\zeta_x|.$$

The bound depends only on the fixed coefficient field and ρ.

D2: Short proof

Answer 6.21. Bound the source field derivatives

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.9, 6.10, 6.19, 6.20.

Task. Use step 6.19 and the bounded corrector multipliers to bound $\partial_x\sigma_\zeta$ and $\partial_y\sigma_\zeta$ by constants times $|\zeta_x|$. Explain why the constants remain valid for every stripe count N in the approximation of Module 8.

Solution. The formal derivative of the Fourier series for $\partial_x u$ is obtained by applying its multiplier to $\partial_\nu X$. Step 6.19 makes these coefficients absolutely summable. Both the original series and this derivative series converge uniformly, so the coordinate-integration argument of step 6.19 identifies the latter as the classical derivative $\partial_\nu\partial_x u$. Thus, for $\nu = x, y$,

$$\|\partial_\nu\partial_x u\|_\infty \le 6\|\partial_\nu X\|_{\mathcal{A}}|\zeta_x|.$$

The constant mean ζ_x disappears under differentiation of σ_ζ. Using $\|PP^\top\| \le 3/2$ and applying step 6.19 yields

$$\|\partial_x\sigma_\zeta\|_\infty \le \frac{9\rho}{(1-q)^2}|\zeta_x|,$$

$$\|\partial_y\sigma_\zeta\|_\infty \le \frac{27\rho}{(1-q)^2}|\zeta_x|.$$

Together with step 6.20, the sum of the three field norms is at most $C_\rho|\zeta_x|$, where one may take

$$C_\rho = \frac{3}{2} + \frac{9\rho}{1-q} + \frac{36\rho}{(1-q)^2}.$$

The constants involve only t, P, D, and the fixed contrast ρ, so they hold for every stripe count N in Module 8.

D3: Guided technical

Answer 6.22. Prove smoothness of the fields

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3, 6.9, 6.10, 6.12, 6.18, 6.19.

Task. Show that differentiating X_p any fixed number of times adds only a polynomial factor in p. Deduce that X, u, and $e = \zeta + \nabla u$ are smooth. For nonnegative integers r, s, establish $\|\partial_x^r \partial_y^s X_p\|_{\mathcal{A}} \le 3^s p^{r+s} \|X_p\|_{\mathcal{A}}$. For any derivative of u of positive order, leave one x- or y-derivative in the bounded multiplier from step 6.9 and transfer the remaining derivatives to X.

Solution. For nonnegative integers r, s, step 6.18 gives

$$\|\partial_x^r \partial_y^s X_p\|_{\mathcal{A}} \le 3^s p^{r+s} (3/2)^{p-1}.$$

After multiplication by ρ^p, the sum is bounded by $3^s \rho \sum_{p\ge 1} p^{r+s} q^{p-1}$, which converges for $q < 1$; the ratio of successive terms tends to q. Hence every differentiated series converges uniformly. Repeated application of the coordinate-integration argument from step 6.19 proves that X is C^∞.

For the potential, denote by K_ν the coefficient multiplier

$$\widehat{K_\nu M}(k) = k_\nu k_x \mathcal{R}(k) P \widehat{M}(k) P^\top \zeta_x \quad (k \ne 0), \qquad \widehat{K_\nu M}(0) = 0.$$

Step 6.9 gives $\|K_\nu M\|_{\mathcal{A}} \le 6|\zeta_x| \|M\|_{\mathcal{A}}$. For a derivative $\partial_x^r \partial_y^s u$ of positive total order, if $r \ge 1$ its Fourier series is $K_x(\partial_x^{r-1} \partial_y^s X)$; if $r = 0$ and $s \ge 1$, it is $K_y(\partial_y^{s-1} X)$. These series all converge uniformly by the bounds just proved. Using step 6.12 for the starting potential and repeating the coordinate-integration argument identifies them as the actual derivatives. Thus X, u, and $e = \zeta + \nabla u$ are smooth periodic fields for $0 < \rho < 2/3$.

7 The first nonzero energy difference

Prerequisites. Generator identities: Step 3.5–Step 3.6; the energy detector: Step 4.7; the convergent expansion and mean flux: Step 6.6–Step 6.8, Step 6.15.

Definition lookup. The next setups recall all matrices, frequencies, and the expansion. The lookup paragraph defines the Fourier convention, coefficient extraction, permutation sign, and determinant; the identity before Step 7.19 is supplied.

Supplied background. B3, B5, B7. Each key links to its statement, scope, and reference.

Setup 7.1. Work on $Y = (0, 2\pi)^2$ with normalized average $\langle\cdot\rangle$. The real symmetric matrices $\Gamma_1, \dots, \Gamma_4$ satisfy $\Gamma_i^2 = I_8$, $\Gamma_i\Gamma_j = -\Gamma_j\Gamma_i$ for $i \neq j$, and $\Gamma_1\Gamma_2\Gamma_3\Gamma_4 = \Omega = \operatorname{diag}(I_4, -I_4)$. Each Γ_j has zero 4×4 diagonal blocks. Put $\mathcal{S} = \operatorname{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$ and $d(M) = M_{11} - M_{55}$. Earlier modules identify the measured energy difference as $\ell(B) = d(H_*)$.

Setup 7.2. For $k \neq 0$, use

$$T(k) = \frac{1}{4}\sum_{j=1}^{4} \frac{jk_x^2}{jk_x^2 + k_y^2}(I_8 + \Gamma_j), \qquad T(0) = 0.$$

Thus $T(k) \in \mathcal{S}$, $T(-k) = T(k)$, $T(k) = 0$ when $k_x = 0$, and $\|T(k)\| \le 3/2$. Let $a = (1, 0)$, $b = (0, 1)$, $c = (1, 3)$, and $t = \frac{1}{2} + \frac{1}{6}[\cos x + \cos y + \cos(x + 3y)]$. Its Fourier coefficients are $\widehat{t}(0) = 1/2$, $\widehat{t}(\pm a) = \widehat{t}(\pm b) = \widehat{t}(\pm c) = 1/12$, and zero otherwise. The convergent expansion is

$$H_* = \sum_{p \ge 1} \rho^p \langle X_p \rangle, \qquad X_1 = tI_8, \qquad X_{p+1} = t\mathcal{T}X_p, \quad \widehat{\mathcal{T}M}(k) = T(k)\widehat{M}(k).$$

For $0 < \rho < 2/3$, its remainder $R_6 = H_* - \sum_{p=1}^{5} \rho^p \langle X_p \rangle$ satisfies $\|R_6\| \le (3/2)^5 \rho^6/(1 - 3\rho/2)$.

Setup 7.3. Lookup: $\langle M \rangle = (2\pi)^{-2} \int_Y M$ and $\widehat{M}(k) = \langle M(x, y)e^{-\mathrm{i}k\cdot(x,y)} \rangle$ for $k \in \mathbb{Z}^2$; hence $\widehat{M}(0) = \langle M \rangle$. Matrix norms are operator norms, $\|M\| = \sup_{|v|=1} |Mv|$, and e_i is a coordinate unit vector. A skew-symmetric real matrix satisfies $A^\top = -A$. The sign $\operatorname{sgn}(\pi)$ is $(-1)^m$ if a permutation is put in order by m adjacent swaps, and $\det A = \sum_\pi \operatorname{sgn}(\pi) \prod_i A_{i,\pi(i)}$. Interchanging two rows changes its sign; equal rows make it zero. The notation $[\rho^p]f$ denotes the coefficient of ρ^p in the given convergent power series. A multiset records its elements with multiplicity while ignoring order. A closed path is an ordered list of frequency steps with sum zero; its intermediate vertices are the partial sums K_j. Products keep their written order. All frequency sums range over $\mathbb{Z}^2$. At each fixed order the Fourier sums are finite: X_1 has finite support, a multiplier preserves support, and multiplication by the finite Fourier series of t preserves finiteness.

D1: Direct

Answer 7.1. A norm bound

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

Task. Show $|d(M)| \le 2\|M\|$ for every 8×8 matrix M. Use the unit coordinate vectors e_1, e_5.

Solution. Write each selected entry as a quadratic pairing: $M_{ii} = e_i^\top M e_i$. The operator norm gives

$$|e_i^\top M e_i| \le |e_i|\,|Me_i| \le \|M\| \qquad (i = 1, 5).$$

The triangle inequality now yields $|M_{11} - M_{55}| \le 2\|M\|$. We will use this bound to control the scalar energy remainder.

D1: Direct

Answer 7.2. The basic values of the detector

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

Task. Compute $d(I_8)$, $d(\Gamma_j)$, and $d(\Omega)$ from the displayed block information.

Solution. The first and fifth diagonal entries of I_8 are both 1, so $d(I_8) = 0$. Both diagonal blocks of Γ_j vanish, so both selected entries are zero and $d(\Gamma_j) = 0$. Finally $\Omega_{11} = 1$ and $\Omega_{55} = -1$, hence $d(\Omega) = 2$. Thus the detector ignores the scalar and single-generator directions but detects the product of all four generators.

D2: Short proof

Answer 7.3. Two short skew-symmetric products

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

Task. For distinct indices, prove that $\Gamma_i\Gamma_j$ and $\Gamma_i\Gamma_j\Gamma_k$ are skew-symmetric. Explain why their diagonal entries are zero.

Solution. For two generators, symmetry of the factors and anticommutation give

$$(\Gamma_i\Gamma_j)^\top = \Gamma_j\Gamma_i = -\Gamma_i\Gamma_j.$$

For three distinct generators, reversing their order requires three swaps. Each swap changes the sign, so

$$(\Gamma_i\Gamma_j\Gamma_k)^\top = \Gamma_k\Gamma_j\Gamma_i = (-1)^3\Gamma_i\Gamma_j\Gamma_k = -\Gamma_i\Gamma_j\Gamma_k.$$

For a real skew-symmetric matrix A, $A_{ii} = (A^\top)_{ii} = -A_{ii}$, hence $A_{ii} = 0$. Therefore d vanishes on both kinds of products.

D2: Short proof

Answer 7.4. Cancellation for at most three factors

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.2, 7.3.

Task. Let $M_1, \dots, M_q \in \mathcal{S}$ with $q \in \{0, 1, 2, 3\}$. Expand the product and use $\Gamma_j^2 = I_8$ to show $d(M_1 \cdots M_q) = 0$; the empty product is I_8.

Solution. Expand each factor into its scalar part and its four generator parts. Each resulting summand is a scalar times a word of length at most three in the generators. If an index is repeated, anticommute the repeated generators until they are adjacent and replace their square by I_8. The word becomes, up to sign, a word in distinct generators of length at most three.

A length-zero word is I_8 and a length-one word is one Γ_j; Exercise 7.2 handles these. Length-two and length-three words are handled by Exercise 7.3. Every expanded summand therefore has zero detector value, so their sum does too.

D2: Short proof

Answer 7.5. A permuted four-generator word

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.2.

Task. For a permutation π of $\{1,2,3,4\}$, show $d(\Gamma_{\pi(1)}\Gamma_{\pi(2)}\Gamma_{\pi(3)}\Gamma_{\pi(4)}) = 2\,\mathrm{sgn}(\pi)$. Count the sign changes when adjacent generators are exchanged.

Solution. The ordered word $\Gamma_1\Gamma_2\Gamma_3\Gamma_4$ equals Ω. To arrange a permuted word in this order, exchange adjacent distinct generators. Each exchange contributes a factor -1 by anticommutation. The parity of the number of exchanges is precisely the permutation sign. Therefore

$$\Gamma_{\pi(1)}\Gamma_{\pi(2)}\Gamma_{\pi(3)}\Gamma_{\pi(4)} = \mathrm{sgn}(\pi)\,\Omega.$$

Apply the linear functional d and use $d(\Omega) = 2$ from Exercise 7.2. This is the signed-word contribution that will be summed in the next exercise.

D2: Short proof

Answer 7.6. Four factors produce a determinant

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.4, 7.5.

Task. Write $M_i = \alpha_{i0}I_8 + \sum_{j=1}^{4}\alpha_{ij}\Gamma_j$. Use Exercises 7.4–7.5 to show

$$d(M_1M_2M_3M_4) = 2\det[\alpha_{ij}]_{i,j=1}^{4}.$$

Hint: only words containing all four different generators survive.

Solution. An expanded term with a scalar part contains at most three generators, so Exercise 7.4 makes its detector zero. A term with four generators but a repeated index reduces to a word of length at most two and also has zero detector. We need keep only the terms with four different indices.

Such terms are indexed by permutations π of $\{1,2,3,4\}$. Exercise 7.5 computes the detector of each word. Therefore the surviving sum is

$$2\sum_{\pi}\mathrm{sgn}(\pi)\prod_{i=1}^{4}\alpha_{i,\pi(i)}.$$

This is exactly twice the permutation definition of the determinant. The scalar coefficients α_{i0} never enter it.

D2: Short proof

Answer 7.7. Why fourth order is still zero

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 6.6, 7.4.

Task. A term in $\langle X_p\rangle$ contains $p-1$ multiplier matrices T. Use Exercise 7.4 to show that the coefficients of $\rho, \rho^2, \rho^3, \rho^4$ in $\ell(B)$ vanish.

Solution. The recurrence starts with $X_1 = tI_8$, which contains no multiplier. Each passage $X_p \mapsto t\mathcal{T}X_p$ inserts one multiplier, so each Fourier summand at order p has exactly $p-1$ factors from $\mathcal{S}$. When $p \le 4$, their number is at most three and Exercise 7.4 applies. Linearity of averaging and d yields $d(\langle X_p\rangle) = 0$ for $p = 1,2,3,4$.

The cancellation concerns the selected scalar coefficient $d(\langle X_p\rangle)$; the averaged matrices themselves may be nonzero.

D1: Direct

Answer 7.8. One convolution step

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

Task. Use $X_{p+1} = t\mathcal{T}X_p$ and the finite-sum Fourier product rule $\widehat{tM}(q) = \sum_s \widehat{t}(s)\widehat{M}(q-s)$ to show

$$\widehat{X}_{p+1}(q) = \sum_s \widehat{t}(s)T(q-s)\widehat{X}_p(q-s)$$

while keeping each multiplier on the left of $\widehat{X}_p$.

Solution. For a matrix trigonometric polynomial M, the scalar-matrix product satisfies

$$\widehat{tM}(q) = \sum_s \widehat{t}(s)\widehat{M}(q-s).$$

Apply this product rule with $M = \mathcal{T}X_p$. By the definition of $\mathcal{T}$, its coefficient at frequency $q-s$ is $T(q-s)\widehat{X}_p(q-s)$. Thus

$$\widehat{X}_{p+1}(q) = \sum_s \widehat{t}(s)T(q-s)\widehat{X}_p(q-s).$$

The definition of $\mathcal{T}$ places the multiplier on the left. This order is retained in each further substitution.

D3: Guided technical

Answer 7.9. The ordered fifth coefficient

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.8.

Task. Iterate Exercise 7.8 at $q = 0$, labeling the outermost frequency k_1. With $K_j = k_1 + \cdots + k_j$ and $k_5 = -K_4$, verify

$$\widehat{X}_5(0) = \sum_{k_1,\dots,k_4} \left(\prod_{i=1}^{5} \widehat{t}(k_i)\right) T(-K_1)T(-K_2)T(-K_3)T(-K_4).$$

Start by writing $\widehat{X}_2(0)$; each further substitution adds one factor on the right.

Solution. Since $\widehat{X}_1(q) = \widehat{t}(q)I_8$, the first iteration at zero is

$$\widehat{X}_2(0) = \sum_{k_1} \widehat{t}(k_1)T(-k_1)\widehat{t}(-k_1)I_8.$$

The next iteration gives

$$\widehat{X}_3(0) = \sum_{k_1,k_2} \widehat{t}(k_1)\widehat{t}(k_2)T(-k_1)T(-k_1-k_2)\widehat{t}(-k_1-k_2)I_8.$$

This shows the pattern: each substitution into the remaining X factor adds the next multiplier on the right and replaces its frequency by the previous frequency minus the new step. After four substitutions, put $K_j = k_1 + \cdots + k_j$ and $k_5 = -K_4$. The result is

$$\widehat{X}_5(0) = \sum_{k_1,\dots,k_4} \left(\prod_{i=1}^{5} \widehat{t}(k_i)\right) T(-K_1)T(-K_2)T(-K_3)T(-K_4).$$

The scalar Fourier coefficients can be moved next to one another, but the matrix factors keep their original order. All sums are finite because the profile has finite Fourier support.

D1: Direct

Answer 7.10. Rewrite it as a closed path

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.9.

Task. Use $T(-k) = T(k)$, $\langle X_5 \rangle = \widehat{X}_5(0)$, and $\ell(B) = d(H_*)$ in Exercise 7.9 to obtain

$$[\rho^5]\ell(B) = \sum_{k_1+\cdots+k_5=0} \left(\prod_{i=1}^{5} \widehat{t}(k_i) \right) d(T(K_1)T(K_2)T(K_3)T(K_4)).$$

Explain why the five steps have total frequency zero.

Solution. In Exercise 7.9, evenness changes each $T(-K_j)$ to $T(K_j)$ while preserving the factor order. The definition $k_5 = -K_4$ is equivalent to $k_1 + \cdots + k_5 = 0$, so one can index the sum by all closed five-step paths. By the Fourier average convention, $\langle X_5 \rangle = \widehat{X}_5(0)$. The coefficient of ρ^5 in $\ell(B) = d(H_*)$ is therefore $d(\widehat{X}_5(0))$. Applying d to the previous formula gives

$$[\rho^5]\ell(B) = \sum_{k_1+\cdots+k_5=0} \left(\prod_{i=1}^{5} \widehat{t}(k_i) \right) d(T(K_1)T(K_2)T(K_3)T(K_4)).$$

The total frequency is zero because averaging keeps only the constant Fourier mode. The four partial sums are the intermediate vertices at which the multiplier acts.

D1: Direct

Answer 7.11. Zero endpoint steps

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.10.

Task. Show that a summand in Exercise 7.10 vanishes if $k_1 = 0$ or $k_5 = 0$.

Solution. If $k_1 = 0$, then $K_1 = 0$ and the first multiplier is $T(0) = 0$. If $k_5 = 0$, the closed-path condition gives $K_4 = -k_5 = 0$, so the last multiplier is zero. A matrix product with a zero factor is zero, and applying d still gives zero.

D2: Short proof

Answer 7.12. Zero interior steps

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.6, 7.10.

Task. Show that a summand vanishes if one of k_2, k_3, k_4 is zero. Use the determinant rule and the repeated partial sums.

Solution. If $k_j = 0$ for $j \in \{2, 3, 4\}$, then $K_j = K_{j-1}$. Write each of the four multiplier matrices in the form used in Exercise 7.6. The rows of coefficients for the equal matrices $T(K_j)$ and $T(K_{j-1})$ are equal. A determinant with two equal rows is zero. Hence the detector of the product is zero.

D2: Short proof

Answer 7.13. The signed-count equations

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.10, 7.11, 7.12.

Task. For five steps in $\{\pm a, \pm b, \pm c\}$, let n_a be the number of a steps minus the number of $-a$ steps, and define n_b, n_c similarly. Show that a closed path satisfies $n_a = -n_c$, $n_b = -3n_c$, and $5 \geq 5|n_c|$. For the inequality, compare the total number of steps in each direction with the magnitude of its signed count.

Solution. The total step vector is $n_a a + n_b b + n_c c$. Its horizontal and vertical coordinates are

$$n_a + n_c = 0, \qquad n_b + 3n_c = 0.$$

Therefore $n_a = -n_c$ and $n_b = -3n_c$. If a direction has p positive and m negative steps, its total number $p + m$ is at least its signed count's magnitude $|p - m|$. Adding this fact for the three directions gives

$$5 \geq |n_a| + |n_b| + |n_c| = |n_c| + 3|n_c| + |n_c| = 5|n_c|.$$

In particular n_c is one of $-1, 0, 1$.

D2: Short proof

Answer 7.14. Only two multisets remain

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.13.

Task. Show that $n_c = 0$ is impossible by parity. Deduce that the multiset of steps is $\{a, b, b, b, -c\}$ or its negative.

Solution. If $n_c = 0$, the signed-count equations make all three signed counts zero. Each direction then has the same number of positive and negative occurrences, so each direction contributes an even number of steps. Their total would be even, contradicting five nonzero steps.

Thus $|n_c| = 1$. The inequality in Exercise 7.13 is then an equality. Equality forces every direction to contain no cancelling positive-negative pair. When $n_c = -1$, the counts are $n_a = 1$, $n_b = 3$, $n_c = -1$, producing exactly $\{a, b, b, b, -c\}$. When $n_c = 1$, all signs reverse and produce its negative.

D2: Short proof

Answer 7.15. Only four ordered paths remain

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.10, 7.14.

Task. Use $T(\pm b) = 0$ to show that the only possibly nonzero paths are $(a, b, b, b, -c)$, $(-c, b, b, b, a)$ and their negatives.

Solution. If the first step is b or $-b$, the first vertex is vertical, so $T(K_1) = 0$. If the last step is vertical, the fourth vertex is its negative because $K_4 = -k_5$, so $T(K_4) = 0$.

Each remaining multiset contains exactly two nonvertical steps. To avoid these zeros, both must occupy the first and last positions. The three identical vertical steps fill the middle positions. For $\{a, b, b, b, -c\}$ the orders are therefore $(a, b, b, b, -c)$ and $(-c, b, b, b, a)$. The negative multiset gives their negatives. These are the only four possible nonzero contributions.

D1: Direct

Answer 7.16. The four intermediate directions

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.15.

Task. For $(a, b, b, b, -c)$, compute K_1, K_2, K_3, K_4 and their squared slopes $(K_{j,y}/K_{j,x})^2$.

Solution. Adding the steps one at a time gives

$$K_1 = (1, 0), \quad K_2 = (1, 1), \quad K_3 = (1, 2), \quad K_4 = (1, 3).$$

All horizontal coordinates are 1, so their squared slopes are $0^2, 1^2, 2^2, 3^2$, namely $r = (0, 1, 4, 9)$. The final step $-c = (-1, -3)$ returns to the origin. These four different slopes will give four different determinant rows.

D2: Short proof

Answer 7.17. The signs agree

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.6, 7.15, 7.16.

Task. Show that exchanging the two endpoint steps reverses the four squared slopes, and that negating all steps leaves T unchanged. Conclude that the four surviving contributions have the same sign and magnitude.

Solution. For $(-c, b, b, b, a)$ the intermediate vertices are $(-1,-3), (-1,-2), (-1,-1), (-1,0)$. Their squared slopes are $9, 4, 1, 0$, the reverse of the first list. The explicit formula for $T(k)$ depends only on squared slope when $k_x \neq 0$. Thus the coefficient rows in the determinant are reversed.

Reversing four rows requires $4 \cdot 3/2 = 6$ swaps, an even number, so its determinant sign is positive. Negating all steps negates every intermediate vertex and changes no multiplier because $T(-k) = T(k)$. The Fourier amplitudes are also unchanged because each nonzero supported frequency has coefficient $1/12$. Therefore all four scalar contributions agree, including their sign.

D2: Short proof

Answer 7.18. Count the numerical factors

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.6, 7.10, 7.11, 7.12, 7.15, 7.16, 7.17.

Task. Let $r = (0, 1, 4, 9)$. Account separately for four paths, five Fourier coefficients, and the determinant rule to prove

$$[\rho^5]\ell(B) = \frac{8}{12^5\, 4^4} \det\left[\frac{j}{j + r_i}\right]_{i,j=1}^{4}.$$

Solution. There are four paths. Each has five nonzero step frequencies, and therefore the product of its Fourier amplitudes is 12^{-5}. At squared slope r_i, the coefficient of Γ_j in T is $j/[4(j + r_i)]$. The determinant rule contributes

$$2 \det\left[\frac{j}{4(j + r_i)}\right] = 2\, 4^{-4} \det\left[\frac{j}{j + r_i}\right].$$

The factor 4^{-4} comes from factoring $1/4$ out of each of four rows. Multiplying the three contributions gives $4 \cdot 12^{-5} \cdot 2 \cdot 4^{-4}$, which is the stated factor $8/(12^5 4^4)$.

Setup 7.4. For the next two exercises you may use the Cauchy determinant identity

$$\det\left[\frac{1}{x_i + y_j}\right]_{i,j=1}^{4} = \frac{\prod_{i<k}(x_k - x_i)\prod_{j<l}(y_l - y_j)}{\prod_{i,j}(x_i + y_j)}.$$

The parameters used below are distinct and all denominators are positive (B7).

D1: Direct

Answer 7.19. Determine the determinant's sign

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 7.16.

Task. Apply the supplied identity with $x_i = r_i$ and $y_j = j$. Show that $\det[j/(j + r_i)] > 0$ without expanding a 4×4 determinant.

Solution. First factor j out of column j. This gives

$$\det\left[\frac{j}{j + r_i}\right] = \left(\prod_{j=1}^{4} j\right) \frac{\prod_{i<k}(r_k - r_i)\prod_{j<l}(l - j)}{\prod_{i,j}(r_i + j)}.$$

The row parameters $0, 1, 4, 9$ are strictly increasing, as are the column parameters $1, 2, 3, 4$. Every difference in the numerator is positive, every denominator is positive, and $\prod_j j = 24 > 0$. Therefore the determinant is strictly positive.

D1: Direct

Answer 7.20. Evaluate the constant

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 7.18, 7.19.

Task. Use $\prod_{i<k}(r_k - r_i) = 4320$ and $\prod_{j<l}(l - j) = 12$ to verify

$$\det\left[\frac{j}{j + r_i}\right] = \frac{3}{200200}, \qquad \kappa := \frac{4}{12^5\, 4^4}\frac{3}{200200} = \frac{1}{1062744883200}.$$

Solution. The row-difference product is $1 \cdot 4 \cdot 9 \cdot 3 \cdot 8 \cdot 5 = 4320$, and the column-difference product is $1 \cdot 2 \cdot 3 \cdot 1 \cdot 2 \cdot 1 = 12$. The four row-products of denominators are

$$\prod_{j=1}^{4}(r_i + j) = 24,\ 120,\ 1680,\ 17160.$$

Consequently the determinant is

$$\frac{24 \cdot 4320 \cdot 12}{24 \cdot 120 \cdot 1680 \cdot 17160} = \frac{432}{1680 \cdot 17160} = \frac{3}{200200}.$$

Now $12^5 = 248832$ and $4^4 = 256$, so

$$\kappa = \frac{4 \cdot 3}{248832 \cdot 256 \cdot 200200} = \frac{1}{1062744883200}.$$

Thus Exercise 7.18 gives $[\rho^5]\ell(B) = 2\kappa$.

D1: Direct

Answer 7.21. Bound the energy remainder

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 6.8, 7.1, 7.7, 7.18, 7.20.

Task. Use Exercise 7.1 and the supplied bound on R_6 to show

$$\ell(B) \geq 2\kappa\rho^5 - \frac{2(3/2)^5\rho^6}{1 - 3\rho/2}.$$

Solution. The coefficients through order four vanish and the fifth coefficient is 2κ. Hence $\ell(B) = 2\kappa\rho^5 + d(R_6)$. Exercise 7.1 gives $d(R_6) \geq -2\|R_6\|$. Substituting the supplied tail bound yields

$$\ell(B) \geq 2\kappa\rho^5 - \frac{2(3/2)^5\rho^6}{1 - 3\rho/2}.$$

The denominator is positive because $\rho < 2/3$. This bounds the total contribution of every order from six onward.

D1: Direct

Answer 7.22. Fix a contrast with a strict difference

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 7.20, 7.21.

Task. Set $\rho = \kappa/32$. Use $\rho < 1/3$ in Exercise 7.21 to prove $\ell(B) \geq (269/256)\kappa\rho^5 > 0$.

Solution. The positive number κ is less than 1, so $\rho = \kappa/32 < 1/32 < 1/3$. Therefore $1 - 3\rho/2 > 1/2$, and the tail term in Exercise 7.21 is at most

$$4(3/2)^5\rho^6 = \frac{243}{8}\rho^6.$$

Use $\rho^6 = (\kappa/32)\rho^5$ to get

$$\begin{aligned}\ell(B) &\geq \left(2 - \frac{243}{8 \cdot 32}\right)\kappa\rho^5 \\ &= \frac{512 - 243}{256}\kappa\rho^5 = \frac{269}{256}\kappa\rho^5 > 0.\end{aligned}$$

The rational contrast is $\rho = 1/34007836262400$.

D1: Direct

Answer 7.23. Why the smooth medium is not binary

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 4.4, 7.22.

Task. The smooth coefficient is $\overline{L} = (1 - t)L_0 + tL_1$. Explain in one or two sentences why its positive energy difference does not yet provide a composite using only the two coefficient values L_0, L_1.

Solution. Since $t(\pi, \pi) = 1/3$, continuity gives an open region on which $\overline{L}(x, y)$ equals neither of the two distinct phases. Module 8 constructs coefficients taking only L_0, L_1 and proves that the strict energy difference survives at a finite stripe count.

8 Replace the smooth medium by binary stripes

Prerequisites. The corrector and mean flux: Step 1.4, Step 1.6, Step 1.9, Step 1.12; smooth-field bounds: Step 6.20–Step 6.22; strict positivity at fixed contrast: Step 7.22.

Definition lookup. The setups below define the stripe geometry, corrector fields, and residual constants. The lookup recalls the loads and spatial-row notation, and supplies the weak-derivative and Lipschitz facts needed at the interfaces.

Supplied background. B1, B6, B8, B11. Each key links to its statement, scope, and reference.

Setup 8.1. On $Y = (0, 2\pi)^2$, use $t = \frac{1}{2} + \frac{1}{6}[\cos x + \cos y + \cos(x+3y)]$ and fix the positive contrast from Exercise 7.22. The smooth coefficient is $\overline{L} = \operatorname{diag}(I_{16} - \rho t P P^\top, D)$, its effective tensor is B, and $\Delta = \ell(B) > 0$, where $\ell(C) = \eta \cdot C\eta - \xi \cdot C\xi$. The two phases are $L_0 = \operatorname{diag}(I_{16}, D)$ and $L_1 = \operatorname{diag}(I_{16} - \rho P P^\top, D)$. Their bounds are $\frac{1}{4} I \preceq L_i \preceq I$. For either selected mean gradient $\zeta \in \{\xi, \eta\}$, let $e = \zeta + \nabla u_\zeta$ be the smooth corrector gradient, $j = \overline{L} e$, and $\sigma = P P^\top e_x$. Thus $\operatorname{div} j = 0$ and $\langle j \rangle = B\zeta$. The preceding regularity argument supplies finite constants $S_0 = \|\sigma\|_\infty$, $S_x = \|\partial_x \sigma\|_\infty$, $S_y = \|\partial_y \sigma\|_\infty$, depending on ζ but independent of the stripe count. The fields e, j, σ are periodic.

Setup 8.2. For an integer $N \geq 2$, put $h = 2\pi/N$ and $y_k = kh$ for $0 \leq k \leq N$. For $k = 0, \ldots, N-1$, set $t_k(x) = t(x, y_k)$. In $y_k \leq y < y_{k+1}$, set $\tau = (y - y_k)/h$, $\chi_N = \mathbf{1}_{\{\tau < t_k(x)\}}$, and $L_N = (1 - \chi_N) L_0 + \chi_N L_1$; extend periodically. Let B_N denote its effective tensor. All L^2 norms below are normalized: $\|f\|_2^2 = \langle |f|^2 \rangle$.

Setup 8.3. Lookup: $P \in \mathbb{R}^{16 \times 8}$ is the stacked matrix of Step 3.11, $S = P^\top P$, and $D = \operatorname{diag}(1, 1/2, 1/3, 1/4) \otimes I_4$, where $\otimes I_4$ repeats each listed scalar as a 4×4 block. The loads from Step 4.7 have $\xi_x = P S^{-1} e_1$, $\eta_x = P S^{-1} e_5$, and $\xi_y = \eta_y = 0$. A field $E \in \mathbb{R}^{2 \times 16}$ has transposed row vectors $E_x, E_y \in \mathbb{R}^{16}$, norm $|E|^2 = |E_x|^2 + |E_y|^2$, and product $E \cdot J = E_x \cdot J_x + E_y \cdot J_y$. The block tensor $\operatorname{diag}(A, D)$ maps these rows to AE_x, DE_y. The order $A \preceq B$ means $v \cdot Av \leq v \cdot Bv$ for every v, and coefficient norms are operator norms. The indicator $\mathbf{1}_A$ is 1 on A and 0 elsewhere; $0 \leq t \leq 1$ follows from $|\cos| \leq 1$. The notation a.e. means outside a set of area zero, and $\|f\|_\infty$ is its essential supremum.

Setup 8.4. Weak-derivative lookup and supplied background: $g = \partial_\nu f$ means $\langle g \cdot \varphi \rangle = -\langle f \cdot \partial_\nu \varphi \rangle$ for every smooth periodic test φ. The distributional derivative of f is the linear functional $\varphi \mapsto -\langle f \cdot \partial_\nu \varphi \rangle$, even when this functional is not represented by an ordinary function. Applying a second derivative moves both derivatives onto the test with positive sign. Periodic H^1 consists of L^2 functions whose weak first derivatives are in L^2, with matching opposite traces; a trace is the boundary value in this Sobolev sense. Smooth periodic functions are dense in this space (B1, B6). Supplied Lipschitz facts: a Lipschitz function has a.e. first derivatives that are its weak derivatives; the minimum of two Lipschitz functions is Lipschitz; the usual product rule holds with a smooth factor (B8). Stripwise integration by parts is valid for these functions. For continuous g, the graph $y = g(x)$ is measurable and has area zero: each vertical section has length zero, so Fubini's theorem applies (B11). These background facts need not be reproved.

D1: Direct

Answer 8.1. Count the phase inside one strip

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* local setup only.

Task. For each fixed x, show that the integral of $\chi_N(x,\cdot)$ across strip k is $ht_k(x)$. Conclude that L_N takes only the original two phase values.

Solution. For fixed x, $0 \le t_k(x) \le 1$. The set $\{y \in [y_k, y_{k+1}) : \tau < t_k(x)\}$ is the interval $[y_k, y_k + ht_k(x))$, of length $ht_k(x)$. Integrating its indicator gives that length. By definition χ_N is either zero or one, so $L_N = L_0$ or L_1 at almost every point. Choices on strip boundaries have measure zero and do not affect cell energies or fractions.

D2: Short proof

Answer 8.2. Count the total fraction exactly

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.1.

Task. Integrate Exercise 8.1 in x and use $\sum_{k=0}^{N-1} \cos(2\pi k/N) = 0$ to prove $\langle \chi_N \rangle = 1/2$.

Solution. Integrate the phase-one heights in x and add the strips. The normalized fraction is

$$\langle \chi_N \rangle = \frac{1}{N} \sum_{k=0}^{N-1} \frac{1}{2\pi} \int_0^{2\pi} t(x, y_k)\, \mathrm{d}x.$$

The x averages of $\cos x$ and $\cos(x + 3y_k)$ vanish. Thus the remaining average is $1/2 + \cos y_k/6$. For $N \ge 2$, $z = e^{2\pi \mathrm{i}/N}$ satisfies $z \ne 1$ and $\sum_{k=0}^{N-1} z^k = (1 - z^N)/(1 - z) = 0$. Taking real parts shows $\sum_k \cos y_k = 0$. The displayed fraction is therefore exactly $1/2$.

D1: Direct

Answer 8.3. Integrate the zero-mean oscillation

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.1.

Task. Define $w_N(x,y) = \int_{y_k}^{y} (\chi_N(x,s) - t_k(x))\, \mathrm{d}s$ in strip k. Show that $w_N = h[\min\{\tau, t_k\} - \tau t_k]$ and that it vanishes at both strip edges.

Solution. Use $s = y_k + hv$ in the integral. Since t_k is constant with respect to the integration variable,

$$\begin{aligned} w_N(x,y) &= h \int_0^{\tau} (\mathbf{1}_{\{v < t_k(x)\}} - t_k(x))\, \mathrm{d}v \\ &= h[\min\{\tau, t_k(x)\} - \tau t_k(x)]. \end{aligned}$$

At the lower edge $\tau = 0$, both terms vanish. At the upper edge $\tau = 1$, their difference is $t_k - t_k = 0$. The zero upper-edge value reflects the exact cancellation of $\chi_N - t_k$ over each strip.

D1: Direct

Answer 8.4. Bound the primitive

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.3.

Task. For $0 \le s, \tau \le 1$, show $0 \le \min\{\tau, s\} - \tau s \le s(1-s) \le 1/4$. Deduce $|w_N| \le h/4$.

Solution. Fix $s \in [0,1]$ and define $f(\tau) = \min\{\tau, s\} - \tau s$. If $0 \le \tau \le s$, then $f(\tau) = \tau(1-s)$, which increases from zero to $s(1-s)$. If $s \le \tau \le 1$, then $f(\tau) = s(1-\tau)$, which decreases back to zero. Hence $0 \le f \le s(1-s)$. Finally $s(1-s) = 1/4 - (s-1/2)^2 \le 1/4$. Substitute $s = t_k(x)$ and multiply by h.

D2: Short proof

Answer 8.5. Differentiate inside a strip

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.3.

Task. Differentiate the two formulas on $\tau < t_k(x)$ and $\tau > t_k(x)$ to prove a.e. that $\partial_y w_N = \chi_N - t_k$ and $|\partial_x w_N| \le h\|t_x\|_\infty$. The omitted equality set is the graph $y = y_k + ht_k(x)$ and has area zero. Use the supplied Lipschitz facts to identify the derivatives as weak derivatives inside a strip.

Solution. Away from the curve $\tau = t_k(x)$, differentiate the piecewise formula. The derivative of $\min\{\tau, t_k\}$ in τ is $\mathbf{1}_{\{\tau<t_k\}}$. Since $\partial_y \tau = 1/h$,

$$\partial_y w_N = \mathbf{1}_{\{\tau<t_k\}} - t_k = \chi_N - t_k.$$

For the horizontal derivative, τ is independent of x. If $\tau < t_k$, then $w_N = h\tau(1-t_k)$ and $\partial_x w_N = -h\tau t_k'$. If $\tau > t_k$, then $w_N = ht_k(1-\tau)$ and $\partial_x w_N = h(1-\tau)t_k'$. Both coefficients have magnitude at most 1, and $|t_k'| \le \|t_x\|_\infty$. Thus $|\partial_x w_N| \le h\|t_x\|_\infty$.

For each fixed x, the equality $\tau = t_k(x)$ selects at most one y, namely $y_k + ht_k(x)$. Its vertical section has length zero, so Fubini's theorem shows that this moving boundary has area zero. Thus the two regions cover the strip a.e. Both $\min\{\tau, t_k(x)\}$ and $\tau t_k(x)$ are Lipschitz on the strip, so w_N is Lipschitz. The computed a.e. derivatives are therefore its weak derivatives there (B8).

D1: Direct

Answer 8.6. Bound the sampling error

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* local setup only.

Task. Use the mean value theorem in the y coordinate to show $|t_k(x) - t(x,y)| \le h\|t_y\|_\infty$ throughout strip k.

Solution. For $y_k \le y < y_{k+1}$, the fundamental theorem of calculus gives

$$t(x,y) - t(x,y_k) = \int_{y_k}^{y} t_y(x,s)\,\mathrm{d}s.$$

Its magnitude is at most $|y - y_k|\|t_y\|_\infty \le h\|t_y\|_\infty$. The error is proportional to the strip height.

D3: Guided technical

Answer 8.7. Check the derivatives across strips

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.3, 8.4, 8.5.

Task. Explain why the matching zero traces of w_N at strip edges make the derivatives in Exercise 8.5 its weak derivatives on the whole periodic cell. Test against a smooth periodic φ: integrate $w_N\partial_y\varphi$ strip by strip and check every horizontal boundary term, including the top and bottom cell edges. For ∂_x, integrate each strip in x and use periodicity. The bounded piecewise derivatives then show $w_N \in H^1_{\mathrm{per}}$.

Solution. Take a smooth periodic scalar test function φ. Integrate $w_N\partial_y\varphi$ by parts on every strip. The interior terms give $-\int(\partial_y w_N)\varphi$. Each boundary term involves the trace of w_N, which is zero on both edges by Exercise 8.3. Hence there are no interface contributions. The same statement holds at the top and bottom of the cell because the traces match periodically.

For the horizontal derivative, integrate in x at each fixed y in a strip, using its Lipschitz formula and the matching periodic values at $x = 0, 2\pi$. There are no boundary terms after periodic cancellation. The moving phase interface is already included in this Lipschitz calculation; it creates no jump in w_N. Adding the finitely many strip identities gives the global horizontal testing identity. The resulting first derivatives are bounded by Exercise 8.5, and w_N is bounded by Exercise 8.4, so all three functions are in L^2. These identities and the matching outer traces prove $w_N \in H^1_{\mathrm{per}}$.

D1: Direct

Answer 8.8. Compute the flux mismatch

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* local setup only.

Task. At the unchanged gradient e, show $(j - L_N e)_x = \rho(\chi_N - t)\sigma$ and $(j - L_N e)_y = 0$.

Solution. The horizontal coefficient of $\overline{L}$ is $I_{16} - \rho t PP^\top$, while that of L_N is $I_{16} - \rho\chi_N PP^\top$. Subtracting their actions on the same e_x gives

$$(j - L_N e)_x = \rho(\chi_N - t)PP^\top e_x = \rho(\chi_N - t)\sigma.$$

The vertical block is the identical constant D in both fields, so the vertical mismatch is zero. The horizontal mismatch contains the oscillating factor $\chi_N - t$.

D3: Guided technical

Answer 8.9. Preserve equilibrium

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 6.22, 8.7.

Task. Define the flux with transposed row vectors

$$j_N = j + \rho\begin{pmatrix}[-\partial_y(w_N\sigma)]^\top \\ [\partial_x(w_N\sigma)]^\top\end{pmatrix}.$$

Prove $\operatorname{div} j_N = 0$ in distributions by cancellation of mixed derivatives. Test the correction against a smooth periodic vector function, move each derivative onto the test, and use $\partial_{xy}\varphi = \partial_{yx}\varphi$.

Solution. Put $G_N = w_N\sigma$. Exercise 8.7 and the weak product rule give $G_N \in H^1_{\mathrm{per}}(Y;\mathbb{R}^{16})$: both factors are bounded, w_N has bounded weak first derivatives, and σ is smooth with bounded first derivatives (B8). Thus the correction is an L^2 field. For each potential component, its divergence is

$$\rho[-\partial_x\partial_y G_N + \partial_y\partial_x G_N] = 0.$$

For a smooth periodic $\mathbb{R}^{16}$-valued test φ, weak integration by parts gives the explicit cancellation

$$\langle -\partial_x\varphi \cdot \partial_y G_N + \partial_y\varphi \cdot \partial_x G_N\rangle = \langle G_N \cdot (\partial_{yx}\varphi - \partial_{xy}\varphi)\rangle = 0.$$

The test is smooth, so its mixed derivatives commute. This proves weak equilibrium for the correcting flux; adding the equilibrium flux j gives $\operatorname{div} j_N = 0$.

D2: Short proof

Answer 8.10. Preserve the mean flux

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.9.

Task. Show $\langle j_N\rangle = B\zeta$ by averaging the periodic derivatives in Exercise 8.9.

Solution. The average of either weak derivative of a periodic function is zero. Componentwise, this follows by testing the derivative against the constant 1. The two correcting derivatives therefore have zero average, so $\langle j_N\rangle = \langle j\rangle = B\zeta$. Keeping this exact mean will allow us to compare the mean flux of the true binary solution directly with that of the smooth medium.

D1: Direct

Answer 8.11. Cancel the horizontal oscillation

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.5, 8.8, 8.9.

Task. For $r_N = j_N - L_N e$, use $\partial_y w_N = \chi_N - t_k$ to derive the a.e. identity

$$(r_N)_x = \rho[(t_k - t)\sigma - w_N\partial_y\sigma].$$

Solution. Use the mismatch from Exercise 8.8 and expand the horizontal correcting derivative:

$$\begin{aligned}(r_N)_x &= \rho(\chi_N - t)\sigma - \rho\partial_y(w_N\sigma)\\ &= \rho[(\chi_N - t)\sigma - (\partial_y w_N)\sigma - w_N\partial_y\sigma]\\ &= \rho[(\chi_N - t - \chi_N + t_k)\sigma - w_N\partial_y\sigma]\\ &= \rho[(t_k - t)\sigma - w_N\partial_y\sigma].\end{aligned}$$

The order-one oscillation $\chi_N - t_k$ has cancelled exactly. The two remaining coefficients, $t_k - t$ and w_N, are both of size at most a constant times h.

D1: Direct

Answer 8.12. Compute the vertical residual

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.8, 8.9, 8.11.

Task. Use the product rule to show a.e. that $(r_N)_y = \rho[(\partial_x w_N)\sigma + w_N\partial_x\sigma]$.

Solution. Exercise 8.8 gives no original vertical mismatch. Therefore the whole vertical residual is the vertical correcting term. The weak product rule gives

$$(r_N)_y = \rho\partial_x(w_N\sigma) = \rho[(\partial_x w_N)\sigma + w_N\partial_x\sigma].$$

Both product terms are well-defined and bounded: $\partial_x w_N$ is bounded by Exercise 8.5 and $\sigma, \partial_x\sigma$ are bounded by the smooth-field regularity estimate.

D1: Direct

Answer 8.13. Make the residual small

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.4, 8.5, 8.6, 8.11, 8.12.

Task. Put $C_\zeta = (\|t_x\|_\infty + \|t_y\|_\infty)S_0 + (S_x + S_y)/4$. Use Exercises 8.4–8.6 to prove $\|r_N\|_2 \le \rho h C_\zeta$.

Solution. Bound the two rows separately using the previous estimates:

$$|(r_N)_x| \le \rho h \left(\|t_y\|_\infty S_0 + \frac{1}{4} S_y \right),$$
$$|(r_N)_y| \le \rho h \left(\|t_x\|_\infty S_0 + \frac{1}{4} S_x \right).$$

The Euclidean norm of the two-row matrix is at most the sum of the row norms. Hence $|r_N| \le \rho h C_\zeta$ a.e. Because the average is normalized, the same constant bounds its L^2 norm.

The constant depends on the fixed contrast and the selected loading through S_0, S_x, S_y, but these smooth-field quantities do not depend on N. The residual bound therefore tends to zero as $N \to \infty$.

D3: Guided technical

Answer 8.14. Obtain the error orthogonality

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 1.4, 1.6, 8.9, 8.11.

Task. Let $e_N = \zeta + \nabla u_{N,\zeta}$ be the exact cell gradient for L_N, and put $\delta e = e_N - e$. Show $\langle \delta e \cdot (L_N \delta e - r_N) \rangle = 0$ using equilibrium of $L_N e_N$ and j_N. Their difference is an L^2 flux; first test it with smooth periodic potentials, then approximate $u_{N,\zeta} - u_\zeta$ in H^1 and pass to the limit by Cauchy–Schwarz.

Solution. Subtracting the two corrector representations gives $\delta e = \nabla v$ with $v = u_{N,\zeta} - u_\zeta \in H^1_{\mathrm{per}}$. Existence of the binary corrector follows from the measurable coefficient bound $\frac{1}{4} I \preceq L_N \preceq I$ and Exercise 1.4. The flux $L_N e_N$ is divergence-free by its cell equation; j_N is divergence-free by Exercise 8.9. Hence $q = L_N e_N - j_N \in L^2$ obeys $\langle \nabla \varphi \cdot q \rangle = 0$ for every smooth periodic φ.

Choose smooth periodic $v_m \to v$ in H^1 using the supplied density fact (B1). Cauchy–Schwarz gives

$$\left| \langle (\nabla v_m - \nabla v) \cdot q \rangle \right| \le \|\nabla v_m - \nabla v\|_2 \|q\|_2 \longrightarrow 0.$$

Passing from the smooth tests to v therefore gives

$$\langle \delta e \cdot (L_N e_N - j_N) \rangle = 0.$$

Use $r_N = j_N - L_N e$ and $e_N = e + \delta e$ to rewrite the flux difference as $L_N \delta e - r_N$. This gives the required identity.

D1: Direct

Answer 8.15. Control the gradient error

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.14.

Task. Use Exercise 8.14, $L_N \succeq \frac{1}{4} I$, and Cauchy–Schwarz to prove $\|e_N - e\|_2 \le 4 \|r_N\|_2$. Include the case $\delta e = 0$.

Solution. The orthogonality identity rearranges to $\langle \delta e \cdot L_N \delta e \rangle = \langle \delta e \cdot r_N \rangle$. Uniform ellipticity and Cauchy–Schwarz give

$$\frac{1}{4} \|\delta e\|_2^2 \le \langle \delta e \cdot L_N \delta e \rangle = \langle \delta e \cdot r_N \rangle \le \|\delta e\|_2 \|r_N\|_2.$$

If $\|\delta e\|_2 = 0$, the conclusion is immediate. Otherwise divide by $\|\delta e\|_2$ and multiply by 4. This is a stability estimate: a small constitutive residual of a compatible-equilibrium pair forces its gradient to be close to the exact solution.

D2: Short proof

Answer 8.16. Control the selected mean flux

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 1.9, 8.10, 8.13, 8.15.

Task. Average $L_N e_N - j_N = L_N \delta e - r_N$ and use $\|L_N(x,y)\| \le 1$ a.e. to show $|(B_N - B)\zeta| \le 5\|r_N\|_2 \le 10\pi\rho C_\zeta / N$.

Solution. The exact binary flux has mean $B_N\zeta$, and Exercise 8.10 gives $\langle j_N \rangle = B\zeta$. Therefore

$$(B_N - B)\zeta = \langle L_N e_N - j_N \rangle = \langle L_N \delta e - r_N \rangle.$$

For normalized averages, $|\langle f \rangle| \le \|f\|_2$. Also $\|L_N(x,y)\| \le 1$ a.e., so

$$|(B_N - B)\zeta| \le \|\delta e\|_2 + \|r_N\|_2 \le 5\|r_N\|_2.$$

Insert Exercise 8.13 and $h = 2\pi/N$ to obtain the final bound.

D1: Direct

Answer 8.17. Control the measured energy difference

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.16.

Task. Use Exercise 8.16 at ξ and η to prove $|\ell(B_N) - \Delta| \le K_\rho / N$, where $K_\rho = 10\pi\rho(|\xi| C_\xi + |\eta| C_\eta)$.

Solution. By definition of ℓ and $\Delta = \ell(B)$,

$$\ell(B_N) - \Delta = \eta \cdot (B_N - B)\eta - \xi \cdot (B_N - B)\xi.$$

The triangle inequality and Cauchy–Schwarz give an upper bound $|\eta|\,|(B_N - B)\eta| + |\xi|\,|(B_N - B)\xi|$ for its magnitude. Apply Exercise 8.16 twice:

$$|\ell(B_N) - \Delta| \le \frac{10\pi\rho}{N}(|\eta| C_\eta + |\xi| C_\xi) = \frac{K_\rho}{N}.$$

Both constants arise from the same fixed smooth medium. They may differ because the two selected gradients differ.

D1: Direct

Answer 8.18. Choose one finite stripe count

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 4.7, 5.16, 5.17, 7.22, 8.2, 8.17.

Task. Choose an integer $N \ge 2$ with $N > 2K_\rho/\Delta$ and prove $\ell(B_N) > \Delta/2$. Explain why the contrast is held fixed while making this choice.

Solution. Since K_ρ is finite and $\Delta > 0$, choose for example

$$N = \max\left\{2, \left\lfloor \frac{2K_\rho}{\Delta} \right\rfloor + 1\right\}.$$

Here $\lfloor s \rfloor$ denotes the greatest integer not exceeding s. This choice gives $N > 2K_\rho/\Delta$, hence $K_\rho/N < \Delta/2$. Exercise 8.17 yields

$$\ell(B_N) \ge \Delta - |\ell(B_N) - \Delta| > \Delta/2 > 0.$$

This is a single finite periodic binary pattern, with fraction exactly $1/2$ by Exercise 8.2. The contrast was fixed first, so both K_ρ and Δ are fixed numbers when N is enlarged. By Step 4.7 and Step 5.16–Step 5.17, every tensor in the unrestricted laminate closure has $\ell = 0$. Thus this positive value also gives separation from that entire closure.

9 Prescribed fractions and more potentials

Prerequisites. Binary witness: Step 8.18; arithmetic horizontal layers: Step 2.10; detector continuity and laminate limits: Step 4.8, Step 5.17.

Definition lookup. The setup defines C_θ's ingredients and the supplied recovery inputs. The lookup below defines cell gradients, exact-fraction closure, and direct sums; Step 9.1–Step 9.2 define C_θ.

Supplied background. B4, B9, B11, B12. Each key links to its statement, scope, and reference.

Setup 9.1. Manuscript: Section 5.3. Let $0 < \theta < 1$ be the prescribed phase-one fraction. The preceding modules give constituents L_0, L_1 and a periodic binary composite with tensor B_N, phase-one fraction $1/2$, and $\ell(B_N) = g_N > 0$. All three tensors have the same constant vertical block D, so horizontal layering gives their arithmetic mean. The detector satisfies $\ell(L_i) = 0$ and $|\ell(A) - \ell(B)| \le (8/3)\|A - B\|$.

Use the following background results. **Separated-scale periodic homogenization (B4):** layering periodic composites and inserting increasingly fine copies of their original phase patterns produces periodic binary coefficients whose effective tensors converge to the laminate tensor and whose fractions converge to the weighted fraction. Iteration gives recovery for finite hierarchies; rational-direction approximation and continuity of the laminate formula allow arbitrary layer normals. **Higher integrability (B9):** for periodic self-adjoint coefficients on $\mathbb{R}^{2\times 16}$ with $\mu I \preceq A \preceq MI$, $0 < \mu \le M$, there are $\delta > 0$ and C depending only on these bounds and the dimensions such that $\|F + \nabla u_{A,F}\|_{2+\delta} \le C|F|$. The steps below use these results to correct the fraction exactly.

Setup 9.2. Lookup: $Y = (0, 2\pi)^2$. In Step 9.4–Step 9.6, A, B are measurable periodic self-adjoint coefficient fields on $\mathbb{R}^{2\times 16}$ with common bounds $\mu I \preceq A, B \preceq MI$, $0 < \mu \le M$. Write $e_{A,F} = F + \nabla u_{A,F}$ for the minimizing cell gradient (Step 1.4–Step 1.5) and A_* for its effective tensor, so $F \cdot A_* F = \langle e_{A,F} \cdot A e_{A,F} \rangle$ (Step 1.10). The indicator $\mathbf{1}_E$ equals one on E and zero elsewhere; $\|f\|_p = (\langle |f|^p \rangle)^{1/p}$. The set G_θ is the operator-norm closure of periodic binary effective tensors at exact fraction θ (Step 2.11–Step 2.12). A direct sum $C \oplus I$ acts by C on the original sixteen potential columns and by the identity on the extra columns. For positive-definite C_1, C_2, a unit normal n, and $0 \le \lambda \le 1$, $\mathrm{Lam}_n(C_1, C_2; \lambda)$ denotes the laminate tensor of Step 2.7.

D1: Direct

Answer 9.1. Lower the fraction

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.10, 8.18.

Task. For $0 < \theta \le 1/2$, mix a fraction λ of B_N with phase 0. Find λ and the resulting tensor C_θ.

Solution. The composite contributes phase-one fraction $\lambda/2$ and the pure phase contributes zero. Thus $\lambda/2 = \theta$, or $\lambda = 2\theta$. This belongs to $(0, 1]$. Horizontal arithmetic averaging gives

$$C_\theta = 2\theta B_N + (1 - 2\theta) L_0.$$

Here C_θ is the effective tensor of the ideal separated-scale layering. A later step obtains an actual finite periodic witness.

D1: Direct

Answer 9.2. Raise the fraction

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.10, 8.18.

Task. For $1/2 < \theta < 1$, mix a fraction λ of B_N with phase 1. Find λ and C_θ.

Solution. The weighted fraction is $\lambda/2 + (1-\lambda) = 1 - \lambda/2$. Solving $1 - \lambda/2 = \theta$ gives $\lambda = 2(1-\theta) \in (0,1)$. Consequently

$$C_\theta = 2(1-\theta)B_N + (2\theta - 1)L_1.$$

Both coefficients are nonnegative and sum to one. At $\theta = 1/2$ the lower-fraction formula gives $C_{1/2} = B_N$, so the two constructions match.

D1: Direct

Answer 9.3. Keep a positive difference

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.1, 9.2, 4.8.

Task. Use linearity of ℓ to show $\ell(C_\theta) = 2\min(\theta, 1-\theta)g_N > 0$ for $0 < \theta < 1$.

Solution. In the first case, $\ell(C_\theta) = 2\theta\ell(B_N)$ because $\ell(L_0) = 0$. In the second, it equals $2(1-\theta)\ell(B_N)$. Combining the cases gives the claimed formula. The result is positive because $0 < \theta < 1$ and $g_N > 0$.

D2: Short proof

Answer 9.4. Compare two cell energies

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 1.5, 1.10.

Task. Let the periodic coefficients A, B obey the common bounds above and agree outside a measurable set $E \subset Y$. Using the B-corrector as a trial for A, prove $F \cdot (A_* - B_*)F \le 2M\langle \mathbf{1}_E | e_{B,F}|^2\rangle$.

Solution. Write $e_{B,F} = F + \nabla u_{B,F}$. It is an admissible trial gradient for the cell problem with coefficient A, so

$$F \cdot A_* F \le \langle e_{B,F} \cdot A e_{B,F}\rangle,$$
$$F \cdot B_* F = \langle e_{B,F} \cdot B e_{B,F}\rangle.$$

Subtract. The integrand of the difference is zero outside E. Moreover $\|A - B\| \le \|A\| + \|B\| \le 2M$, so its value is at most $2M|e_{B,F}|^2$ on E. This proves the one-sided estimate.

D2: Short proof

Answer 9.5. Estimate an integral over a small set

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.4.

Task. Put $\alpha = \delta/(2+\delta)$. Use Hölder's inequality and the given higher-integrability bound to show $\langle \mathbf{1}_E |e_{B,F}|^2\rangle \le C^2|F|^2(|E|/|Y|)^\alpha$.

Solution. Use conjugate exponents $p = (2+\delta)/2$ and $p' = (2+\delta)/\delta$. With normalized integration, Hölder gives

$$\langle \mathbf{1}_E |e|^2\rangle \le \left(\langle |e|^{2+\delta}\rangle\right)^{2/(2+\delta)} \left(\langle \mathbf{1}_E\rangle\right)^{\delta/(2+\delta)}.$$

The first factor is $\|e\|_{2+\delta}^2 \le C^2|F|^2$; the second is $(|E|/|Y|)^\alpha$. The positive exponent α is the reason a small fraction correction has a small effect even along varying microstructures.

D2: Short proof

Answer 9.6. Obtain a uniform tensor estimate

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.4, 9.5, 1.9.

Task. Exchange A and B in the last two steps. Deduce $\|A_* - B_*\| \le 2MC^2(|E|/|Y|)^\alpha$.

Solution. The preceding estimates give an upper bound on $F \cdot (A_* - B_*)F$. Exchanging the coefficients gives the same bound for its negative, with the same constant since the ellipticity bounds agree. Thus

$$|F \cdot (A_* - B_*)F| \le 2MC^2|F|^2(|E|/|Y|)^\alpha.$$

The difference of the two effective tensors is self-adjoint. For a self-adjoint finite-dimensional map T, the spectral theorem gives $\|T\| = \sup_{|F|=1} |F \cdot TF|$. Take this supremum to get the result. In particular, the constant is uniform over the sequence of geometries.

D2: Short proof

Answer 9.7. Correct a binary fraction exactly

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.11.

Task. A binary indicator has fraction θ_n. Explain how to change it on a set of normalized measure $|\theta_n - \theta|$ so the new fraction is exactly θ, even when θ is irrational. Hint: slice the phase to be removed by $\{0 < x < a\}$ and use continuity of its measure in a (B11, B12).

Solution. If $\theta_n > \theta$, choose a subset of the phase-one region with measure $|Y|(\theta_n - \theta)$ and switch it to phase zero. There is enough phase one since $\theta_n - \theta \le \theta_n$. If $\theta_n < \theta$, choose a subset of the phase-zero region with measure $|Y|(\theta - \theta_n)$ and switch it to phase one. There is enough phase zero since $\theta - \theta_n \le 1 - \theta_n$.

A measurable region $S \subset Y$ has a subset of every measure between zero and $|S|$: the function $a \mapsto |S \cap \{0 < x < a\}|$ is continuous, nondecreasing, and has endpoint values $0, |S|$ (continuity follows by dominated convergence). Apply the intermediate value theorem. Extend the new indicator periodically; cell-edge choices affect only a null set.

D2: Short proof

Answer 9.8. Pass to exact-fraction recovery

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.6, 9.7.

Task. Separated-scale recovery supplies tensors $A_{n,*} \to C_\theta$ and fractions $\theta_n \to \theta$. Use the last two steps to show $C_\theta \in G_\theta(L_0, L_1)$.

Solution. Change each binary coefficient A_n to a binary coefficient $\widetilde{A}_n$ of exact fraction θ as above. Both still take only the fixed values L_0, L_1, so they share the same ellipticity constants. They differ on a set E_n of normalized measure $|\theta_n - \theta|$. The uniform estimate gives

$$\|\widetilde{A}_{n,*} - A_{n,*}\| \le 2MC^2|\theta_n - \theta|^\alpha \longrightarrow 0.$$

The triangle inequality now implies $\widetilde{A}_{n,*} \to C_\theta$. By definition, G_θ is the closure of tensors of periodic binary coefficients at this exact fraction. This proves membership.

D1: Direct

Answer 9.9. Select a periodic witness

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.8, 9.3, 4.8.

Task. Let $g_\theta = \ell(C_\theta) > 0$. Show any exact-fraction periodic approximant B_χ with $\|B_\chi - C_\theta\| < 3g_\theta/16$ satisfies $\ell(B_\chi) > g_\theta/2$.

Solution. The continuity estimate for the functional gives

$$|\ell(B_\chi) - g_\theta| \le \tfrac{8}{3}\|B_\chi - C_\theta\| < \tfrac{8}{3}\tfrac{3g_\theta}{16} = \tfrac{1}{2}g_\theta.$$

Therefore $\ell(B_\chi) > g_\theta/2 > 0$. The exact-fraction sequence from the preceding step eventually satisfies this tolerance. Choose one such index; its periodic coefficient field is the required witness.

D1: Direct

Answer 9.10. Exclude every laminate limit

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.9, 5.17.

Task. All finite laminate tensors satisfy $\ell = 0$. Explain why the tensor B_χ just chosen cannot be a limit of them, even if their fractions vary.

Solution. If laminate tensors T_k converged to B_χ in operator norm, continuity would give $\ell(B_\chi) = \lim_k \ell(T_k) = 0$. This contradicts the strictly positive value established above. The proof used no condition on the total fraction of T_k, so it excludes the closure of all finite laminates. In particular, it excludes the smaller closure at the prescribed fraction θ.

D2: Short proof

Answer 9.11. Add uncoupled potentials in the cell problem

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 1.1, 1.5.

Task. For $m > 16$ put $L_i^{(m)} = L_i \oplus I$ on the original and extra columns. Show the effective tensor of the same indicator is $B_\chi \oplus I$.

Solution. Split the mean gradient as (F, G) and the corrector as (u, v) in the two column groups. The energy is the sum

$$\langle (F + \nabla u) \cdot L_\chi (F + \nabla u) \rangle + \langle |G + \nabla v|^2 \rangle.$$

The minimizations are independent. The first minimum is $F \cdot B_\chi F$. In the second, the mean of ∇v is zero, so it equals $|G|^2 + \langle |\nabla v|^2 \rangle$ and is minimized at $\nabla v = 0$. The total minimum is $F \cdot B_\chi F + |G|^2$, exactly the quadratic form of $B_\chi \oplus I$.

D2: Short proof

Answer 9.12. Add uncoupled potentials in a laminate

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.2, 2.7, 9.11.

Task. Show a laminate of $C_1 \oplus I, C_2 \oplus I$ is $\mathrm{Lam}_n(C_1, C_2; \lambda) \oplus I$ for every normal n.

Solution. Split the compatible jump variable as (z, w) in the two column groups. For a unit n, the extra part of the weighted layer energy is

$$\begin{aligned} &\lambda |G + (1-\lambda) n \otimes w|^2 + (1-\lambda) |G - \lambda n \otimes w|^2 \\ &\qquad = |G|^2 + \lambda(1-\lambda)|w|^2. \end{aligned}$$

Its minimum is $|G|^2$, independently of z. The original block gives the original laminate tensor. The argument includes the endpoint proportions by direct evaluation. Iteration proves the statement for every finite hierarchy; passage to tensor limits preserves it. Extending ξ, η by zero preserves their energy difference, so the same separation holds for every $m \geq 16$ without changing the spatial dimension.

10 Complementary energy and planar duality

Prerequisites. The ordered phases and detecting loads: Step 4.4–Step 4.7; layer compatibility/equilibrium: Step 2.1–Step 2.7; Fourier weak derivatives: Step 6.10–Step 6.12.

Definition lookup. The setup defines rotation, curl, the planar dual, and the two quadratic branches. The lookup below recalls the exact phase matrices and explains Fourier and operator notation.

Supplied background. B3, B4, B6, B10. Each key links to its statement, scope, and reference.

Setup 10.1. Manuscript: Section 6.1 and Proposition 6.2. Tensors act on $E = \mathbb{R}^{2\times 16}$. For a self-adjoint positive-definite tensor C, define its planar dual by

$$R_\perp = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C^{\mathrm{d}} = R_\perp C^{-1} R_\perp^\top.$$

Rotation acts on spatial rows and satisfies $R_\perp^\top R_\perp = I$. The fields j, w below belong to $L^2_{\mathrm{per}}(Y;E)$. Define curl columnwise by $\operatorname{curl} w = \partial_x w_y - \partial_y w_x$, with derivatives understood in distributions. The Fourier convention is $\widehat{w}(k) = \langle w(x) e^{-\mathrm{i}k\cdot x}\rangle$. Parseval and Fourier differentiation are supplied in B6.

For $F = (f, g) \in \mathbb{R}^{2\times 32}$, define $q_i(F) = f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g$ and $W = \min(q_0, q_1)$; the same index $i \in \{0, 1\}$ chooses both terms. A laminate has a unit normal n and proportions $\lambda, 1-\lambda$, $0 \le \lambda \le 1$. Its primal gradients have continuous tangential rows and its fluxes have continuous normal rows. Denote its effective tensor by C_λ.

Setup 10.2. Lookup: $Y = (0, 2\pi)^2$, $k \in \mathbb{Z}^2$, and the dot product $k \cdot \widehat{w}(k)$ contracts spatial rows separately in every potential column. Curl and divergence are distributional derivatives as in Module 8's lookup (B8). The statement $\nabla v = w$ will be verified by finite Fourier truncations and an H^1 limit; $\widehat{v}(0) = 0$ fixes the mean. An expression such as $R_\perp C^{-1} R_\perp^\top$ denotes composition of operators on the spatial-first gradient space, in the written order. From Step 3.11–Step 4.7, $S = P^\top P$, P has full column rank, $L_0 = \operatorname{diag}(I_{16}, D)$, $L_1 = \operatorname{diag}(I_{16} - \rho P P^\top, D)$, $0 < \rho < 2/3$, $D = \operatorname{diag}(1, 1/2, 1/3, 1/4) \otimes I_4$, and $\xi_x = P S^{-1} e_1$, $\xi_y = 0$. Thus $P^\top \xi_x = e_1 =: e_+$.

D1: Direct

Answer 10.1. See why the original minimum is trivial

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 4.4.

Task. Given $L_1 \preceq L_0$, prove $\min(f \cdot L_0 f, f \cdot L_1 f) = f \cdot L_1 f$.

Solution. The order means $f \cdot (L_0 - L_1) f \ge 0$ for every f. Thus the second quadratic is always the smaller one. The minimum is a single positive quadratic, hence convex and therefore both rank-one convex and quasiconvex. Its two envelopes coincide with itself. A second, competing preference is needed, which is the purpose of complementary energy.

D1: Direct

Answer 10.2. Express complementary energy as a gradient energy

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 5.14.

Task. For positive-definite C, show C^{d} is positive definite and $(R_\perp p) \cdot C^{\mathrm{d}}(R_\perp p) = p \cdot C^{-1} p$.

Solution. For any nonzero g, the vector $R_\perp^\top g$ is nonzero because rotation is invertible. Hence

$$g \cdot C^{\mathrm{d}} g = (R_\perp^\top g) \cdot C^{-1} (R_\perp^\top g) > 0.$$

Symmetry follows by transposition. Substituting $g = R_\perp p$ and using $R_\perp^\top R_\perp = I$ gives the claimed identity. It converts the energy written in the flux variable p into a quadratic energy of the second gradient block.

D1: Direct

Answer 10.3. Rotate equilibrium into zero curl

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 1.6, 1.12.

Task. If $\operatorname{div} j = 0$ and $p = \langle j \rangle$, show $w = R_\perp(j - p)$ has zero curl and zero mean.

Solution. Columnwise, $w_x = j_y - p_y$ and $w_y = -j_x + p_x$. Since p is constant,

$$\operatorname{curl} w = -\partial_x j_x - \partial_y j_y = -\operatorname{div} j = 0.$$

These identities hold in distributions for L^2 fields. The mean is $\langle w \rangle = R_\perp(\langle j \rangle - p) = 0$. The zero mean removes the obstruction to a periodic potential; a nonzero constant vector field would have zero curl but could not be the gradient of a periodic potential.

D2: Short proof

Answer 10.4. Construct the stream potential in Fourier variables

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.3, 6.10.

Task. For a mean-zero, curl-free $w \in L^2_{\mathrm{per}}(Y; E)$, set $\widehat{v}(k) = -\mathrm{i}\, k \cdot \widehat{w}(k)/|k|^2$ for $k \neq 0$ and $\widehat{v}(0) = 0$. Show $\mathrm{i}k\widehat{v}(k) = \widehat{w}(k)$ for all k. Hint: zero curl makes each column of $\widehat{w}(k)$ parallel to k.

Solution. The curl equation says $k_x \widehat{w}_y - k_y \widehat{w}_x = 0$. For nonzero real k, this means the two-component vector $\widehat{w}(k)$ is a (possibly complex) scalar multiple of k in each potential column. Equivalently, $\widehat{w}(k) = k(k \cdot \widehat{w}(k))/|k|^2$. Multiplying the proposed coefficient by $\mathrm{i}k$ gives exactly this projection. At frequency zero, both sides are zero. Thus the candidate gradient coefficients equal those of w. The next step constructs the corresponding H^1 potential.

D3: Guided technical

Answer 10.5. Check regularity and reality of the stream potential

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.4, 6.11, 6.12.

Task. For real $w \in L^2$ in the preceding step, show the constructed v is real and belongs to H^1_{per}. Use Parseval with weight $1 + |k|^2$; finite Fourier truncations justify the weak derivative.

Solution. For $k \neq 0$, Cauchy–Schwarz gives $|\widehat{v}(k)| \leq |\widehat{w}(k)|/|k|$. Since nonzero integer frequencies have $|k| \geq 1$, Parseval yields

$$\sum_k (1 + |k|^2)|\widehat{v}(k)|^2 \leq 2 \sum_k |\widehat{w}(k)|^2 < \infty.$$

This is an H^1 Fourier bound: finite Fourier truncations are Cauchy together with their derivatives in L^2. Their limit has those weak derivatives, since testing against a smooth periodic function and applying Cauchy–Schwarz passes the integration-by-parts identity to the limit (B6). Since w is real, $\widehat{w}(-k) = \overline{\widehat{w}(k)}$; the factor $-\mathrm{i}k$ changes to its complex conjugate at $-k$. Hence $\widehat{v}(-k) = \overline{\widehat{v}(k)}$, which makes v real. Consequently $R_\perp j = R_\perp p + \nabla v$ is an admissible affine-plus-periodic gradient in the second block.

D1: Direct

Answer 10.6. Verify the dual constitutive law

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.2.

Task. For a primal layer with $j = Ce$, set $e^{\mathrm{d}} = R_\perp j$ and $j^{\mathrm{d}} = R_\perp e$. Check $j^{\mathrm{d}} = C^{\mathrm{d}} e^{\mathrm{d}}$.

Solution. Substitute the definitions, preserving the order of the operators:

$$C^{\mathrm{d}} e^{\mathrm{d}} = R_\perp C^{-1} R_\perp^\top R_\perp j = R_\perp C^{-1} j = R_\perp e = j^{\mathrm{d}}.$$

The cancellation uses $R_\perp^\top R_\perp = I$ and $C^{-1} C = I$ in the written order.

D2: Short proof

Answer 10.7. Rotate the interface conditions

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 2.5, 10.3, 10.6.

Task. Put $\tau = R_\perp n$. Show continuity of $n^\top j$ and $\tau^\top e$ implies continuity of $\tau^\top e^{\mathrm{d}}$ and $n^\top j^{\mathrm{d}}$.

Solution. Because $\tau = R_\perp n$ and $R_\perp^\top R_\perp = I$,

$$\tau^\top e^{\mathrm{d}} = (R_\perp n)^\top R_\perp j = n^\top j.$$

Also $n^\top R_\perp = -(R_\perp n)^\top = -\tau^\top$, so $n^\top j^{\mathrm{d}} = -\tau^\top e$. Thus the two desired continuous quantities are exactly the given continuous quantities, up to a harmless minus sign. The same layers, normal, and proportions work in the dual problem.

D2: Short proof

Answer 10.8. Identify the effective tensor of the dual layers

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 1.9, 2.1, 2.4, 2.5, 2.6, 10.6, 10.7.

Task. Primal layer means are $\langle e \rangle = f$ and $\langle j \rangle = C_\lambda f$. Show the effective tensor of the dual layers is C_λ^{d}.

Solution. The rotated layer gradients have a normal jump by Step 10.7, so Step 2.1 and Step 2.4 give a compatible periodic potential. Their fluxes have continuous normal rows, hence satisfy weak equilibrium by Step 2.5. The dual constitutive law and Step 2.6 therefore identify these fields as dual cell minimizers. They have mean gradient $G = R_\perp C_\lambda f$ and mean flux $R_\perp f$. The map $f \mapsto G$ is invertible: its inverse is $G \mapsto C_\lambda^{-1} R_\perp^\top G$, since C_λ is positive definite. Therefore, for every dual mean gradient G, its mean flux is

$$R_\perp C_\lambda^{-1} R_\perp^\top G = C_\lambda^{\mathrm{d}} G.$$

This identifies the effective tensor, and proves that taking the planar dual commutes with lamination with the *same* normal and proportions.

D1: Direct

Answer 10.9. Check what the two branches mean

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.2, 4.2.

Task. Show each $q_i(f,g) = f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g$ is a positive-definite homogeneous quadratic with its only minimum at zero.

Solution. Both tensors in the two terms are symmetric positive definite. Thus the sum is positive whenever $(f,g) \neq (0,0)$ and is zero at $(0,0)$. Bilinearity gives $q_i(sf, sg) = s^2 q_i(f,g)$ for every real s. It is a quadratic form on the full direct sum of the two gradient blocks, namely $\mathbb{R}^{2\times 32}$. The term "two-well" in this manuscript refers to two quadratic branches, not to two different minimizing matrices; both branches minimize at the origin.

D1: Direct

Answer 10.10. The primal term prefers phase one

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 4.4, 4.5.

Task. Use $L_0 - L_1 = \operatorname{diag}(\rho P P^\top, 0)$ and $P^\top \xi_x = e_+$ to show $q_1(\xi, 0) < q_0(\xi, 0)$.

Solution. The difference is $q_0(\xi,0) - q_1(\xi,0) = \rho |P^\top \xi_x|^2 = \rho > 0$, because $P^\top \xi_x = e_+$ is a unit vector and $\rho > 0$. With the second block zero there is no complementary term, so the ordered constituents have exactly their original phase preference.

D2: Short proof

Answer 10.11. The dual term prefers phase zero

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 3.12, 4.5, 10.2.

Task. On range P, $PP^\top$ has eigenvalues $1/2, 3/2$. Use $\xi_x \in \operatorname{range} P \setminus \{0\}$ and $A_1 = I - \rho P P^\top$ to show $q_0(0, R_\perp \xi) < q_1(0, R_\perp \xi)$.

Solution. Use the complementary-energy identity:

$$q_1(0, R_\perp \xi) - q_0(0, R_\perp \xi) = \xi_x \cdot (A_1^{-1} - I)\xi_x.$$

To transfer the spectrum of S to the physical range, note $PP^\top(Pe) = PSe$. If $Se = \lambda e$, then $Pe \neq 0$ is a $PP^\top$ eigenvector with eigenvalue λ. An eigenbasis of S maps to a basis of range P because P is injective. Hence on that range the eigenvalues of $PP^\top$ are $1/2$ and $3/2$. Thus $A_1^{-1} - I$ has eigenvalues $(1-\rho\lambda)^{-1} - 1 > 0$ there, since $0 < \rho\lambda < 1$. Our ξ_x is a nonzero vector in that range, so the difference is strictly positive. The dual term reverses the preferred phase in some directions.

D1: Direct

Answer 10.12. Keep the phase index shared

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.2, 10.9.

Task. For the scalar example $L_0 = 2I_2, L_1 = I_2$ and $|f| = |g| = 1$, compare $\min_i [f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g]$ with $\min_i f \cdot L_i f + \min_i g \cdot L_i^{\mathrm{d}} g$.

Solution. Here $L_0^{\mathrm{d}} = \frac{1}{2} I_2$ and $L_1^{\mathrm{d}} = I_2$. The shared-index branches have values $2 + 1/2 = 5/2$ and $1 + 1 = 2$, so their minimum is 2. Independent choices give $\min(2,1) + \min(1/2, 1) = 3/2$. They define different functions: independent minimization permits the two blocks to use different phases at the same point. The manuscript uses one physical phase choice for both terms.

D1: Direct

Answer 10.13. Keep the coefficients rational

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 4.4, 10.2.

Task. If L_i has rational entries and is invertible, explain why q_i has rational coefficients.

Solution. The inverse formula $L_i^{-1} = \operatorname{adj}(L_i)/\det L_i$ consists of rational operations on rational entries, with a nonzero denominator. Thus L_i^{-1} is rational. Rotation has integer entries, so $L_i^{\mathrm{d}} = R_\perp L_i^{-1} R_\perp^\top$ is rational as well. Both matrices defining the block quadratic q_i are rational.

11 A rank-one convex lower bound

Prerequisites. Compact family and invariant: Step 4.2–Step 4.8; lamination stability: Step 5.13–Step 5.17; dual lamination: Step 10.6–Step 10.8.

Definition lookup. The setup defines E_C, V, rank-one convexity, and the envelope $\mathrm{RC}\, W$. Rotation and the dual are recalled below.

Supplied background. B3, B10, B12. Each key links to its statement, scope, and reference.

Setup 11.1. Manuscript: Proposition 6.2. Write $F = (f, g) \in \mathbb{R}^{2\times 32}$, with $f, g \in \mathbb{R}^{2\times 16}$. The symbol p denotes a constant matrix in $\mathbb{R}^{2\times 16}$. Modules 4–5 give a nonempty compact family $\mathcal{K}$ containing L_0, L_1, stable under simple lamination, and satisfying $\mu I \preceq C \preceq MI$ with $0 < \mu \le M$. Its energy invariant is $\xi \cdot C\xi = \eta \cdot C\eta$. Define

$$E_C(f, g) = f \cdot Cf + g \cdot C^{\mathrm{d}} g, \qquad V(F) = \min_{C\in\mathcal{K}} E_C(F).$$

Step 11.2 verifies attainment of this minimum. A finite real-valued function h is rank-one convex if

$$h(\lambda F_1 + (1-\lambda)F_2) \le \lambda h(F_1) + (1-\lambda)h(F_2)$$

for $0 \le \lambda \le 1$ whenever $F_1 - F_2 = n \otimes z$, where $|n| = 1$ and $z \in \mathbb{R}^{32}$. These jumps include zero ($z = 0$); a nonzero rank-one factorization is normalized by replacing n, z with $n/|n|, |n|z$. The envelope $\mathrm{RC}\, W$ is the pointwise supremum of finite real-valued rank-one convex minorants of W. Module 10 shows that the laminate of the duals equals the dual of the primal laminate with the same normal and proportions.

Setup 11.2. Lookup: $F = (f, g)$ denotes two 2×16 column blocks, so $|F|^2 = |f|^2 + |g|^2$. Rotation is $R_\perp = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ on spatial rows and $C^{\mathrm{d}} = R_\perp C^{-1} R_\perp^\top$ (Step 10.2). The obstacle is $W(F) = \min_{i=0,1} E_{L_i}(F)$ (Step 10.9). Compactness and the extreme value theorem are the elementary background in B12; continuity of inversion is explicitly proved in Step 11.2. The norm of a tensor is its operator norm, and all order comparisons are between self-adjoint positive-definite tensors.

D1: Direct

Answer 11.1. Bound the paired matrices uniformly

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.2, 10.2, 5.14.

Task. Put $K_0 = \max(M, \mu^{-1})$. Show the matrix defining E_C has operator norm at most K_0, uniformly over $C \in \mathcal{K}$.

Solution. The eigenvalues of C^{-1} lie in $[1/M, 1/\mu]$. Orthogonal conjugation by $R_\perp$ preserves eigenvalues and operator norm, so $\|C^{\mathrm{d}}\| \le 1/\mu$. The paired matrix is the direct sum $\mathrm{diag}(C, C^{\mathrm{d}})$ on the two blocks. The norm of such a direct sum is the larger of the two block norms, at most $\max(M, 1/\mu) = K_0$. There is also the uniform lower bound $\min(\mu, 1/M)|F|^2 \le E_C(F)$.

D2: Short proof

Answer 11.2. Show the minimum is attained

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.3, 11.1.

Task. Show that $V(F)$ is attained by a tensor $C_F \in \mathcal{K}$.

Solution. The map $C \mapsto C^{-1}$ is continuous on the uniformly positive family. One direct estimate is

$$C_1^{-1} - C_2^{-1} = C_1^{-1}(C_2 - C_1)C_2^{-1}, \qquad \|C_1^{-1} - C_2^{-1}\| \le \mu^{-2}\|C_1 - C_2\|.$$

It follows that $C \mapsto E_C(F)$ is continuous. A continuous real-valued function on the nonempty compact set $\mathcal{K}$ attains its minimum by the extreme value theorem. This permits the minimizing tensor choices used in the rank-one convexity proof.

D1: Direct

Answer 11.3. Check homogeneity

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.2.

Task. Prove $V(sF) = s^2 V(F)$ for every real s.

Solution. For every C, the quadratic energy satisfies $E_C(sF) = s^2 E_C(F)$. If $s \neq 0$, multiplying every value in a minimization by the positive constant s^2 multiplies the minimum by that same constant. Thus $V(sF) = s^2 \min_C E_C(F) = s^2 V(F)$. At $s = 0$, both sides are zero.

D2: Short proof

Answer 11.4. Continuity of the minorant

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.1, 11.2.

Task. Show $|V(F) - V(G)| \le K_0(|F| + |G|)|F - G|$. Hint: use a minimizing tensor at one endpoint, then exchange the endpoints.

Solution. For the symmetric paired matrix A_C, expand

$$E_C(F) - E_C(G) = (F - G) \cdot A_C F + G \cdot A_C(F - G).$$

Its absolute value is at most $K_0(|F|+|G|)|F-G|$ by the uniform norm bound. Choose a minimizer C_G at G. Since $V(F) \le E_{C_G}(F)$,

$$V(F) - V(G) \le E_{C_G}(F) - E_{C_G}(G) \le K_0(|F| + |G|)|F - G|.$$

Exchange F, G for the opposite bound. This proves the claimed estimate and continuity.

D1: Direct

Answer 11.5. Compare the minorant with the obstacle

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.4, 10.9.

Task. Use $L_0, L_1 \in \mathcal{K}$ to prove $V \le W$.

Solution. The minimization defining V includes each of the two choices $C = L_i$. Therefore $V(F) \le E_{L_i}(F) = q_i(F)$ for $i = 0, 1$. Taking their minimum gives $V(F) \le \min(q_0(F), q_1(F)) = W(F)$. The next steps establish rank-one convexity using lamination.

D1: Direct

Answer 11.6. Split an arbitrary rank-one jump

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 2.1, 2.2.

Task. If $F_1 - F_2 = n \otimes z$ in $\mathbb{R}^{2\times 32}$ with $|n| = 1$, write $z = (a, b) \in \mathbb{R}^{16} \times \mathbb{R}^{16}$. Identify the jumps and weighted means in the f and g blocks.

Solution. Splitting columns gives $f_1 - f_2 = n \otimes a$ and $g_1 - g_2 = n \otimes b$. Both have the same spatial normal n. For $f_\lambda = \lambda f_1 + (1-\lambda) f_2$, solve these two linear relations to get

$$f_1 = f_\lambda + (1-\lambda) n \otimes a, \qquad f_2 = f_\lambda - \lambda n \otimes a.$$

The identical formulas hold for g with b in place of a. Thus both pairs are compatible trial layer gradients with their respective prescribed mean, for one and the same laminate geometry.

D2: Short proof

Answer 11.7. Apply the two laminate trial inequalities

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.6, 2.7, 10.8.

Task. Set $C_\lambda = \mathrm{Lam}_n(C_1, C_2; \lambda)$ for $C_1, C_2 \in \mathcal{K}$. Put $F_\lambda = \lambda F_1 + (1-\lambda) F_2$. For the compatible fields in the preceding step, prove $E_{C_\lambda}(F_\lambda) \le \lambda E_{C_1}(F_1) + (1-\lambda) E_{C_2}(F_2)$.

Solution. The primal laminate minimum is no larger than the value at the given trial gradients f_1, f_2, hence

$$f_\lambda \cdot C_\lambda f_\lambda \le \lambda f_1 \cdot C_1 f_1 + (1-\lambda) f_2 \cdot C_2 f_2.$$

For the g block, Module 10 identifies the effective tensor of the dual layers as C_λ^{d}. Their normal and proportions are the same. Therefore

$$g_\lambda \cdot C_\lambda^{\mathrm{d}} g_\lambda \le \lambda g_1 \cdot C_1^{\mathrm{d}} g_1 + (1-\lambda) g_2 \cdot C_2^{\mathrm{d}} g_2.$$

Adding proves the desired inequality. The two blocks need not minimize at the trial fields: admissibility alone gives the upper bounds.

D2: Short proof

Answer 11.8. Conclude rank-one convexity

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.2, 11.5, 11.7, 5.13, 5.15.

Task. Choose C_i attaining $V(F_i)$ in the preceding step. Show V is rank-one convex and hence $V \le \mathrm{RC}\, W$.

Solution. The laminate tensor C_λ belongs to $\mathcal{K}$ by its stability under lamination. Thus it is an available competitor in $V(F_\lambda)$:

$$\begin{aligned} V(F_\lambda) &\le E_{C_\lambda}(F_\lambda) \\ &\le \lambda E_{C_1}(F_1) + (1-\lambda) E_{C_2}(F_2) \\ &= \lambda V(F_1) + (1-\lambda) V(F_2). \end{aligned}$$

This holds for every rank-one jump in the full 2×32 space, including jumps with both blocks nonzero. Together with $V \le W$, it means that V is one member of the class whose supremum defines $\mathrm{RC}\, W$. Therefore $V \le \mathrm{RC}\, W$ pointwise.

D2: Short proof

Answer 11.9. Complete the square

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.7, 10.2.

Task. Use $\xi \cdot C\xi = \eta \cdot C\eta$ to prove

$$E_C(\xi, R_\perp p) = 2\eta \cdot p + (p - C\eta) \cdot C^{-1}(p - C\eta).$$

Solution. The complementary-energy identity gives $E_C(\xi, R_\perp p) = \xi \cdot C\xi + p \cdot C^{-1}p$. Expand the square on the proposed right side:

$$\begin{aligned}(p - C\eta) \cdot C^{-1}(p - C\eta) &= p \cdot C^{-1}p - p \cdot \eta - \eta \cdot p + \eta \cdot C\eta \\ &= p \cdot C^{-1}p - 2\eta \cdot p + \eta \cdot C\eta.\end{aligned}$$

The tensor is self-adjoint, so the cross terms have the indicated form. Replace $\eta \cdot C\eta$ by $\xi \cdot C\xi$, using precisely the laminate-family energy invariant. Adding $2\eta \cdot p$ proves the identity.

D1: Direct

Answer 11.10. Obtain a lower bound for the rank-one convex envelope

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.8, 11.9.

Task. Deduce $\mathrm{RC}\, W(\xi, R_\perp p) \geq V(\xi, R_\perp p) \geq 2\eta \cdot p$ for every p.

Solution. Positive definiteness of C^{-1} makes the square in the preceding identity nonnegative for every $C \in \mathcal{K}$. Consequently every value being minimized in $V(\xi, R_\perp p)$ is at least $2\eta \cdot p$, so its minimum has the same lower bound. The first inequality follows from $V \leq \mathrm{RC}\, W$, already proved globally.

D2: Short proof

Answer 11.11. Understand equality for the minorant

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.2, 11.9, 11.10.

Task. Show $V(\xi, R_\perp p) = 2\eta \cdot p$ if and only if some $C \in \mathcal{K}$ satisfies $p = C\eta$.

Solution. If such C exists, its square is zero in the completed-square identity. It gives an upper bound $V(\xi, R_\perp p) \leq 2\eta \cdot p$, and the preceding lower bound forces equality. Conversely, choose a tensor attaining the minimum of V. If that minimum equals $2\eta \cdot p$, its nonnegative square must be zero. Positive definiteness of C^{-1} then forces $p - C\eta = 0$.

12 A periodic trial and the strict envelope gap

Prerequisites. Exact witness: Step 9.9; stream potential: Step 10.3–Step 10.5; lower bound: Step 11.10.

Definition lookup. The setup defines quasiconvex minorants, the test convention, $\mathrm{QC}\,W, p, g_\chi, \overline{F}$, and the two cell fields. The lookup below recalls the obstacle and the meaning of strong convergence.

Supplied background. B1, B6, B10, B11. Each key links to its statement, scope, and reference.

Setup 12.1. Manuscript: Section 6.3. Keep $W = \min(q_0, q_1)$ on $\mathbb{R}^{2\times 32}$. A finite real-valued, Borel measurable, locally bounded function h is quasiconvex if

$$h(F) \le |Q|^{-1} \int_Q h(F + \nabla\psi), \qquad Q = (0,1)^2,$$

for every $F \in \mathbb{R}^{2\times 32}$ and every Lipschitz $\psi : Q \to \mathbb{R}^{32}$ that vanishes on the boundary. The envelope $\mathrm{QC}\,W$ is the pointwise supremum of these functions with $h \le W$. Such functions are locally Lipschitz and rank-one convex, and their Jensen inequality holds for periodic $W^{1,\infty}$ tests (B10). In particular,

$$h(F) \le \langle h(F + \nabla\varphi)\rangle$$

for smooth periodic $\varphi : Y \to \mathbb{R}^{32}$. The test field has values in a compact set, so the integrand is bounded. Periodic smoothing (B1) gives smooth $\varphi_k \to \varphi$ strongly in H^1 for every $\varphi \in H^1_{\mathrm{per}}(Y;\mathbb{R}^{32})$.

Choose the binary witness χ from Module 9, with effective tensor B_χ and $g_\chi = \ell(B_\chi) > 0$. For $\zeta \in \{\xi, \eta\}$ let $e_\zeta = \zeta + \nabla u_\zeta$ be its cell gradient. Set $j_\eta = L_\chi e_\eta$, $p = B_\chi \eta$, and $\overline{F} = (\xi, R_\perp p)$. Module 11 gives $\mathrm{RC}\,W(\overline{F}) \ge 2\eta \cdot p$.

Setup 12.2. Lookup: $W(F) = \min(q_0(F), q_1(F))$ with $q_i(f,g) = f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g$; these are the branches in Step 10.9. In Step 12.3, A, B are gradient matrices, whereas in Step 12.4 they are fields of such matrices. Strong L^2 convergence means $\|A_k - A\|_2 \to 0$; strong H^1 convergence additionally includes convergence of both weak derivatives. The constant K_0 bounds both branch matrices, as in Step 11.1. The two loads and detector are in Step 4.7–Step 4.8; e_ζ, j_ζ are the cell gradient and equilibrium flux of Step 1.6.

D1: Direct

Answer 12.1. Bound one quasiconvex minorant

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 10.9.

Task. For a smooth periodic φ and any quasiconvex minorant $h \le W$, show $h(F) \le \langle W(F + \nabla\varphi)\rangle$.

Solution. Apply the periodic Jensen inequality supplied in B10 to get $h(F) \le \langle h(F + \nabla\varphi)\rangle$. At each point $h(F + \nabla\varphi) \le W(F + \nabla\varphi)$. Integrate this pointwise inequality to obtain the claim. The required integrability of h on the smooth test follows from its local Lipschitz continuity and the compact range of the test; W is continuous with quadratic growth, so the smooth test has finite energy.

D1: Direct

Answer 12.2. Take the supremum over minorants

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 12.1.

Task. Deduce $\mathrm{QC}\,W(F) \le \langle W(F+\nabla\varphi)\rangle$ for every smooth periodic φ.

Solution. The right side of the preceding inequality is independent of the chosen minorant h. If every number in a set is bounded above by the same number, its supremum has that upper bound as well. Taking the supremum of $h(F)$ gives the assertion.

D2: Short proof

Answer 12.3. Control the change in the obstacle

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 11.4, 10.9.

Task. If the matrices of q_0, q_1 have norms at most K_0, prove $|W(A)-W(B)| \le K_0(|A|+|B|)|A-B|$.

Solution. For either symmetric quadratic matrix Q_i,

$$|q_i(A) - q_i(B)| = |(A-B)\cdot Q_i A + B\cdot Q_i(A-B)| \le K_0(|A|+|B|)|A-B|.$$

Choose an index minimizing at B. It bounds $W(A)-W(B)$ by the corresponding quadratic difference. Exchange A, B to bound the negative difference. The displayed estimate follows. The argument is the same as the continuity proof for V and requires no smoothness of the minimum.

D2: Short proof

Answer 12.4. Pass energies through strong L^2 convergence

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 12.3.

Task. If $A_k \to A$ in $L^2(Y)$, show $\langle W(A_k)\rangle \to \langle W(A)\rangle$.

Solution. Integrate the preceding pointwise estimate and apply Cauchy–Schwarz:

$$\langle |W(A_k) - W(A)|\rangle \le K_0(\|A_k\|_2 + \|A\|_2)\|A_k - A\|_2.$$

Strong convergence makes the second factor tend to zero and keeps the first factor bounded, by the triangle inequality. The absolute integral difference therefore tends to zero.

D2: Short proof

Answer 12.5. Allow H^1 periodic trials

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 12.2, 12.4.

Task. Use Step 12.4 and periodic smoothing to extend the trial inequality to every $\varphi \in H^1_{\mathrm{per}}$.

Solution. Choose smooth periodic $\varphi_k \to \varphi$ strongly in H^1. Then $F+\nabla\varphi_k \to F+\nabla\varphi$ strongly in L^2. For each k the smooth trial inequality gives

$$\mathrm{QC}\,W(F) \le \langle W(F+\nabla\varphi_k)\rangle.$$

The right side converges to $\langle W(F+\nabla\varphi)\rangle$ by the previous step. Taking the limit proves the desired inequality.

D2: Short proof

Answer 12.6. Make the joint field admissible

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 1.4, 1.9, 10.3, 10.4, 10.5.

Task. Show $(e_\xi, R_\perp j_\eta) = \overline{F} + \nabla(u_\xi, v)$ for a periodic pair of H^1 potentials.

Solution. The first block is $e_\xi = \xi + \nabla u_\xi$ by the cell problem. The second field j_η is in L^2 because the coefficient is bounded and $e_\eta \in L^2$. It is divergence-free and has mean $p = B_\chi \eta$. The stream-potential construction of Module 10 gives $R_\perp j_\eta = R_\perp p + \nabla v$ with periodic $v \in H^1$. Putting the blocks together gives the stated identity with mean gradient $\overline{F} = (\xi, R_\perp p)$. Both correctors use the same coefficient field L_χ.

D1: Direct

Answer 12.7. Identify the complementary energy pointwise

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 10.2.

Task. Use $j_\eta = L_\chi e_\eta$ to show $j_\eta \cdot L_\chi^{-1} j_\eta = e_\eta \cdot L_\chi e_\eta$ almost everywhere.

Solution. At almost every point L_χ equals one of the invertible symmetric constituents. Substitute the flux and use its inverse:

$$j_\eta \cdot L_\chi^{-1} j_\eta = (L_\chi e_\eta) \cdot e_\eta = e_\eta \cdot L_\chi e_\eta.$$

Both sides are integrable by boundedness of the tensors and square integrability of the cell gradients. Thus their cell averages agree as well.

D1: Direct

Answer 12.8. Use one branch at each point

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 10.2, 10.9, 12.6.

Task. For $e = (e_\xi, R_\perp j_\eta)$, prove $W(e) \le e_\xi \cdot L_\chi e_\xi + j_\eta \cdot L_\chi^{-1} j_\eta$ almost everywhere.

Solution. Where $\chi = i$, the minimum $W(e) = \min(q_0(e), q_1(e))$ is no greater than $q_i(e)$. The first term of q_i is $e_\xi \cdot L_i e_\xi$. The complementary-energy identity converts its second term into $j_\eta \cdot L_i^{-1} j_\eta$. Since the same indicator chooses i for both blocks, their sum is exactly the proposed right side.

D2: Short proof

Answer 12.9. Evaluate the averaged trial energy

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 1.10, 12.5, 12.6, 12.7, 12.8.

Task. Deduce $\mathrm{QC}\, W(\overline{F}) \le 2\eta \cdot p - g_\chi$.

Solution. The joint field is an admissible H^1 trial, so the envelope trial inequality and the preceding branch comparison give

$$\begin{aligned} \mathrm{QC}\, W(\overline{F}) &\le \langle W(e) \rangle \\ &\le \langle e_\xi \cdot L_\chi e_\xi \rangle + \langle e_\eta \cdot L_\chi e_\eta \rangle \\ &= \xi \cdot B_\chi \xi + \eta \cdot B_\chi \eta. \end{aligned}$$

The last equality is the cell-energy identity at the two loads. Since $g_\chi = \eta \cdot B_\chi \eta - \xi \cdot B_\chi \xi$ and $p = B_\chi \eta$, this sum equals $2\eta \cdot p - g_\chi$.

D1: Direct

Answer 12.10. Finish the strict gap

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 11.10, 12.9.

Task. Combine the last estimate with the Module 11 lower bound to prove $\mathrm{RC}\,W(\overline{F}) - \mathrm{QC}\,W(\overline{F}) \geq g_\chi > 0$.

Solution. The lower bound is $\mathrm{RC}\,W(\overline{F}) \geq 2\eta \cdot p$. The upper bound is $\mathrm{QC}\,W(\overline{F}) \leq 2\eta \cdot p - g_\chi$. Subtracting yields the inequality. Since $g_\chi > 0$, it also gives the strict ordering

$$\mathrm{QC}\,W(\overline{F}) < \mathrm{RC}\,W(\overline{F}).$$

All values are finite: zero is an admissible minorant because $W \geq 0$, and both envelopes lie below the finite obstacle W.

D2: Short proof

Answer 12.11. Check the order of choices

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 7.22, 8.18, 9.8, 9.9.

Task. List the order in which ρ, N, the prescribed fraction θ, and its final periodic approximation are chosen in the composite proof.

Solution. First choose the explicit matrices and profile. Choose and fix the rational contrast $\rho = \kappa/32$ so the smooth difference is positive. With this ρ fixed, choose a sufficiently large finite N so the binary fraction-$1/2$ composite keeps the positive difference. These constituents and detecting gradients are now fixed. For an arbitrary prescribed $0 < \theta < 1$, form C_θ and select one exact-fraction periodic approximant sufficiently close to retain its positive difference. That last approximation may depend on θ.

D1: Direct

Answer 12.12. Recognize the scope of the conclusion

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 9.10, 9.11, 12.10.

Task. Does this proof show that sixteen potentials are necessary, or that the two envelopes have been evaluated exactly? Explain briefly.

Solution. No. It constructs a counterexample at sixteen potentials and extends it to larger numbers, but gives no impossibility result at smaller numbers. Similarly, it places the two envelopes on opposite sides of the same number with a positive margin; it does not identify their exact values.

The two proofs assembled

The composite theorem. Modules 3–5 construct fixed constituents and a continuous linear functional ℓ which vanishes on the compact lamination-stable family $\mathcal{K}$. Induction and continuity extend that equality to every finite hierarchy and every tensor limit, with unrestricted total fractions. Modules 6–7 construct a smooth coefficient field in $\mathcal{K}$ whose effective tensor has strictly positive ℓ: four Clifford factors first appear at order five, the four surviving Fourier paths have the same positive sign, and the convergent series controls the remainder.

With the contrast chosen in Module 7, Module 8 takes sufficiently many binary stripes. An equilibrium flux correction has an L^2 constitutive residual of order N^{-1}. Coercivity converts that residual into errors in the two selected mean fluxes, so a finite stripe pattern keeps positive ℓ. Module 9 layers this fraction-1/2 material with a pure phase, recovers periodic patterns at the exact prescribed fraction, and selects one finite approximant retaining the strict sign. It lies outside the entire laminate closure. Direct sums with an identity block give every $m \geq 16$. This is Theorem 1.1 with the constituent pair fixed before θ.

The energy theorem. Module 10 pairs gradient energy and rotated complementary energy into two positive-definite quadratics on $\mathbb{R}^{2\times 32}$. The same phase index chooses both terms. Planar duality commutes with a laminate using the same normal and proportions. In Module 11 this fact and stability of $\mathcal{K}$ show that $V = \min_{C\in\mathcal{K}} E_C$ is rank-one convex on the full gradient space. The energy equality on $\mathcal{K}$ then gives, for every p,

$$\mathrm{RC}\, W(\xi, R_\perp p) \geq V(\xi, R_\perp p) \geq 2\eta \cdot p.$$

For the actual binary witness take $p = B_\chi \eta$. Module 12 uses its corrector at ξ and its rotated equilibrium flux at η as one admissible periodic trial. The shared indicator gives

$$\mathrm{QC}\, W(\xi, R_\perp p) \leq 2\eta \cdot p - \ell(B_\chi) < 2\eta \cdot p.$$

Combining the estimates proves Theorem 6.1 and the quantitative bound $\mathrm{RC}\, W - \mathrm{QC}\, W \geq \ell(B_\chi) > 0$ at the selected mean gradient.

Answers to the optional oral checkpoints

1. The two energies define the continuous linear detector ℓ. It is zero on every finite laminate and therefore on every tensor limit. A strictly positive value contradicts such approximation, regardless of fractions.
2. The normalization gives $P^\top P S^{-1} e = e$. The two loads select corresponding entries in two equal diagonal blocks of $S^{-1} - H$, making their energies agree throughout $\mathcal{K}$.
3. The shifted laminate formula uses only sums and inverses inside $\mathcal{S}$. Fourier interaction instead involves ordered ordinary products; four distinct generators can produce $\Omega \notin \mathcal{S}$.
4. Four independent generator coefficients permit a four-row determinant. The chosen representation acts on eight auxiliary coordinates; four four-row factors form the sixteen-potential material. Pairing primal and dual gradients doubles the potentials to thirty-two. These are sufficient construction dimensions, not proved minimal dimensions.
5. The detector annihilates products of at most three generator-bearing factors. A term of perturbation order p contains $p - 1$ multiplier factors, so the first detectable one is order five. Tensor corrections of lower order need not vanish.
6. Its partial sums have squared slopes $0, 1, 4, 9$. Zero steps produce zero multipliers or repeated determinant rows; signed counts restrict the remaining multisets; vertical endpoint steps eliminate all but four paths.

7. Weak convergence of a coefficient or of its phase indicator does not identify a homogenized tensor: the minimizing gradients also change. The corrected-flux residual controls that change quantitatively at the two loads actually used in the proof.
8. The smooth positive margin and the constants in the residual estimate depend on ρ. Once ρ is fixed, the error tends to zero with N, so it can be made less than half that fixed margin.
9. Membership in G_θ is membership in a norm closure. One must choose an exact-fraction periodic tensor close enough that continuity preserves the strict inequality. That selected approximant is the witness.
10. The constituents are ordered, $L_1 \preceq L_0$, so phase one wins pointwise everywhere. Complementary energy reverses the preference in some directions and produces competition.
11. It is not the minimum operation alone. A rank-one jump gives compatible trials in both blocks with the same normal; lamination stability and planar duality provide a competitor in $\mathcal{K}$ obeying the required convexity inequality.
12. At $p = B_\chi \eta$, the lower bound is $\mathrm{RC}\, W \geq 2\eta \cdot p$; the periodic trial gives $\mathrm{QC}\, W \leq 2\eta \cdot p - \ell(B_\chi)$. The positive difference between the two loads supplies the strict gap.

Background results and precise references

These are the general results supplied by the course. Each entry states the version used, identifies its applications, and points to a book section or a named theorem. The numbered steps work out their applications to the coupled system used in the construction.

B1. Periodic Sobolev spaces, completeness, Poincaré, and density

Input. A Hilbert space is a complete inner-product space. H^1 with its usual inner product is Hilbert, as are its closed linear subspaces. On the torus, smooth periodic functions are dense in H^1_{per}. For a zero-mean periodic v, $\|v\|_2 \le C_P\|\nabla v\|_2$. **Periodic version.** Extend periodically and convolve with a smooth approximate identity. Convolution preserves periodicity and converges in L^2 for the function and each weak derivative; subtract its mean if necessary. Poincaré also follows directly from Parseval: the zero Fourier mode is absent and $|k| \ge 1$ on every remaining mode. The finite number of potential columns causes no change: apply scalar statements columnwise. **Where used.** Module 1, Step 2.6, Step 8.14, and Step 12.5. **Reference.** Evans [4], Chapter 5, especially §§5.2–5.3 (Sobolev spaces and approximation), §5.5 (traces), and §5.8.1 (Poincaré inequalities). The preceding convolution and Fourier arguments explain the periodic adaptations; see also B6.

B2. Lax–Milgram

Input. If $\mathcal{H}$ is a real Hilbert space, a is a bounded bilinear form with $a(v,v) \ge c\|v\|^2_{\mathcal{H}}$ for $c > 0$, and b is a bounded linear functional, there is a unique $u \in \mathcal{H}$ with $a(u,v) = b(v)$ for all $v \in \mathcal{H}$. **Where used.** Step 1.1–Step 1.4; the bounds and the coercivity constant needed for this application are checked there. **Reference.** Evans [4], §6.2.1, the Lax–Milgram theorem.

B3. Symmetric matrices, operator norms, and order

Input. A real symmetric matrix has an orthonormal eigenbasis; its trace is the eigenvalue sum, its operator norm is the largest absolute eigenvalue, and its largest Rayleigh quotient is the largest eigenvalue. A positive-definite matrix has a positive square root and inverse square root, obtained by applying the scalar functions to that eigenbasis. For symmetric A, $\|A\| = \sup_{|z|=1} |z^\top A z|$. The identity $z^\top P^\top P z = |Pz|^2$ explains Gram matrices and rank. Congruence preserves order: $A \preceq B \Longrightarrow X^\top A X \preceq X^\top B X$. The last statement follows by evaluating the two forms at Xz. For an invertible matrix, $A^{-1} = \operatorname{adj}(A)/\det A$; thus a rational invertible matrix has a rational inverse. **Where used.** Modules 1, 3–5, 9–11. Order reversal under inversion is proved, rather than assumed, in Step 5.14. **Reference.** Horn–Johnson [6], Chapter 0 (basic matrix algebra and determinants), §2.5 (spectral theorem for normal matrices), §§4.1–4.2 (Hermitian matrices and variational characterizations), §5.6 (matrix norms), §§7.1–7.2 (positive definiteness), and §7.7 (matrix order).

B4. Separated scales, laminate composition, and the G-closure

Input. For uniformly coercive, bounded, self-adjoint quadratic gradient systems, first homogenize the fine periodic materials in each part of a coarser periodic pattern, then homogenize the coarse pattern. There are actual patterns in the original phases whose tensors converge to this iterated effective tensor. Fractions converge to the weighted fractions. The same statement holds for a finite hierarchy. With exact fraction correction it gives

$$C_i \in G_{\theta_i} \quad \Longrightarrow \quad \mathrm{Lam}_n(C_1, C_2; \lambda) \in G_{\lambda\theta_1 + (1-\lambda)\theta_2}.$$

Rational layer directions admit periodic patterns in a square cell; general directions follow by

approximation and continuity of the laminate formula. An aligned cell is used when deriving that formula in Module 2. **Scope.** The needed class consists of full-gradient quadratic systems with finitely many potential columns and a common coercivity constant. Milton's abstract field framework covers coupled systems; Allaire develops the locality and closure mechanism in conductivity and elasticity settings. **Where used.** Step 2.12, Step 9.8, and the interpretation of finite hierarchies. Step 9.4–Step 9.8 prove the exact-fraction correction from the weaker convergent-fraction recovery statement. **References.** Milton [10], Chapters 9, 12–13 and §§30.1–30.2; Allaire [2], §2.1.2, Theorem 2.1.2 and Lemmas 2.1.5–2.1.7 (locality, stability, and volume-fraction issues).

B5. Absolutely convergent Fourier series

Input. Summable coefficient sequences form a complete space under their ℓ^1 norm. For finite matrices use the operator norm in each summand. Absolute convergence permits rearrangement and gives a uniformly convergent Fourier series. Convolution estimates follow by the triangle inequality and $\|AB\| \le \|A\|\|B\|$; the order of matrix factors must be retained. These estimates are proved in Step 6.3–Step 6.7. Completeness of matrix-valued ℓ^1 follows by applying scalar completeness to each entry; all norms on a finite-dimensional matrix space are equivalent. **Where used.** Modules 6–7, including the geometric series and its differentiated versions. **References.** Katznelson [7], Chapter I, §6 (absolutely convergent Fourier series) and §9 (the higher-dimensional torus); Folland [5], Chapter 6, for completeness of L^p, applied to counting measure.

B6. Parseval, Fourier differentiation, and H^1 limits

Input. On $Y = (0, 2\pi)^2$ with normalized measure,

$$\|f\|_2^2 = \sum_{k \in \mathbb{Z}^2} |\widehat{f}(k)|^2.$$

For vector and matrix fields, use the Euclidean and Frobenius norms, respectively. A square-summable coefficient sequence is the Fourier sequence of an L^2 field. If $\sum_k (1 + |k|^2)|\widehat{v}(k)|^2 < \infty$, its finite Fourier truncations converge in H^1, and the weak derivative has coefficients $ik_j\widehat{v}(k)$. Conversely, testing a weak derivative against exponentials gives that coefficient relation. Finite sums verify differentiation first; their H^1 limit verifies it for the resulting function. For real fields, $\widehat{f}(-k) = \overline{\widehat{f}(k)}$. **Where used.** Step 6.10–Step 6.14, Module 7's frequency expansions, and Step 10.3–Step 10.5. **References.** Katznelson [7], Chapter I, §5 (square-summable Fourier series) and §9; Evans [4], §5.2, for weak derivatives. The explicit truncation argument is given in the answers to Step 6.12 and Step 10.5.

B7. The Cauchy determinant identity

Input. For distinct x_i and distinct y_j, with all $x_i + y_j \neq 0$,

$$\det\left[\frac{1}{x_i + y_j}\right]_{i,j=1}^{n} = \frac{\prod_{i<k}(x_k - x_i)\prod_{j<l}(y_l - y_j)}{\prod_{i,j}(x_i + y_j)}.$$

Where used. Step 7.19–Step 7.20, with $x = (0, 1, 4, 9)$ and $y = (1, 2, 3, 4)$. Pulling j out of column j accounts for the extra column factors in the course's determinant. **Reference.** Krattenthaler [8], §2.1, equation (2.7), the Cauchy double alternant. Its two Vandermonde factors may both be written with reversed differences without changing the sign.

B8. Lipschitz weak derivatives, traces, and distributions

Input. A Lipschitz function has bounded weak first derivatives agreeing with its derivatives almost everywhere. Its product with a smooth function obeys the weak product rule. The minimum of two Lipschitz functions is Lipschitz, since $\min(a,b) = (a+b-|a-b|)/2$ and absolute value is Lipschitz. On a cell divided into finitely many strips, Lipschitz formulas with matching traces have the piecewise weak derivatives: integration by parts cancels the interface terms. Distributional mixed partial derivatives commute, since both differentiate the smooth test function twice with the same result. **Where used.** Step 2.4, Step 8.5–Step 8.10, and Step 10.3. The answers carry out the interface cancellation and mixed-derivative tests. **References.** Evans [4], §5.2 (weak derivatives), §5.3 (approximation), §5.5 (traces), and §5.8.3 (Lipschitz functions); Folland [5], Chapter 9, for distributions.

B9. Uniform higher integrability for elliptic systems

Input. If A is periodic, measurable, self-adjoint, and $\mu I \preceq A \preceq MI$ with $0 < \mu \le M$ on $\mathbb{R}^{2\times m}$, there are $\delta > 0$ and $C < \infty$, depending only on the ellipticity bounds and the fixed dimensions, such that every periodic cell minimizer satisfies

$$\|F + \nabla u_{A,F}\|_{2+\delta} \le C|F|.$$

The constants are independent of phase geometry. This is the periodic form of local higher integrability for uniformly elliptic systems. **Bridge to this version.** Extend A periodically and apply the local system estimate to $U(x) = F^{\top}x + u_{A,F}(x)$ on a finite covering of the torus by balls with larger concentric balls in the periodic extension. Its gradient is the cell field, and the cell energy bound gives $\|F + \nabla u_{A,F}\|_2 \le \sqrt{M/\mu}\,|F|$. Periodicity controls the larger balls by finitely many cells. This yields the displayed estimate with constants uniform in A. **Where used.** Step 9.5–Step 9.8; Hölder's inequality then supplies the explicit modulus for small-set coefficient changes. **References.** Allaire [2], Theorem 1.3.41 and Remark 1.3.42 (the latter explicitly discusses elliptic systems); Meyers–Elcrat [9] for system higher-integrability theory.

B10. Quasiconvexity, rank-one convexity, and periodic testing

Input and hypotheses. Use finite real-valued, Borel measurable, locally bounded quasiconvex functions, with quasiconvexity defined by the Jensen inequality for zero-boundary $W^{1,\infty}$ tests. Such a function is rank-one convex and locally Lipschitz. For a Borel measurable, locally bounded integrand the usual quasiconvex inequality is equivalent to the periodic $W^{1,\infty}$ version. In particular the inequality holds for smooth periodic tests. Rescaling $\psi(x) = \varphi(2\pi x)/(2\pi)$ transfers the unit-cell assertion to $Y = (0,2\pi)^2$. In Module 12, first apply it to $h \le W$, then use the L^2 continuity of the obstacle W to allow H^1 trials (Step 12.3–Step 12.5). **Convention.** Texts often put the potential index in the first matrix coordinate. Transposing all matrices gives this course's spatial-first convention; it preserves the rank-one condition and the gradient Jensen inequality. **Where used.** Definitions in Modules 11–12, Step 12.1–Step 12.2, and the envelope comparison. **Reference.** Dacorogna [3], §5.2.1 (definitions), Theorem 5.3(i),(iv) (implication to rank-one convexity and local Lipschitz continuity), and §5.2.4, Proposition 5.13 (periodic characterization, including its local boundedness hypothesis).

B11. Integration and L^p estimates

Input. Cauchy–Schwarz is Hölder with exponents $2, 2$; Hölder with $p, q > 1$ and $p^{-1} + q^{-1} = 1$ gives $\langle |fg| \rangle \le \|f\|_p \|g\|_q$ for normalized measure. Fubini applies to the integrable functions used here, and dominated convergence applies under a common integrable bound. The graph of a continuous function $y = g(x)$ has area zero by its zero-length vertical sections. For a measurable $E \subset Y$, the function $a \mapsto |E \cap \{0 < x < a\}|$ is continuous: its increment is bounded by the area of the intervening vertical strip. Its endpoints are 0 and $|E|$, so any intermediate area can be selected. **Where used.** Energy estimates throughout; Step 8.5, Step 9.5, Step 9.7, and Step 12.4. **Reference.** Folland [5], Chapter 2 (Lebesgue integration, dominated convergence, Fubini) and Chapter 6 (L^p and Hölder inequalities). The slicing continuity needed in Step 9.7 also follows from the strip bound just given.

B12. Elementary compactness, continuity, and differentiation of series

Input. A closed bounded set in a finite-dimensional real space is compact. A continuous real function on a nonempty compact set attains its extrema. A continuous function on an interval takes all intermediate values. In finite dimensions all norms are equivalent. For a series of C^1 functions on a periodic cell, uniform convergence of the series of each first partial derivative, together with convergence of the function series at one point, permits differentiation term by term. The last fact follows by integrating the uniformly convergent derivative series along coordinate segments. In Step 6.19–Step 6.22 absolute majorants give the required uniform convergence directly. **Where used.** Step 4.3, Step 6.19–Step 6.22, Step 9.7, Step 11.2, and Step 11.11. **Reference.** Rudin [11], Chapters 2 (compactness), 4 (continuity), and 7 (sequences and series of functions, including differentiation).

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